7451A/61A/61P

Digital Multimeter

Operation Manual

MANUAL NUMBER FOE-00000061B01

Applicable Models
7451A
7461A
7461P

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Safety Summary

To ensure thorough understanding of all functions and to ensure efficient use of this instrument, please read the manual carefully before using. Note that ADC Corporation (hereafter referred to as ADC) bears absolutely no responsibility for the result of operations caused due to incorrect or inappropriate use of this instrument.

If the equipment is used in a manner not specified by ADC, the protection provided by the equipment may be impaired.

- **Warning Labels**

  Warning labels are applied to ADC products in locations where specific dangers exist. Pay careful attention to these labels during handling. Do not remove or tear these labels. If you have any questions regarding warning labels, please ask your nearest ADC dealer. Our address and phone number are listed at the end of this manual.

  Symbols of those warning labels are shown below together with their meaning.

  DANGER: Indicates an imminently hazardous situation which will result in death or serious personal injury.

  WARNING: Indicates a potentially hazardous situation which will result in death or serious personal injury.

  CAUTION: Indicates a potentially hazardous situation which will result in personal injury or a damage to property including the product.

- **Basic Precautions**

  Please observe the following precautions to prevent fire, burn, electric shock, and personal injury.

  - Use a power cable rated for the voltage in question. Be sure however to use a power cable conforming to safety standards of your nation when using a product overseas.
  - When inserting the plug into the electrical outlet, first turn the power switch OFF and then insert the plug as far as it will go.
  - When removing the plug from the electrical outlet, first turn the power switch OFF and then pull it out by gripping the plug. Do not pull on the power cable itself. Make sure your hands are dry at this time.
  - Before turning on the power, be sure to check that the supply voltage matches the voltage requirements of the instrument.
  - Connect the power cable to a power outlet that is connected to a protected ground terminal. Grounding will be defeated if you use an extension cord which does not include a protective conductor terminal.
  - Be sure to use fuses rated for the voltage in question.
  - Do not use this instrument with the case open.
  - If there is any abnormality such as abnormal heart, smoke, smell or noise, immediately stop using the instrument, turn off the power and disconnect the plug from the outlet.

- Do not place anything on the product and do not apply excessive pressure to the product. Also, do not place flower pots or other containers containing liquid such as chemicals near this product.
- When the product has ventilation outlets, do not stick or drop metal or easily flammable objects into the ventilation outlets.
- When using the product on a cart, fix it with belts to avoid its drop.
- When connecting the product to peripheral equipment, turn the power off.

- **Caution Symbols Used Within this Manual**

Symbols indicating items requiring caution which are used in this manual are shown below together with their meaning.

DANGER: Indicates an item where there is a danger of serious personal injury (death or serious injury).

WARNING: Indicates an item relating to personal safety or health.

CAUTION: Indicates an item relating to possible damage to the product or instrument or relating to a restriction on operation.

- **Safety Marks on the Product**

The following safety marks can be found on ADC products.

 : ATTENTION - Refer to manual.

 : Protective ground (earth) terminal.

 : DANGER - High voltage.

 : CAUTION - Risk of electric shock.

- **Replacing Parts with Limited Life**

The following parts used in the instrument are main parts with limited life.
Replace the parts listed below before their expected lifespan has expired to maintain the performance and function of the instrument.
Note that the estimated lifespan for the parts listed below may be shortened by factors such as the environment where the instrument is stored or used, and how often the instrument is used.
The parts inside are not user-replaceable. For a part replacement, please contact the ADC sales office for servicing.

Each product may use parts with limited life.
For more information, refer to the section in this document where the parts with limited life are described.

Main Parts with Limited Life

Part name	Life
Unit power supply	5 years
Fan motor	5 years
Electrolytic capacitor	5 years
LCD display	6 years
LCD backlight	2.5 years
Memory backup battery	5 years

- **Hard Disk Mounted Products**

The operational warnings are listed below.

- Do not move, shock and vibrate the product while the power is turned on.
  Reading or writing data in the hard disk unit is performed with the memory disk turning at a high speed. It is a very delicate process.

- Store and operate the products under the following environmental conditions.
  An area with no sudden temperature changes.
  An area away from shock or vibrations. An area free from moisture, dirt, or dust.
  An area away from magnets or an instrument which generates a magnetic field.

- Make back-ups of important data.
  The data stored in the disk may become damaged if the product is mishandled. The hard disc has a limited life span which depends on the operational conditions. Note that there is no guarantee for any loss of data.

- **Precautions when Disposing of this Instrument**

When disposing of harmful substances, be sure dispose of them properly with abiding by the state-provided law.

Harmful substances: (1) PCB (polycarbon biphenyl)
(2) Mercury
(3) Ni-Cd (nickel cadmium)
(4) Other
Items possessing cyan, organic phosphorous and hexadic chromium and items which may leak cadmium or arsenic (excluding lead in solder).

Example: fluorescent tubes, batterie

Environmental Conditions

This instrument should be only be used in an indoor area which satisfies the following conditions:

- An area free from corrosive gas
- An area away from direct sunlight
- A dust-free area
- An area free from vibrations
- Altitude of up to 2000 m

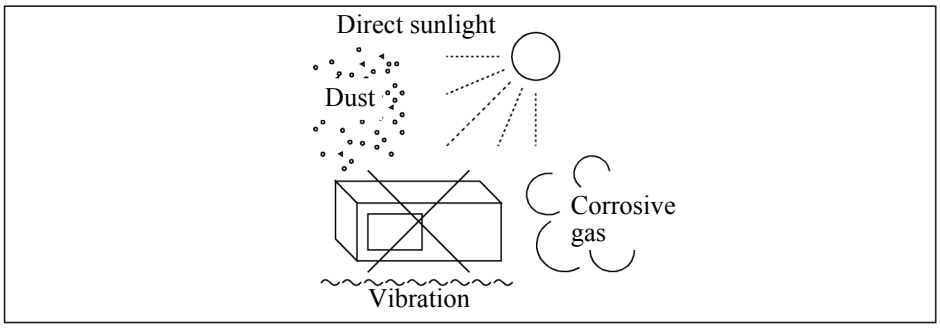


Figure-1 Environmental Conditions

- Operating position

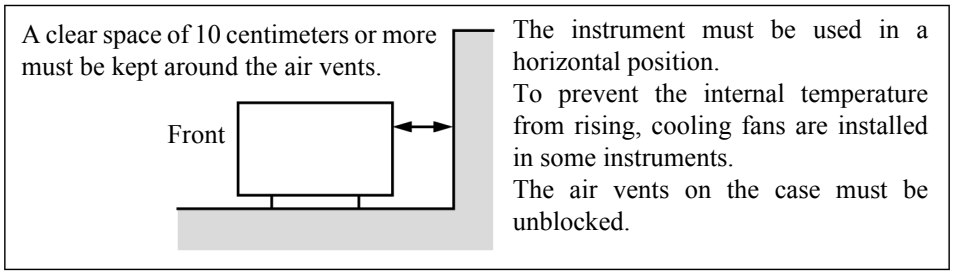


Figure-2 Operating Position

- Storage position

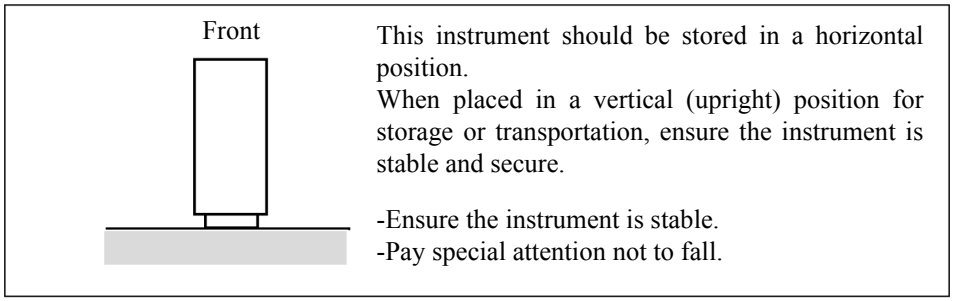


Figure-3 Storage Position

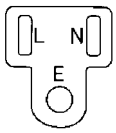
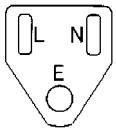
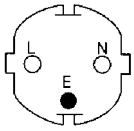
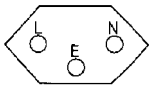
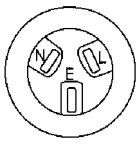
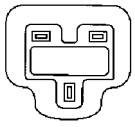
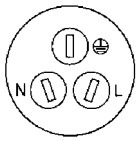
- The classification of the transient over-voltage, which exists typically in the main power supply, and the pollution degree is defined by IEC61010-1 and described below.

  Impulse withstand voltage (over-voltage) category II defined by IEC60364-4-443

  Pollution Degree 2

Types of Power Cable

Replace any references to the power cable type, according to the following table, with the appropriate power cable type for your country.

Plug configuration	Standards	Rating, color and length	Model number (Option number)
	PSE: Japan Electrical Appliance and Material Safety Law	125 V at 7 A Black 2 m (6 ft)	Straight: A01402 Angled: A01412
	UL: United States of America CSA: Canada	125 V at 7 A Black 2 m (6 ft)	Straight: A01403 (Option 95) Angled: A01413
	CEE: Europe DEMKO: Denmark NEMKO: Norway VDE: Germany KEMA: The Netherlands CEBEC: Belgium OVE: Austria FIMKO: Finland SEMKO: Sweden	250 V at 6 A Gray 2 m (6 ft)	Straight: A01404 (Option 96) Angled: A01414
	SEV: Switzerland	250 V at 6 A Gray 2 m (6 ft)	Straight: A01405 (Option 97) Angled: A01415
	SAA: Australia, New Zealand	250 V at 6 A Gray 2 m (6 ft)	Straight: A01406 (Option 98) Angled: ---------
	BS: United Kingdom	250 V at 6 A Black 2 m (6 ft)	Straight: A01407 (Option 99) Angled: A01417
	CCC: China	250 V at 10 A Black 2 m (6 ft)	Straight: A114009 (Option 94) Angled: A114109

Certificate of Conformity

This is to certify, that

Digital Multimeter

7461A, 7451A, 7461P, 7451T

complies with the provisions of the EMC Directive, 2014/30/EC in accordance with EN61326-1, Low Voltage Directive, 2014/35/EC in accordance with EN61010-1/A1, and RoHS Directive, 2011/65/EC in accordance with EN IEC 63000.

Directive: Electro Magnetic Compatibility Directive (EMC):2014/30/EC
Using the following Harmonized Standards
EN 61326-1:2013

Directive: Low Voltage Directive (LVD):2014/35/EC
Using the following Harmonized Standards
EN 61010-1:2010/A1:2019

Directive: RoHS Directive:2011/65/EC
Using the following Harmonized Standards
EN IEC 63000:2018

ADC Corporation

77-1, Miyako Namegawa-machi, Hiki-gun, Saitama 355-0812, JAPAN

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1. INTRODUCTION

This chapter describes the contents of this manual and the product overview of the 7451A/7461A/7461P Digital Multimeter.

1.1 Contents of this Manual

This manual can be used by novices or experienced users of this instrument. This manual can be read from Chapter 1 to learn more about this instrument or it can be used as a reference. Refer to the table of contents, which is found at the beginning of each chapter, and directly jump to the required section.

The contents of each chapter are as follows.

Chapter 1. INTRODUCTION	This chapter describes the contents of this manual and the product overview.
Chapter 2. PRECAUTIONS	This chapter describes precautions when using this instrument. Read this chapter before using this instrument.
Chapter 3. SETUP	This chapter describes how to setup this instrument. After setting up this instrument in an appropriate location, turn on the power and check that this instrument starts correctly.
Chapter 4. QUICK START	This chapter describes the function of each section on the panels of this instrument and how to operate this instrument.
Chapter 5. MENU OPERATION AND FUNCTION DESCRIPTION	This chapter describes the menu structure and functions of this instrument.
Chapter 6. HOW TO USE THE INTERFACE	This chapter describes how to use the interface of this instrument.
Chapter 7. SPECIFICATIONS	This chapter describes the specifications of this instrument.
Chapter 8. MAINTENANCE	This chapter describes how to care for this instrument such as cleaning, calibration, and storage to maintain the high performance and smooth functioning of this instrument. This chapter also describes how to identify problems and the relevant procedures to follow.

1.2 Product Overview

This digital multimeter includes the AD converter of the integration type.

This multimeter is ideal for use as a stand-alone product or in a system.

The main features of this instrument are as follows:

- Simple and easy panel operations, and flexible and various system functions by using menus
- Large and visible fluorescent display
- Wide dynamic range: Maximum display of 1199999 (7461A/7461P) and 319999 (7451A)
- High sensitivity: 0.1 μV (7461A/7461P) and 1 μV (7451A) in the DC voltage measurement
- Wide dynamic range: 1 nA to 3 A (7461A/7461P) and 10 nA to 3 A (7451A) in the DC current measurement
- Low power resistance measurement function that reduces the self-heating effect of the device
- High throughput measurement due to including the DCV-Bch input
- Temperature measurement function with a resolution of 0.001°C using a platinum resistance bulb (Pt100 or JPt100) (7461P only)
- Digital output function that is capable of controlling TTL signals and contact signals by remote commands. (7461P only)
- Wide range of the sampling control: long-interval to high-speed
  Sampling interval: 50 μsec to 1 hour, Integration time: 10 μsec to 10 sec (7461A/7461P)
  Sampling interval: 200 μsec to 1 hour, Integration time: 100 μsec to 10 sec (7451A)
- Level trigger, delta trigger, and pre-trigger functions are included.
- Digitize function and logging function by using the large-capacity memory (7451A/7461A: 10,000 data, 7461P: 20,000 data)
- PC and automatic control devices can be connected easily to this instrument, because the USB, GPIB, trigger input, measurement complete signal output, and comparator calculation output connecters are included as standard with this instrument.
- A large variety of calculation functions that can be combined.
- Four files of setting parameters can be saved.
- Conventional and SCPI command systems are included as standard with this instrument.

2. PRECAUTIONS

This chapter describes precautions when using this instrument. Read this chapter before using this instrument.

CAUTION: ***Using this instrument while it is stood on end or stacked with other products is unstable and dangerous.***
Do not push or place heavy loads on this instrument.

2.1 If a Fault Occurs

If any smoke, smell, or noise emanates from this instrument, disconnect this instrument from the power supply by removing the power cable. Contact an ADC CORPORATION sales representative immediately.

2.2 Removing the Case

The case of this instrument should only be opened by ADC CORPORATION service engineers.

WARNING: ***This instrument contains high-voltage and high temperature parts which may cause electrical shocks or burns.***

2.3 Power Fuse

This instrument is protected from overcurrent by a power fuse.

The power fuse, which is located on the rear panel, cuts the power supply if an overcurrent flows. If the power fuse was blown, remove the power cable and separate between the power supply and this instrument. There may be some problems in this instrument and request an ADC CORPORATION sales representative to repair it.

2.4 Electromagnetic Interference

This instrument may cause electromagnetic interference and affect television and radio. If this instrument's power is turned off and any electromagnetic interference that may be present is reduced, then this instrument has caused the interference.

Electromagnetic interference from this instrument may be prevented by the following precautions.

- Changing the direction of the antenna of the television or radio.
- Placing this instrument on the opposite side of the television or radio.
- Placing this instrument away from the television or radio.
- Using a different power source for the television or radio, and this instrument.

2.5 Note When Turning on the Power

When turning on the power, do not connect any DUTs.

3. SETUP

This chapter describes how to set up this instrument on delivery. The following topics are covered in this chapter.

3.1 Inspection on Delivery

3.2 Installation Environment

3.3 Accessories

3.4 Power Requirements

3.1 Inspection on Delivery

After receiving the product, inspect the outside and the accessories according to the following procedure.

1. Check that the shipping container and the cushioning material are not damaged.

   IMPORTANT: ***If the shipping container or the cushioning material is damaged, keep them until the following inspections are complete.***

2. Check that the outside of the product is not damaged.

   WARNING: ***If any outside components of the product such as the cover, panel (front or rear), power switch, or connector are damaged, do not turn on the power. Doing so may cause an electrical shock.***

3. Check that the standard accessories listed in Table 3-1 are complete and not damaged.

If any of the following occur, contact an ADC CORPORATION sales representative.

- The shipping container or the cushioning material is damaged, or signs of stress are found.
- The outside of the product is damaged.
- The standard accessories are incomplete or are damaged.
- Defects are found in the operation check.

Table 3-1 Standard Accessories

Name	Model	Quantity	Remarks
Power cable	A01402	1	*1
Input cable (Red, Black)	CC010001	Each 1	
Alligator clip adapter (Red, Black)	CC015001	Each 1	
Power Fuse (For 100V/120V)	DFT-AAR25A-1	1	*2 T250mA/250V
Power Fuse (For 220V/240V)	DFT-AAR16A-1		T160mA/250V
Overcurrent protection fuse	DFS-AN3R15A-1	1	F3.15A250V
Operation manual	E7451A/61A/61P	1	This manual

*1: The power cable included in this instrument depends on the option that was specified when this instrument was purchased (see “Safety Summary”).
Quote the model name or option number when ordering the power cable.
*2: Either one is included according to the specified option.

NOTE: Quote the model name of the accessory when ordering.

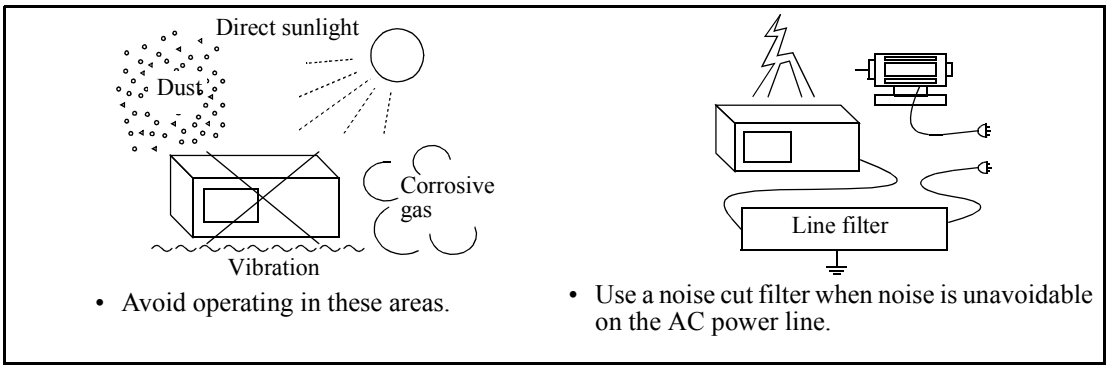
3.2 Installation Environment

This section describes the environment into which this instrument should be installed.

3.2.1 Operating Environment

Install this instrument in an environment in which the following conditions are satisfied.

- Ambient temperature: 0 °C to +50 °C (operating temperature)
  -25 °C to +70 °C (storage temperature)
- Relative humidity: 80 percent or less with no condensation
- An area free from corrosive gas
- An area away from direct sunlight
- A dust-free area
- An area free from vibrations
- A low noise area
  Although this instrument has been designed to withstand a certain amount of noise from the AC power line, it should be used in a low noise area. Use a noise cut filter if ambient noise is unavoidable.



- Avoid operating in these areas.
- Use a noise cut filter when noise is unavoidable on the AC power line.

Figure 3-1 Operating Environment

NOTE: ***Warm-up***
After this instrument reaches room temperature, turn on the power and warm up for 60 minutes or more to perform the high accuracy measurement.

3.2.2 Protecting Against Electrostatic Discharge

To prevent semiconductors from being damaged by electrostatic discharge (ESD), the precautions shown below should be taken. We recommend combining two or more countermeasures to prevent damage from ESD.

(Static electricity can be generated easily by the movement of a person or the friction against insulation.)

Table 3-2 ESD Countermeasures

Human Body	Use a wrist strap (See Figure 3-2).
Work floor	Install a conductive mat, use conductive shoes, and connect to earth. (See Figure 3-2)
Workbench	Install a conductive mat and connect to earth. (See Figure 3-3)

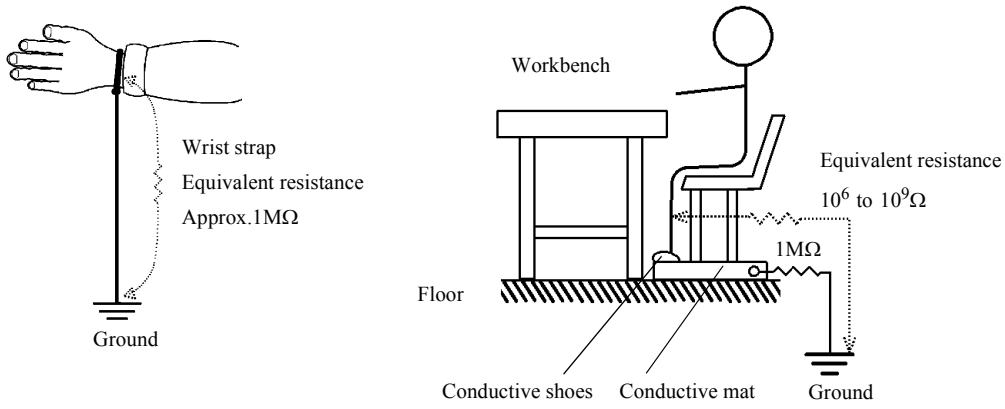


Figure 3-2 Electrostatic Countermeasures for the Work Floor and Human Body

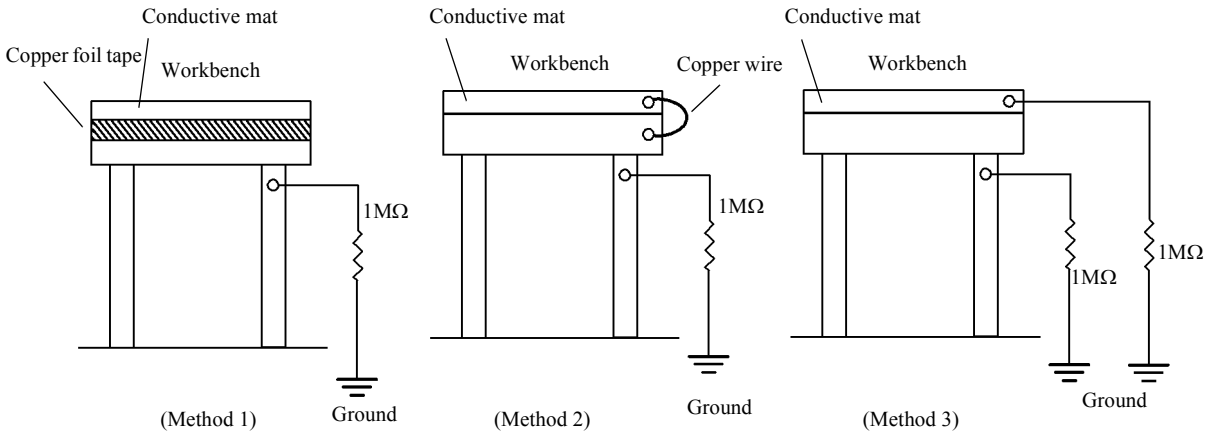


Figure 3-3 Electrostatic Countermeasures for the Workbench

3.3 Accessories

The accessories of this instrument are shown in Table 3-3.

This section describes the accessories necessary to use this instrument.

Table 3-3 Accessories

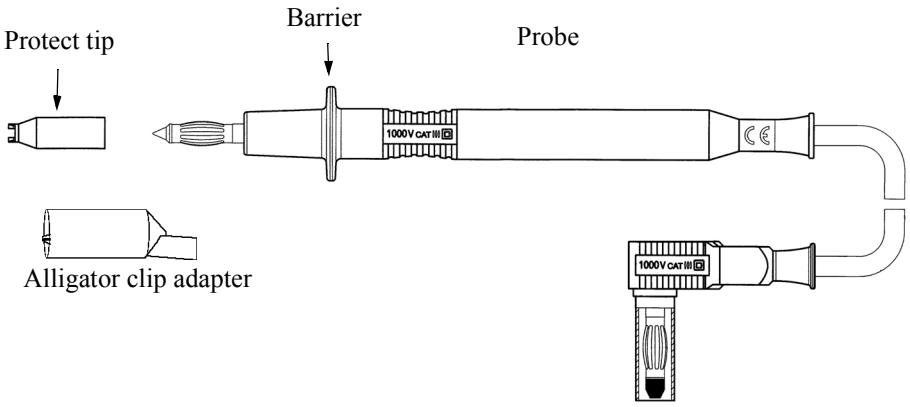
Name	Model	Remarks
Input cable	CC010001	Standard Accessory
	A01001	Shielded cable 33 VAC and 70 VDC or less
	A01006	For 4WΩ measurement
Alligator clip adapter	CC015001	Standard Accessory 33 VAC and 70 VDC or less
Terminal adapter	TR1111	
JIS rack-mount set	A02263	
	A02264	Twins
EIA rack-mount set	A02463	
	A02464	Twins
Panel-mount set	A02039	
	A02040	Twins
Internal temperature detecting sensor	1104-010	JPt100, probe-type
Sheath-type platinum resistance bulb	1104-001	JPt100 Sheath outer diameter: ϕ3.2 mm
	1104-002	JPt100 Sheath outer diameter: ϕ1.0 mm

3.3.1 Input Cable

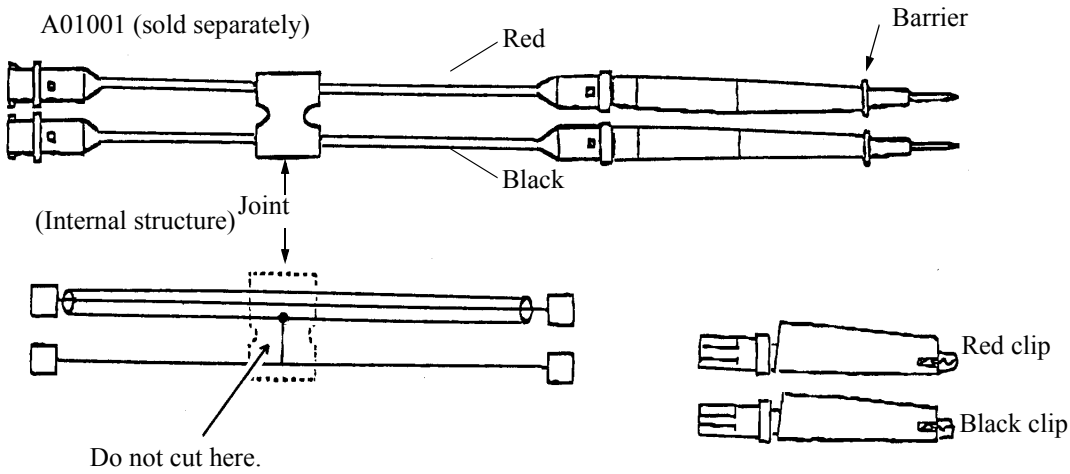
1. The standard input cable (CC010001) consists of red and black and it can be used easily.

CAUTION: ***Handle the input cable with caution because it has a sharp tip. The rating of the alligator clip adapter is 33 VAC and 70 VDC or less.***

WARNING: ***Must not touch the metal part of the input cable to prevent any damage of electrical shock.***



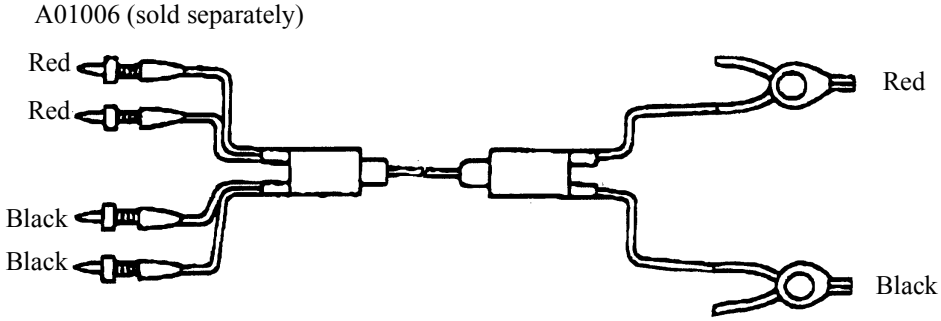
2. To reduce the effects of short-term stability in high resistance (in the order of MΩ) or high sensitivity (in the order of μV) measurements, use the optional input cable A01001 in which the HI side is shielded.



The red wire of A01001 is shielded. Do not cut the joint.

The rating of the probe decreases to 33 VAC and 70 VDC when the alligator clip adapter is attached.

3. Use the dedicated input cable A01006 in the 4WΩ measurement.



3.3.2 Caution when Connecting Peripherals

When making up a system, which consists of two or more devices, be careful of the following:

1. Before connecting this instrument to a controller and peripherals, check that each device operates correctly according to each manual.
2. Do not use too long a cable to connect the instrument or controller.
3. Use shielded cables which are connected between peripherals.

Setting up the USB port

To use the USB interface of this instrument, the USB driver of our ADC instrument must be installed in a personal computer.

The USB driver of the ADC instrument can be downloaded from the following our home page.

URL http://www.adcmt-e.com

For more information on how to install the driver, refer to the operation manual that is included in the driver file.

Compatible Operating Systems:	English-language version of Windows XP Professional
Compatible languages	MS Excel2000 VBA MS Visual Basic 6.0 MS Visual C++6.0

3.4 Power Requirements

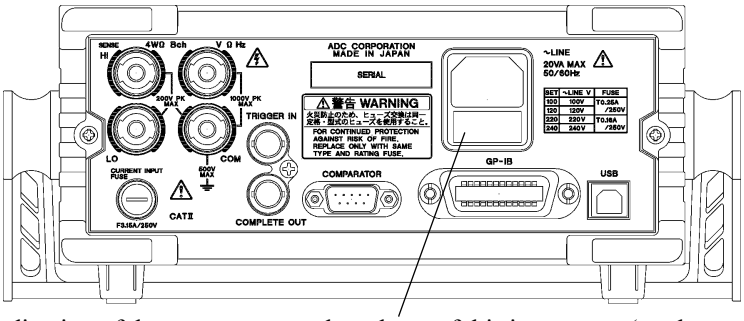
This section describes power requirements and how to connect the power cable.

3.4.1 Checking the Power Supply Voltage

Check that the power supply voltage setting on the rear panel of this instrument accords with the AC power supply voltage to be used.

Table 3-4 Relation Table between AC Power Supply Voltage and Indication of Set Power Supply Voltage

AC power supply voltage	Indication of the set power supply voltage	Specification of the applicable fuse	Frequency	Power consumption
90 V - 110 V	100 V	T250 mA/250 V	48 Hz-66 Hz	20 VA or less
103 V - 132 V	120 V			
198 V - 242 V	220 V	T160 mA/250 V		
207 V - 250 V	240 V			



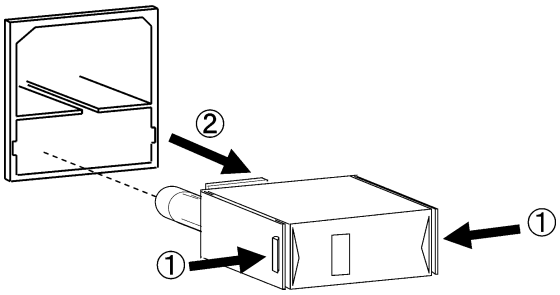
Indication of the set power supply voltage of this instrument (on the rear panel)

Figure 3-4 Indication of the Set Power Supply Voltage

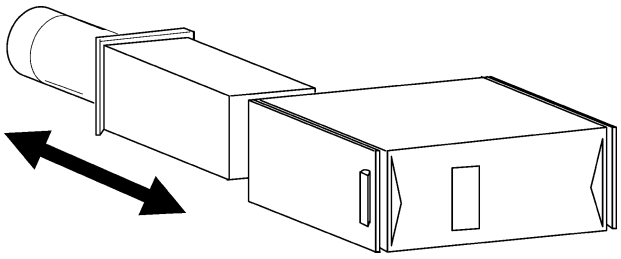
3.4.2 Changing the Power Supply Voltage

1. Remove the fuse holder assembly from the rear panel.

   Push both side of the fuse holder assembly by using flathead screwdrivers (①) and then pull it out (②).

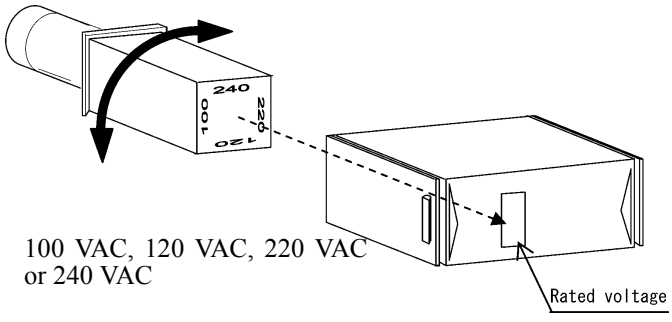


2. Remove the power supply voltage selector from the fuse holder assembly.



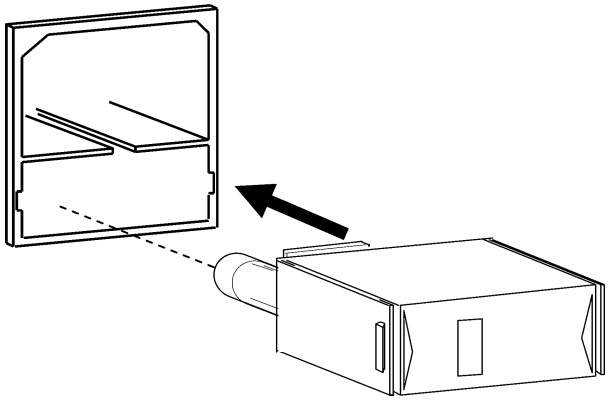
   100 VAC or 120 VAC: T250-mA fuse
   220 VAC or 240 VAC: T160-mA fuse

3. Rotate the power supply selector to be looked the correct voltage from the window.



4. Replace a fuse that is compliant with the specification (see Table 3-1).

5. Replace the fuse holder assembly into the rear panel.



6. Check that the correct power supply voltage is indicated in the window.

3.4.3 Connecting the Power Cable

This instrument includes a three-core power cable with a grounding conductor. To prevent accidents caused by electric shocks, only use the included power cable and securely connect to the ground through a three-pin power outlet.

1. Check that the included power cable is not damaged.

   WARNING: ***Never use a damaged power cable. Doing so may cause an electrical shock.***

2. Connect the AC power connector on the rear panel of this instrument to a three-pin power outlet that has a ground terminal by using the included power cable (See Figure 3-5).

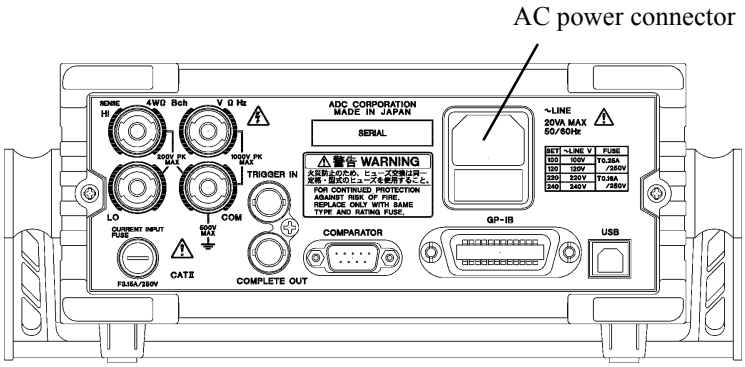


Figure 3-5 Connecting the Power Cable

WARNING:

1. ***Use a suitable power cable for the power supply voltage. Use a power cable that complies with the safety standards in the country where the instrument is used (Refer to "Safety Summary").***
2. ***To prevent any danger of electrical shock, connect the power cable to a three-pin power outlet that is connected to a ground terminal. The instrument will not be grounded if an extension cord, which does not include a ground terminal, is used.***

4. QUICK START

The high speed measurement and high accuracy measurement can be performed in this digital multimeter. To make the performing operations easier, this instrument includes the following functions on the front panel.

Setting the measurement function

Setting the measurement range

Sampling conditions

Setting the NULL and smoothing calculations to ON or OFF.

Chapter 4 describes the front and rear panels and panel operations which do not use MENU. Other settings are set in the layered menus. For more information on function descriptions and how to navigate the menus, refer to Chapter 5.

4.1 Panel Description

4.1.1 Front Panel Description

The front panel consists of the display section, operation keys, and measurement input terminal section. (This is described in the 7461A)

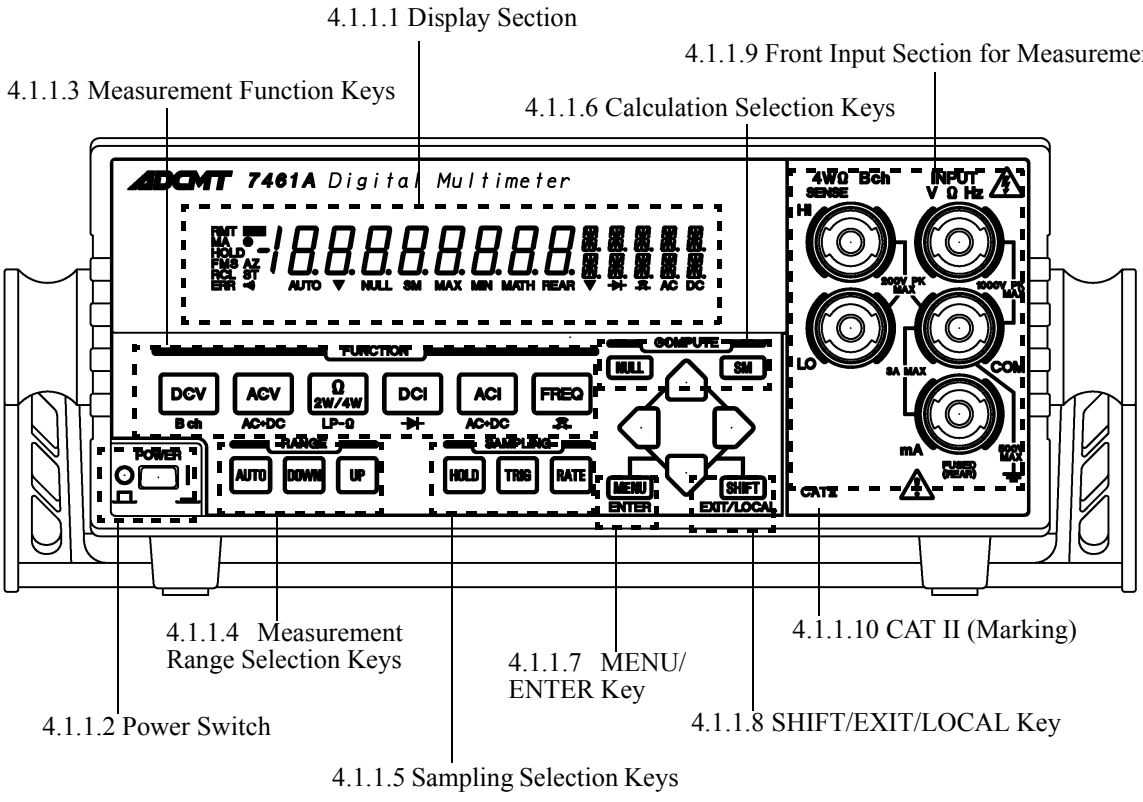


Figure 4-1 Front Panel Description

4.1.1.1 Display Section

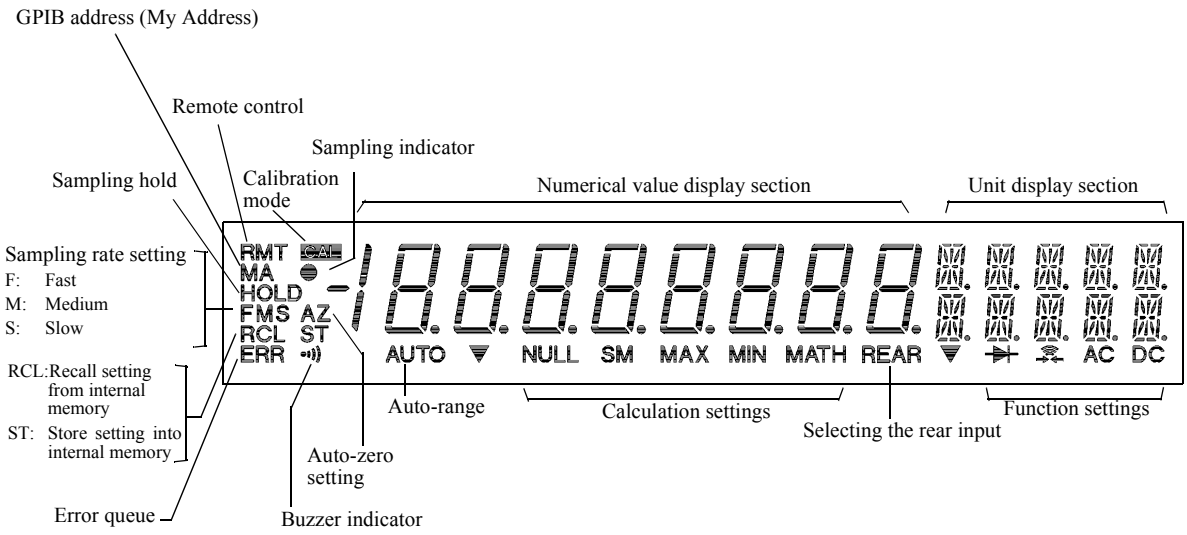


Figure 4-2 Display Section Description

4.1.1.2 Power Switch

1.	POWER switch	Pressing the power switch turns on the power and the power is supplied to this instrument. Re-pressing the power switch turns off the power and the power is shut down.

4.1.1.3 Measurement Function Keys

1.	DCV	Selects the DC voltage measurement (DCV).
2.	ACV	Selects the AC voltage measurement (ACV). The AC filter can be selected from FAST, MED, and SLOW.
3.	2WΩ/4WΩ	Selects the resistance measurement. Either 2WΩ or 4WΩ can be selected alternately.
4.	DCI	Selects the DC current measurement (DCI).
5.	ACI	Selects the AC current measurement (ACI). The AC filter can be selected from FAST, MED, and SLOW.
6.	FREQ	Selects the frequency measurement (FREQ)
7.	Bch (SHIFT+DCV)	Selects the DC voltage measurement (DCV-Bch).
8.	AC + DC (SHIFT + ACV)	Selects the AC+DC coupling or AC coupling in the AC voltage measurement.

9. LP-2WΩ/LP-4WΩ (SHIFT + 2WΩ/4WΩ)
   Selects the low power resistance measurement. Either 2WΩ or 4WΩ can be selected.
10. ⊣▶⊢ (SHIFT + DCI) Selects the diode measurement (⊣▶⊢).
11. AC + DC (SHIFT + ACI) Selects the AC+DC coupling or AC coupling in the AC current measurement.
12. (SHIFT+FREQ) Selects the continuity measurement ( ).
13. TEMP Selects the temperature measurement (TEMP).

4.1.1.4 Measurement Range Selection Keys

1. AUTO Switches the measurement range between auto-range and fixed range.
2. DOWN Decreases the measurement range by one level and fixes the range.
3. UP Increases the measurement range by one level and fixes the range.

4.1.1.5 Sampling Selection Keys

1. HOLD Holds or releases the sampling mode.
2. TRIG Starts the measurement when the sampling mode is set to HOLD.
3. RATE Set the sampling rate to the high (FAST1 or FAST2), medium (MED), or low speed (SLOW1 or SLOW2).

4.1.1.6 Calculation Selection Keys

1. NULL Sets or cancels the NULL calculation.
2. SM Sets or cancels the smoothing calculation.

4.1.1.7 MENU/ENTER Key

1. MENU This key is used to enter into MENU.
2. ENTER Saves the settings when MENU is used.
3. △ ▽ ◁ ▷ Selects the menu and its layer.

4.1.1.8 SHIFT/EXIT/LOCAL Key

1. SHIFT — Sets or cancels the SHIFT mode.

• In the setting mode

2. EXIT — Exits from MENU and returns to the measurement state display. (The data in process of changing the setting is not saved.)

• In the remote operation

3. LOCAL — Changes to the LOCAL operation when the instrument is in the remote state.

NOTE: ***The LOCAL operation cannot be set if the LLO(LOCAL LOCKOUT) command is set from the GPIB interface.***

4.1.1.9 Front Input Section for Measurement

1. INPUT V Ω Hz HI terminal — HI terminal used for the DC voltage, AC voltage, resistance, diode, and frequency measurements (Source HI terminal used for the 4WΩ or temperature measurement)
2. COM terminal — LO terminal used for the DC voltage, AC voltage, DC current, AC current, resistance, diode, and frequency measurements (Source LO terminal used for the 4WΩ or temperature measurement)
3. Bch, 4WΩ SENSE HI terminal — HI terminal used for the DC voltage (DCV-Bch) measurement and SENSE HI terminal used for 4WΩ or temperature measurement
4. Bch, 4WΩ SENSE LO terminal — LO terminal used for the DC voltage (DCV-Bch) measurement and SENSE LO terminal used for 4WΩ or temperature measurement
5. mA terminal — HI terminal used for the DC current and AC current measurements

4.1.1.10 CAT II (Marking)

This display shows that the safety of this instrument satisfies the measurement category II which is compliant with IEC 61010-1 Ed.2. Measurement category II is for measurements performed on circuits directory connected to the low voltage installation.

4.1.2 Rear Panel Description

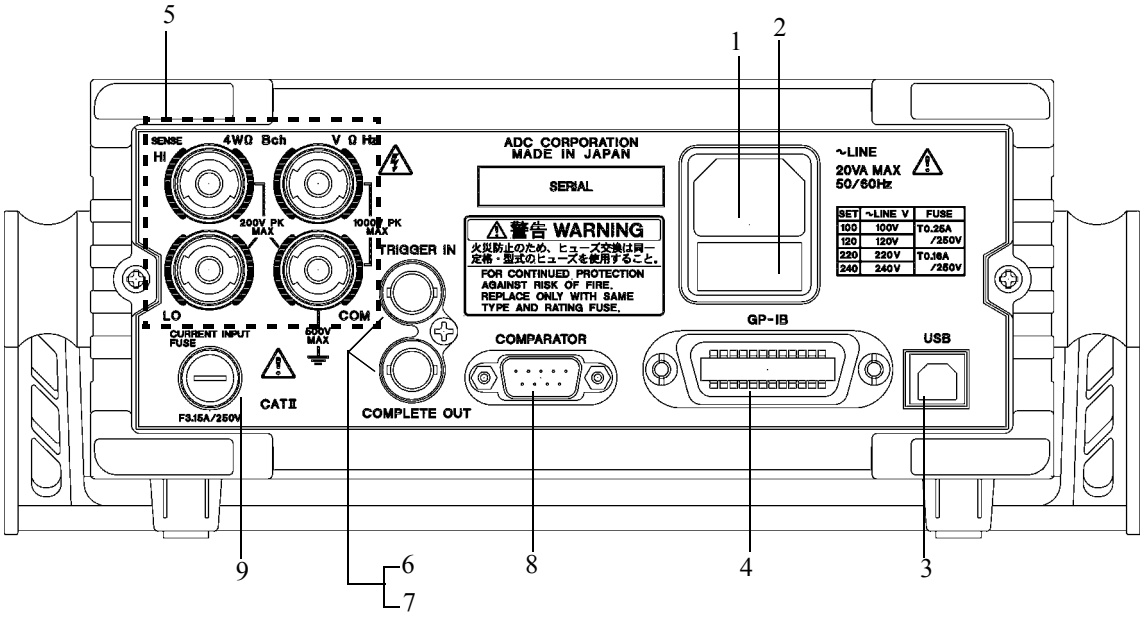


Figure 4-3 Rear Panel Description

1.	Power connector	Used to supply the AC power by connecting the included power cable (A01402).
2.	Fuse holder assembly	The power supply voltage can be selected from 100 V, 120 V, 220 V, and 240 V. A slow-blow fuse is put.
3.	USB connector	Connector for USB Used to output data and control remotely.
4.	GPIB connector	Connector for GPIB Used to output data and control remotely.
5.	Rear input section	Used to connect the input cable. (The current measurement terminal is located on the front panel only.)
6.	TRIGGER IN	Used to input the external trigger.
7.	COMPLETE OUT	Outputs the complete signal of the measurement.
8.	COMPARATOR	Outputs the comparator calculation signal. (The 7461P also outputs the digital signal.)
9.	Overcurrent protection fuse	Protects the overcurrent into the current input terminal.

4.2 Basic Operation

When the power is turned on, this instrument automatically performs the self-test and then proceeds to the measurement sequence.

For more information on the detail of the self-test, refer to Chapter 8, “MAINTENANCE.”

When the self-test is complete correctly, this instrument enters into the measurement state. If any abnormal error occurs, the instrument enters into the error display state. Check the error number by operating MENU and exit from here.

For more information on navigating the menus, refer to Section 5.1, “Menu.”

This instrument saves the set conditions when the power is turned off. The factory defaults are used in the descriptions in Chapter 4. For more information on how to initialize this instrument, refer to Chapter 8.

4.2.1 Measurement Functions

1. DC voltage measurement (DCV)

   Press **DCV**.

Table 4-1 Maximum Allowable Applied Voltage (DCV)

Between V and COM terminals	1000 Vpeak
Between COM terminal and chassis	500 V

Table 4-2 Range and Input impedance (DCV)

Range		Input impedance
7461A/61P	7451A	
100 mV – 10 V	300 mV, 3000 mV	Selecting Hi-Z or 10 MΩ
100 V, 1000 V	30 V – 1000 V	10 MΩ

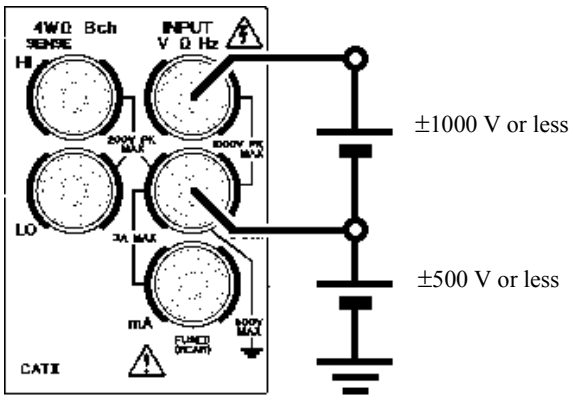


Figure 4-4 DC Voltage Measurement (DCV)

2. DC voltage measurement (DCV-Bch)

1. Press **SHIFT** and **DCV**.
'BDC' is displayed.
The measurement is performed between Bch HI and COM terminals in the fixed range of 10 V.

Table 4-3 Maximum Allowable Applied Voltage (DCV-Bch)

Input terminal	Measurement voltage range	Maximum allowable applied voltage
Between Bch Hi and COM	±12 V	200 Vpeak
Between Bch Lo and COM	±12 V	200 Vpeak
Between Bch Hi and Lo	±12 V	200 Vpeak

Table 4-4 Range and Input impedance

Input terminal	Range	Input impedance
Between Bch HI and LO	10V	Hi-Z
Between Bch HI/LO and COM	10V	Hi-Z

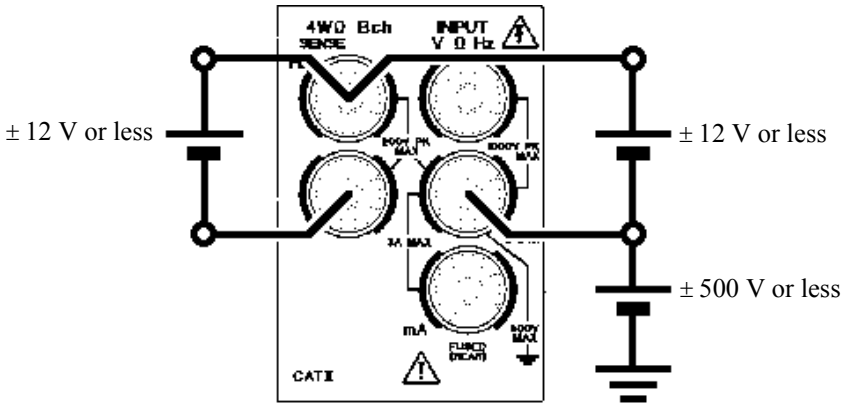


Figure 4-5 DC Voltage Measurement (DCV-Bch)

3. AC voltage measurement (ACV and ACV(AC+DC))

ACV measurement

1. Press **ACV**.
   AC voltage with AC coupling is measured.
2. Pressing continuously **ACV** switches the AC filter in turn.

ACV(AC+DC) measurement

3. Press **SHIFT** and **ACV**.
   ACV and DCV are measured and the root mean square value ($\sqrt{ACV^2+DCV^2}$) is displayed.
   The AC and DC indicators are turned on to show the AC+DC coupling.
4. Pressing continuously **ACV** switches the AC filter in turn.
5. Press **SHIFT** and **ACV** to return to the ACV measurement.

Selecting the AC filter

In this instrument, the AC filter can be selected from three for the accuracy optimization in low frequency and the high speed response.

Select the AC filter in consideration of the measurement signal frequency, necessary measurement accuracy, and response time.

Table 4-5 Selecting the AC Filter

AC filter	Measurement signal frequency	Response time *1	Display
FAST	300 Hz to 300 kHz	100 msec	ACF
MED	20 Hz to 300 kHz	500 msec	ACM
SLOW	20 Hz to 300 kHz	3 sec	ACS

*1: Time until the measurement value reaches within 0.1% of the final value

Table 4-6 Maximum Allowable Applied Voltage (ACV and ACV(AC+DC))

Between V and COM	1000 Vpeak, 700 Vrms, 2.2×10^7 VHz
Between COM and chassis	500 V

Table 4-7 Measurement range

	7461A/61P	7451A
Measurement method	TrueRMS	
Measurement range	5% or more of the range	
Crest factor	5	3

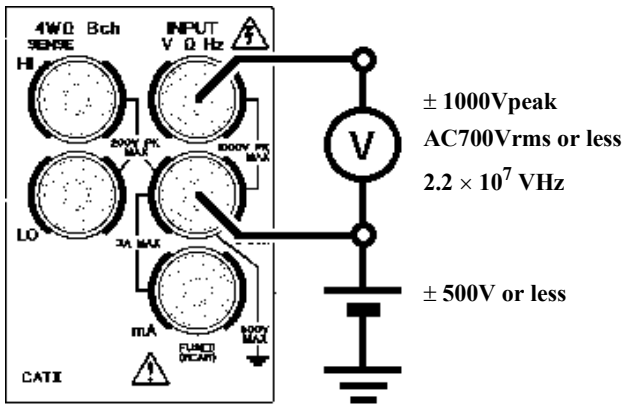


Figure 4-6 AC Voltage Measurement (ACV and ACV(AC+DC))

4. Resistance measurement (2WΩ, 4WΩ, LP-2WΩ, and LP-4WΩ)

1. Press [Ω].
   The resistance is measured by using the 2WΩ or 4WΩ which is set before this measurement.

2. Press [SHIFT] and [Ω].
   The resistance is measured in low-power in which the measurement current is reduced.

3. Press [SHIFT] and [Ω] to return to the 2WΩ and 4WΩ measurement.

4. Press [Ω].
   The 2WΩ and 4WΩ are switched alternately.

NOTE: ***In the following measurement ranges, the LO terminal of 4WΩ SENSE does not recognize the voltage and operates as in 2WΩ measurement.***

7461A: ***1000 kΩ range or more in 4WΩ, 100 kΩ range or more in LP-4WΩ***

7451A: ***3000 kΩ range or more in 4WΩ, 300 kΩ range or more in LP-4WΩ***

Table 4-8 Maximum Allowable Applied Voltage (2WΩ, 4WΩ, LP-2WΩ, LP-4WΩ)

Between Ω and COM	1000 Vpeak *1
Between 4WΩHI and LO	200 Vpeak
Between 4WΩHI/LO and COM	200 Vpeak
Between COM and chassis	500 V

*1: If an over-voltage is applied, the internal circuits may be separated from the input terminals by the protection function.

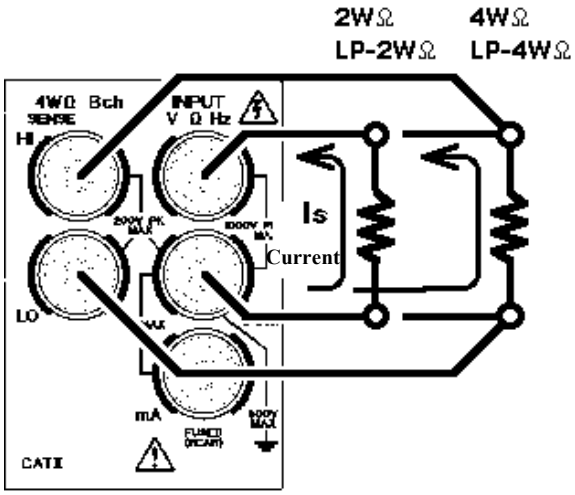


Figure 4-7 Resistance Measurement (2WΩ, 4WΩ, LP-2WΩ, and LP-4WΩ)

5. DC current measurement (DCI)

   Press **DCI**.

   Maximum allowable applied current 3 A

WARNING: ***Must not measure the current source whose potential is more than 250 V. This instrument may be damaged and injury accidents may be caused.***

The mA terminal is protected from the applied overcurrent by using the fast-blow fuse.

For more information on how to replace the overcurrent protection fuse, refer to Chapter 8, "MAINTENANCE."

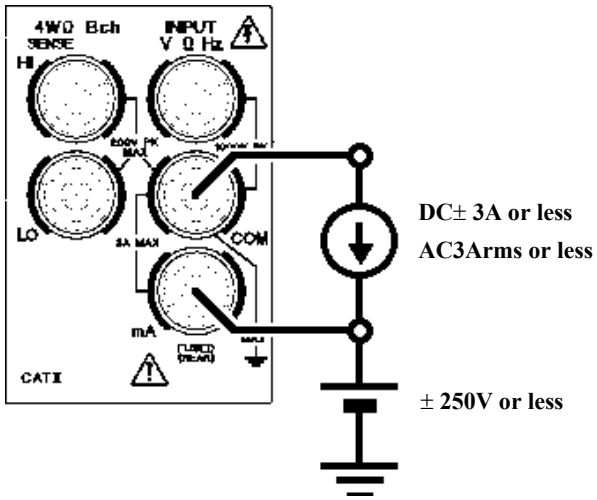


Figure 4-8 DC Current Measurement (DCI)

6. AC current measurement (ACI and ACI(AC+DC))

ACI measurement

1. Press **ACI**.
   AC current with AC coupling is measured.
2. Pressing **ACI** continuously switches the AC filter in turn.

ACI(AC+DC) measurement

3. Press **SHIFT** and **ACI**.
   ACI and DCI are measured and the root mean square value ($\sqrt{ACI^2+DCI^2}$) is displayed.
   The AC and DC indicators are turned on to show the AC+DC coupling.
4. Pressing **ACI** continuously switches the AC filter in turn.
5. Press **SHIFT** and **ACI** to return to the ACI measurement.

Selecting the AC Filter

In this instrument, the AC filter can be selected from three for the accuracy optimization in low frequency and the high speed response.

Select the AC filter in consideration of the measurement signal frequency, necessary measurement accuracy, and response time.

Table 4-9 Selecting the AC Filter

AC filter	Measurement signal frequency	Response time *1	Display
FAST	300 Hz to 5 kHz	100 msec	ACF
MED	20 Hz to 5 kHz	500 msec	ACM
SLOW	20 Hz to 5 kHz	3 sec	ACS

*1: Time until the measurement value reaches to within 0.1% of the final value

Maximum allowable applied current 3 A

WARNING: ***Must not measure a current source whose voltage is more than 250 V. This instrument may be damaged and injury or an accident may occur.***

The mA terminal is protected from the applied overcurrent by using the fast-blow fuse.

For more information on how to replace the overcurrent protection fuse, refer to Chapter 8, “MAINTENANCE.”

Table 4-10 Measurement Range

	7461A/61P	7451A
Measurement method	TrueRMS	
Measurement range	5% or more of the range	
Crest factor *1	5	3

*1: Except for the 3 A range

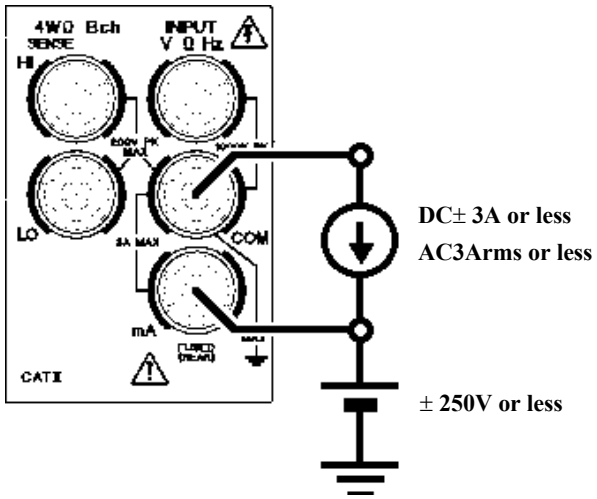


Figure 4-9 AC Current Measurement (ACI and ACI(AC+DC))

7. Continuity measurement ( )

   Press **SHIFT** and **FREQ**.

   The resistance is measured in 2WΩ. If the measured resistance is lower than the continuity threshold constant, the buzzer sounds.

Table 4-11 Continuity Measurement

		7461A/61P	7451A
Measurement range		1000 Ω	3000 Ω
Measurement current		1 mA	1 mA
Continuity threshold constant		1 Ω to 1000 Ω	
Maximum allowable applied voltage	Between Ω and COM terminals	1000 Vpeak *1	
	Between COM terminal and chassis	500 V	

*1: If an over-voltage is applied, the internal circuits may be separated from the input terminals by the protection function.

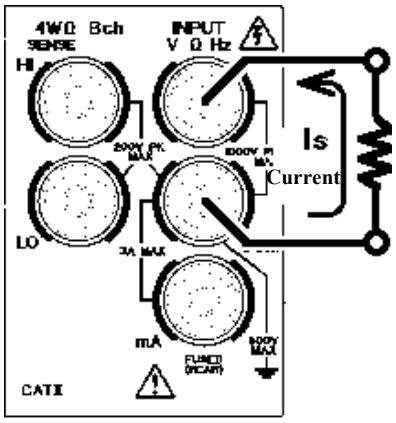


Figure 4-10 Continuity Measurement

8. Diode measurement (⊣▶)

   Press **SHIFT** and **DCI**.

   The voltage between the terminals is measured while the current of about 1 mA is applied to the diode.

Table 4-12 Diode Measurement

		7461A/61P	7451A
Measurement range		1000 mV	3000 mV
Measurement current		1 mA	1 mA
Maximum allowable applied voltage	Between Ω and COM terminals	1000 Vpeak *1	
	Between COM terminal and chassis	500 V	

*1: If an over-voltage is applied, the internal circuits may be separated from the input terminals by the protection function.

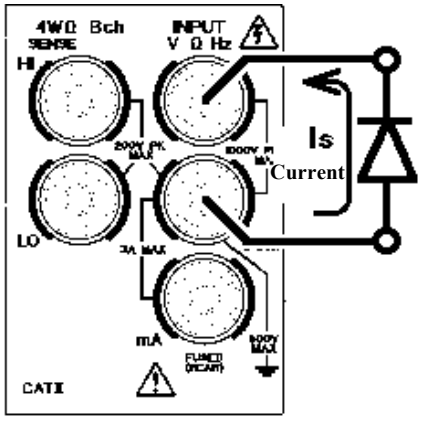


Figure 4-11 Diode Measurement

9. Frequency measurement (FREQ)

Press **FREQ**.
The frequency in the ACV measurement function is displayed.

The DC component of the input signal is cut by using the AC coupling and the frequency is measured at the zero-cross point.

Measurement method: Reciprocal

Input voltage range: 100 mVrms to 700 Vrms and 5% or more of the range (Sinusoidal wave)
The range does not mean the frequency range but the input voltage range.
In the 100 mV range, the input voltage from 120 mV to about 1.2 V can be measured without the "over" display.

Gate time

Sampling rate	Gate time	Measurement frequency range	Maximum measurement period	Maximum display	
				7461A/61P	7451A
SLOW	1000 ms	1 Hz to 300 kHz	2.2 s	9999999	999999
MED	100 ms	10 Hz to 300 kHz	220 ms	999999	99999
FAST	10 ms	100 Hz to 300 kHz	22 ms	99999	9999

If the input signal cannot be detected because the period of the input signal is longer than the gate time, 0 Hz is displayed.

Table 4-13 Maximum Allowable Applied Voltage (FREQ)

Between V and COM terminals	1000 Vpeak, 700 Vrms, 2.2×10^7 VHz
Between COM terminal and chassis	500 V

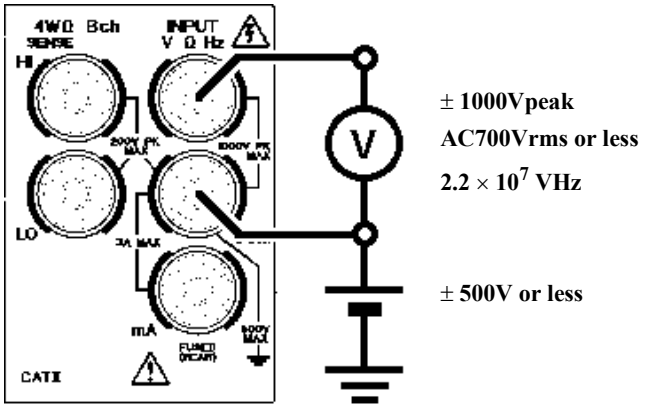


Figure 4-12 Frequency Measurement (FREQ)

Use a suitable measurement range for the input voltage.
If the input voltage exceeds the measurement range, the measurement is performed but the accuracy of the measured value may fail of the specifications.

10. Temperature measurement (TEMP-2W, TEMP-3W, TEMP-4W)

Press **TEMP**.
The temperature is measured by using the 2W, 3W or 4W (formerly set method).

Table 4-14 Temperature Measurement

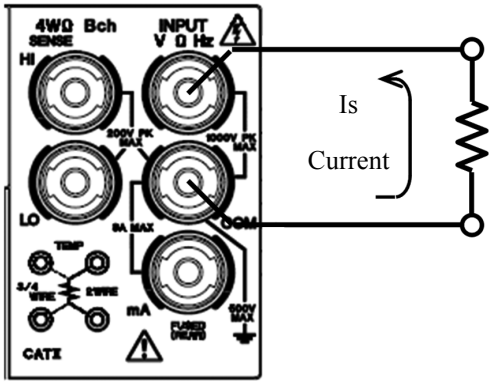
Resistance bulb	Pt100	JPt100
Measurement range	-200 °C to +850 °C	-200 °C to +649 °C
Measurement current	1 mA	

The measurement resolution varies depending on the sampling rate.

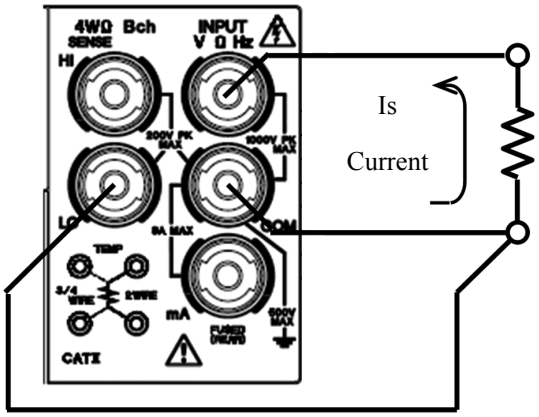
Pressing **RATE** switches the sampling rate in turn.
Select the sampling rate according to the necessary resolution.

Table 4-15 Display Resolution

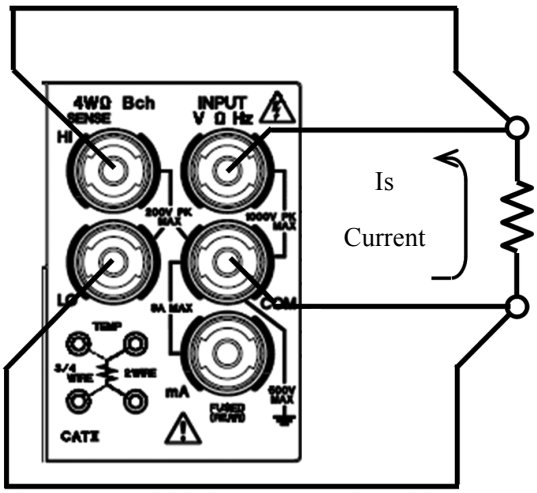
Sampling rate	Integration time	Measurement resolution
FAST1	0.02PLC	0.1 °C
FAST2	0.1PLC	0.1 °C
MED	1PLC	0.01 °C
SLOW1	5PLC	0.001 °C
SLOW2	10PLC	0.001 °C



(a) 2WΩ measurement



(b) 3WΩ measurement



(c) 4WΩ measurement

Figure 4-13 Temperature Measurement

4.2.2 Setting the Measurement Range

UP Increases the measurement range by one level and fixes the range.

DOWN Decreases the measurement range by one level and fixes the range.

AUTO Switches the measurement range between auto-range and fixed range.

4.2.3 Setting the Sampling Rate

Press **RATE**.

The measurement speed and display digit are changed as shown in Table 4-16 and Table 4-17.

In the frequency measurement, the sampling rate is used to set the gate time. For more information on the frequency measurement, refer to Section 4.2.1, "Measurement Functions."

Table 4-16 Sampling Rate, Measurement Speed, and Auto-Zero

Sampling rate	FMS indicator	Measurement speed (In Auto-range OFF)				Integration time	Auto-zero
		DCV Bch-DCV DCI 2WΩ LP-2WΩ	4WΩ LP-4WΩ	ACV ACI Continuity Diode	ACV(AC+DC) ACI(AC+DC)		
FAST1	F (Bright)	1000 times/s *1	110 times/s	400 times/s	4 times/s	0.02 PLC	OFF
FAST2	F (Dark)	300 times/s *1	80 times/s	250 times/s	4 times/s	0.1 PLC	OFF
MED	M	20 times/s	20 times/s	20 times/s	3 times/s	1 PLC	ON
SLOW1	S (Bright)	4.5 times/s	4.5 times/s	4.5 times/s	2 times/s	5 PLC	ON
SLOW2	S (Dark)	2 times/s	2 times/s	2 times/s	1 time/s	10 PLC	ON

*1 The speed of a measurement which starts by one trigger under the following conditions.
Trigger source: BUS, EXT, MAN
Sampling count:2 or more
Complete signal output mode:SINGLE
Display: OFF, Calculation: OFF, Measurement data memory: ON

4.2.3 Setting the Sampling Rate

Table 4-17 Sampling Rate and Display Digits

Function / Sampling rate	DCV/DCI DCV-Bch 2W/4WΩ LP-2W/4WΩ ACV/ACI Diode/Continuity		ACV(AC+DC) ACI(AC+DC)	
	7461A/61P	7451A	7461A/61P	7451A
FAST1, FAST2	4 1/2-digit display	4 1/2-digit display	3 1/2-digit display	
MED, SLOW1, SLOW2	6 1/2-digit display	5 1/2-digit display	4 1/2-digit display	

Also refer to Section 5.7, “Sampling Operation.”

Setting HOLD

HOLD Sets the sampling mode to HOLD or cancels.

TRIG Starts the measurement when the sampling mode is set to HOLD.

5. MENU OPERATION AND FUNCTION DESCRIPTION

5.1 Menu

The functions and conditions in this instrument are set by using layered menus.

5.1.1 How to Navigate the Menus

Menus have a three-layer structure. (Some menus have a four-layer structure.)

First layer	Category layer	Selects the menu category.
Second layer	Selection layer	Selects the setting parameters in the category.
Third layer	Entry/Execution layer	Used to enter the values or select the setting.

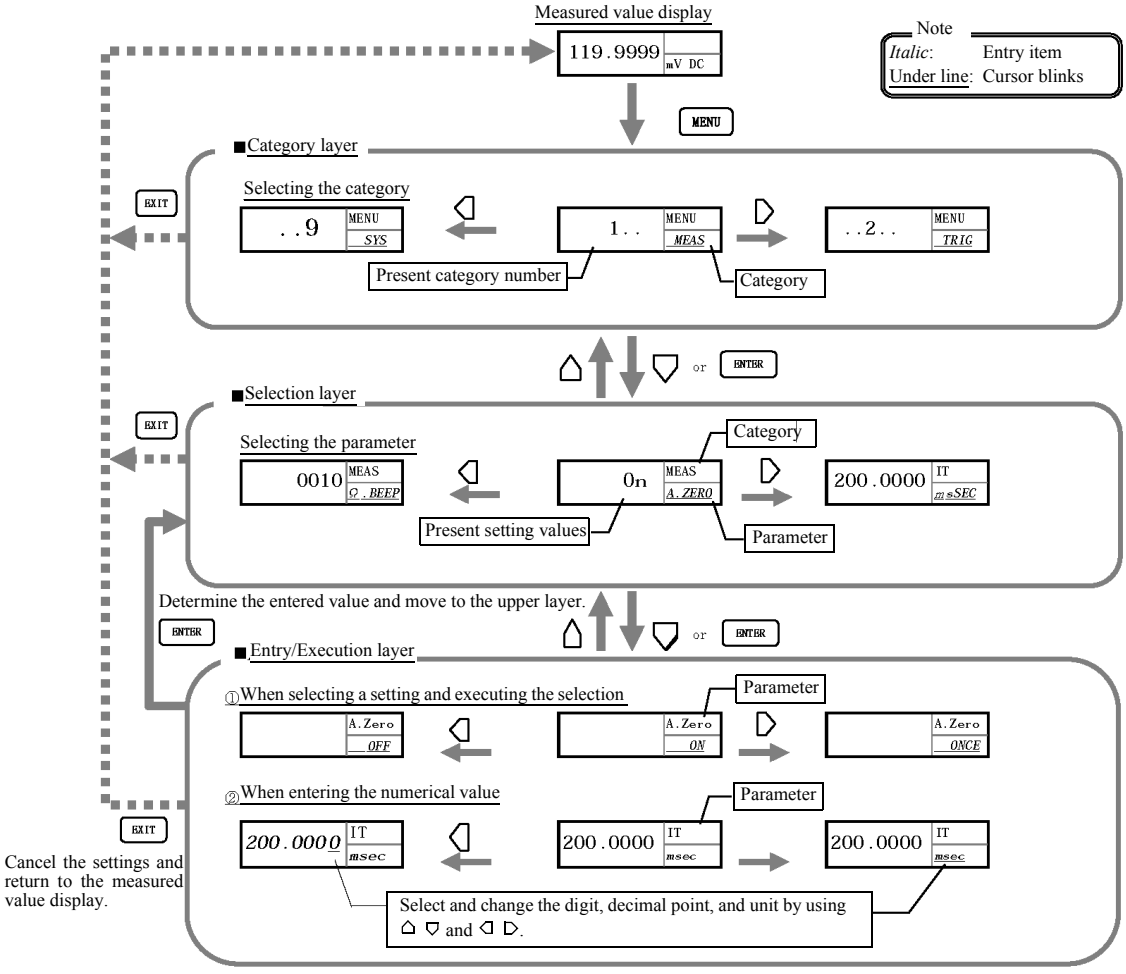


Figure 5-1 Menu Operation

1. Press **MENU**.

   The Category layer in which **1 MEAS** to **9 SYS** be selected is displayed.

2. Select the category by using ◁ and ▷.

3. Press ▽.

   The Selection layer is displayed.

4. Select the parameters by using ◁ and ▷.

5. Press ▽.

   The Entry/Execution layer is displayed.

6. Set the parameters by using △, ▽, ◁, ▷, and the **TRIG** key.

7. Pressing **ENTER** determines the entered parameter settings and moves to the Selection layer.

   If pressing **EXIT/LOCAL** before ENTER, the entered settings are canceled and MENU is quitted.

   "DONE" after "BUSY" is displayed during the process.

Setting the selectable parameters

For the selectable parameter that is set by selecting the item, select the parameter by using ◁ and ▷, and determine it by pressing **ENTER**.

Setting the numerical parameters

For the numerical parameter that is set by entering the numerical value, select either numerical digit, decimal point position, or unit prefix by using ◁ and ▷.

The operation of △ and ▽ varies depending on the target items.

If there is no target item that blinks, the layer is moved to the Selection layer by using △.

If a number blinks, the numerical value can be increased or decreased.

If a decimal point blinks, the decimal point can be moved.

If the unit display section blinks, the unit prefix can be changed in the following order: p, n, μ, m, none, k, M, G, and T

Pressing the **TRIG** key performs the measurement and can set the result as the value of the parameter.

Although any numerical value can be entered into the numerical parameter, it is rounded off to the resolution digit of each parameter.

Table 5-1 shows the key function in each layer.

Table 5-1 Menu Operation Key and Function

	Category layer	Selection layer	Entry/Execution layer		
			Numerical value parameter		Selectable parameter/ execution item
			Parameter: Dark display, No blinking	Target items: Numerical value/decimal point/unit	
⌂	-	Moves to the Category layer.	Moves to the Selection layer.	Increases the value.	Moves to the Selection layer.
▽	Moves to the Selection layer.	Moves to the Entry/Execution layer.	-	Decreases the value.	-
◁	Moves the target item to the left category.	Moves the target item to the left parameter.	Moves the target item to the numerical value.	Moves the target item to the left.	Displays the previous selection.
▷	Moves the target item to the right category.	Moves the target item to the right parameter.	Moves the target item to the unit.	Moves the target item to the right.	Displays the next selection.
TRIG	-	-	Performs the measurement and reflects the result to the parameter. *1		-
ENTER	Moves to the Selection layer.	Moves to the Entry/Execution layer.	Determines the entered value and moves to the Selection layer.		Determines the selection and moves to the Selection layer. *2
EXIT	Leaving the MENU system	Leaving the MENU system	Leaving the MENU system		

*1: Only the constant that is used to calculate the HIGH, LOW, and NULL values
*2: For the execution item, quits MENU.
Example: Parameter SAVE/LOAD

5.1.2 Menu List

Category layer　　Selection layer　　Entry/Execution layer

MENU

1 MEAS

A.ZERO	Auto-zero OFF/ON/ONCE
IT SEC	Integration time (sec) 10 μsec to 10 sec (For 7461A/61P) 100 μsec to 10 sec (For 7451A)
IT PLC	Integration time (PLC) 0.02 PLC to 100 PLC When the integration time is outside this range, it is set in seconds. Although the setting range is the same in IT SEC, the value is converted into the unit of PLC.
BchDC	DC voltage measurement selection (DCV-Bch) H-COM/HI-LO/RATIO
M.DISP	Display mode OFF/ON
DIGIT	Number of display digits. 6 1/2 to 3 1/2 (For 7461A/61P) 5 1/2 to 3 1/2 (For 7451A)
Zin	DCV input impedance selection HiZ/10 MΩ
ΩBEEP	The threshold in the continuity measurement function 1 Ω to 1000 Ω

(a)

Category layer | Selection layer | Entry/Execution layer

(a)

2 TRIG

DELAY	Trigger delay 0 s to 3600 s
N.Smpl	Number of samples 1 to 16000
SI	Sampling interval 50 µs to 3600 s (For 7461A/61P) 200 µs to 3600 s (For 7451A)
SRC	Trigger source Selection IMMED/MAN/EXT/BUS/LEVEL/DELTA
LEVEL	Level count value (For the trigger source is set to only LEVEL) -1199999 to 1199999 (For 7461A/61P) -319999 to 319999 (For 7451A)
SLOPE	Slope (For the trigger source is set to only LEVEL) POSI/NEGA
DELTA	Delta count value (For the trigger source is set to only DELTA) 0 to 1199999 (For 7461A/61P) 0 to 319999 (For 7451A)
SLOPE	Slope (For the trigger source is set to only DELTA) POSI/BOTH/NEGA
PATRN	Pattern (For the trigger source is set to only DELTA) EMISS/CONV
P.Trig	Pre-trigger (For the trigger source is set to LEVEL or DELTA) OFF/25%/50%/75%

3 CALC

COMP	Comparative calculation OFF/ON
MATH	Other calculations OFF/dB/dBm/SCL
MX/MN	MAX/MIN calculation and selecting the display OFF/ON/MAX/MIN
RSLT	Display of the MAX/MIN calculation result SAMPL/MAX/MIN/AVG
PASS	PASS condition settings LGH/__H/_G_/L__/L_H/LG_/_GH/___
BEEP	Calculation Result and buzzer OFF/PASS/FAIL
SM.CNT	Number of times smoothing is performed 2 to 100 times
D.OUT *7461P only	Digital output enabled/disabled OFF/ON

(b)

5.1.2 Menu List

Category layer | Selection layer | Entry/Execution layer

(b)

4 CONST

HIGH	HIGH constant -9999.999E+9 to 9999.999E+9 (For 7461A/61P) -999.999E+9 to 999.999E+9 (For 7451A)
LOW	LOW constant -9999.999E+9 to 9999.999E+9 (For 7461A/61P) -999.999E+9 to 999.999E+9 (For 7451A)
ΔNULL	NULL constant (For only NULL ON) -9999.999E+9 to 9999.999E+9 (For 7461A/61P) -999.999E+9 to 999.999E+9 (For 7451A)
Val_A	Constant A (SCL) -9999.999E+9 to 9999.999E+9 (For 7461A/61P) -999.999E+9 to 999.999E+9 (For 7451A)
Val_B	Constant B (SCL) -9999.999E+9 to 9999.999E+9 (For 7461A/61P) -999.999E+9 to 999.999E+9 (For 7451A)
Val_C	Constant C (SCL) -9999.999E+9 to 9999.999E+9 (For 7461A/61P) -999.999E+9 to 999.999E+9 (For 7451A)
ValDB	Constant D (dB/dBm) 0.000001E-9 to 9999.999E+9 (For 7461A/61P) 0.000001E-9 to 999.999E+9 (For 7451A)

5 PARAM

PON.LD	Selection of parameters which are loaded when the power is turned on P.OFF/USER0
SAVE	Parameter save-file selection USER0/USER1/USER2/USER3
LOAD	Parameter load-file selection USER0/USER1/USER2/USER3/DFLT

6 INIT

PARAM	Initialization DFLT

7 MEM

STORE	Measurement memory OFF/ON
RCL	Measurement memory recall Recall setting for the data number
STAT	Statistical calculations and measurement result readings

(c)

Category layer | Selection layer | Entry/Execution layer

(c)

8 I/F

BUS	Interface selection GPIB/USB
GP.Adr	GPIB address (Only when GPIB is selected for the interface) 0 to 30
USBid	USB ID (Only when USB is selected for the interface) 1 to 127
HEADR	Header OFF/ON
T.Only	Addressable/Talk only (Only when GPIB is selected for the interface) OFF/ON
LANG	Command language selection ADC/SCPI/.... (Different depending on the model)
CONT *1	Initial value selection of the Continue setting OFF/ON (Only when SCPI is selected as the command language.)

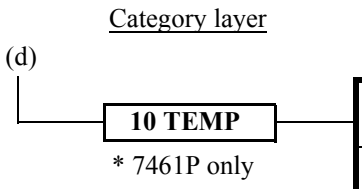
*1: Applied to software revision B00 and later.

9 SYS

BEEP	Buzzer setting OFF/ON
TEST	Self-Test
P/KEY	Display and key tests
TERM	Input terminal selection FRONT/REAR
C.Widt	Complete signal pulse width 5 µs/100 µs
C.Sig	Complete signal output mode SGL/MULTI
F.Inhi	Disable function Each function can be set to ENA (Enabling) or DIS (Disabling).
P.Lock	Panel lock OFF/ON
PWD	Password (For panel lock) 4 digits password
R.CAL	Zero calibration in the rear input terminal (For only CAL mode ON)
CAL	CAL mode OFF/ON
QUEUE	Error queue No.01 to No.20

(d)

5.1.2 Menu List



WIRE	Measurement wire connection 2W/3W/4W
TYPE	Platinum resistance bulb selection PT/JPT
UNIT	Temperature unit °C/°F/K

5.2 Measurement Functions and Parameters

For more information on the measurement function description, refer to Chapter 4.

This chapter describes the parameters set in MENU that is used in the measurement function.

5.2.1 DC Voltage Measurement (DCV)

Selecting the input impedance

The input impedance in the following range can be selected from HiZ and 10 MΩ.

100 mV, 1000 mV, and 10 V ranges (7461A/61P)

300 mV and 3000 mV ranges (7451A)

Select HiZ or 10 MΩ from 'ZIn' in **1 MEAS** category in **MENU**.

Press **ENTER** to determine the setting.

Select HiZ when the signal source impedance is high.

5.2.2 DC Voltage Measurement (DCV-Bch)

Selecting the measurement terminal

The measurement terminal can be selected as shown in below from 'BchDC' in **1 MEAS** category in **MENU**.

HI - COM	Displays the voltage between Bch HI and COM terminals.
HI - LO	Measures the voltage between Bch HI and COM terminals, measures the voltage between Bch Lo and COM terminals, and displays the difference between both voltages.
RATIO	Display the voltage ratio between the following voltages: The voltage between Bch HI and LOW terminals, and the voltage between V and COM terminals

Press **ENTER** to determine the setting.

The measurement voltage of each between terminals is as follows. For more information on, refer to Section 4.2.1, "Measurement Functions."

Measurement terminal	Measurement voltage range
Bch HI - COM	±12 V
Bch LO - COM	±12 V
Bch HI - Bch LO	±12 V

5.2.3 Continuity Measurement

Setting the continuity threshold constant

The continuity threshold constant can be selected in the range between 1 Ω and 1000 Ω from 'Ω.BEEP' in **1 MEAS** category in **MENU**.

Press **ENTER** to determine the setting.

5.2.4 Temperature Measurement

Selecting the measurement wire connection

The measurement wire connection can be selected from 'WIRE' in **10 TEMP** category in **MENU**.

2W	2-wire connection
3W	3-wire connection
4W	4-wire connection

Press **ENTER** to determine the setting.

Selecting the platinum resistance bulb

The platinum resistance bulb can be selected from 'TYPE' in **10 TEMP** category in **MENU**.

PT	Pt100
JPT	JPt100

Press **ENTER** to determine the setting.

Selecting the measurement unit

The measurement unit can be selected from 'UNIT' in **10 TEMP** category in **MENU**.

°C	Celsius
°F	Fahrenheit
K	Kelvin

Press **ENTER** to determine the setting.

5.3 Measurement Range and Auto-Range

5.3.1 Measurement Function and Range Structure

Table 5-2 shows the range structure corresponding to the measurement function.

Table 5-2 Measurement Function and Range Structure

	DCV	DCV Bch	ACV ACV (AC+DC)	2WΩ/ 4WΩ	LP-2WΩ/ 4WΩ	DCI	ACI ACI (AC+DC)	FREQ	Continuity	Diode
7461A/61P	100 mV		100 mV	100 Ω	100 Ω	1000 μA	1000 μA	100 mV		
	1000 mV		1000 mV	1000 Ω	1000 Ω	10 mA	10 mA	1000 mV	1000 Ω	1000 mV
	10 V	10 V	10 V	10 kΩ	10 kΩ	100 mA	100 mA	10 V		
	100 V		100 V	100 kΩ	100 kΩ	1000 mA	1000 mA	100 V		
	1000 V		700 V	1000 kΩ	1000 kΩ	3 A	3 A	700 V		
				10 MΩ	10 MΩ					
				100 MΩ						
7451A	300 mV		300 mV	30 Ω		3000 μA	3000 μA	300 mV		
	3000 mV		3000 mV	300 Ω	300 Ω	30 mA	30 mA	3000 mV		3000 mV
	30 V	10V	30 V	3000 Ω	3000 Ω	300 mA	300 mA	30 V	3000 Ω	
	300 V		300 V	30 kΩ	30 kΩ	3000 mA	3000 mA	300 V		
	1000 V		700 V	300 kΩ	300 kΩ			700 V		
				3000 kΩ	3000 kΩ					
				30 MΩ	30 MΩ					
				300 MΩ						

NOTE: The range settings for each function are saved.
For the frequency function, the range is not switched in the frequency range but switched in the voltage range.

5.3.2 Auto-range

When Auto-range is set to ON, the measurement range of this instrument automatically changes to the optimum range depending on the input voltage.

The judgment value used when the range is changed

Range DOWN	When the displayed value is less than 10% of the displayed range (less than 300 V if the DCV 1000 V range and ACV (AD+DC) 700 V range are both selected)
Range UP	When the displayed value exceeds the maximum display

Auto-range is set to OFF under the following conditions.

- The sampling interval is set to 1 msec or less.
- The trigger source is set to the level trigger or delta trigger.

NOTE: Before using Auto-range

If this instrument is used while Auto-range is set to ON, some measurements cannot be performed. For example, in the following measurements, set the range before performing any measurements.

- ***When measuring the average value of a pulse:***
  Set the measurement range so that the maximum value of the pulse can be measured.
- ***When acquiring the measurement data at a constant interval:***
  The measurement interval varies if the measurement range is changed.
- ***When setting the sampling count to two or more for one trigger***
  The measurement interval varies if the measurement range is changed.
- ***When performing the high-speed measurement:***
  It takes time to set the measurement range to the optimum range.

5.4 Auto-Zero Operation

Auto-zero function eliminates the offset error in the measurement system of this instrument.

Set Auto-zero in 'A.Zero' in **1 MEAS** in **MENU**.

If Auto-zero is set to ON, the internal offset is measured and the offset error is eliminated because the offset is subtracted from the input measurement value. However, the measurement time is doubled.

Auto-zero is performed at the current measurement range in the measurement function and range shown in the following table.

Table 5-3 Performing Auto-Zero

Function	Range	Auto-zero	Indicator
DCV	All	Performed	ON
ACV ACV(AC+DC)	All	Not performed	OFF
2WΩ LP-2WΩ	1 MΩ to 100 MΩ 300 kΩ to 300 MΩ	Not performed *1	ON
2WΩ LP-2WΩ	100 Ω to 100 kΩ 300 Ω to 30 kΩ	Performed	ON
4WΩ LP-4WΩ	All	Not performed	OFF
DCI	All	Performed	ON
ACI ACI(AC+DC)	All	Not performed	OFF
FREQ	All	Not performed	OFF
Diode	1000 mV 3000 mV	Performed	ON
Continuity	1000 Ω 3000 Ω	Performed	ON
TEMP-2W	-	Performed	ON
TEMP-3W TEMP-4W	-	Not performed	OFF

*1 Auto zero is not performed in the high resistance ranges in 2WΩ and LP-2WΩ measurement even if Auto zero is set to ON.

While no measurement is performed, Auto-zero is performed under the following conditions.

- The power is turned on.
- The setting is changed to a function in which Auto-zero is performed.

If the high-speed or high-throughput measurement is required, Auto zero can be performed only once according to the measurement condition or measurement timing.

If ONCE is set and Auto-zero is performed, the offset error is eliminated and then Auto-zero is turned off.

If the resolution and measurement speed are changed by using the **RATE** key, Auto-zero is switched ON or OFF.

For more information, refer to Table 4-16 in Section 4.2.3, “Setting the Sampling Rate.”

5.5 Display Mode Setting

Setting the display mode

Select either ON or OFF in ‘M.DISP’ in **1 MEAS** in **MENU**.
Press **ENTER** to apply the setting.
A time of approximately 1 ms is required for the output to be displayed. (The time may vary depending on conditions.)
Select OFF if sampling at high speed.
An error is displayed even if the display is set to OFF.

5.6 Display Digit Setting

Setting the number of display digits

Set the number of display digits in ‘DIGIT’ in the **1 MEAS** category in **MENU**.
Press **ENTER** to apply the setting.
This setting is disabled in remote output but display.

5.7 Sampling Operation

Figure 5-2 shows the measurement sequence of this instrument.

Measurements are performed when the trigger event is received.

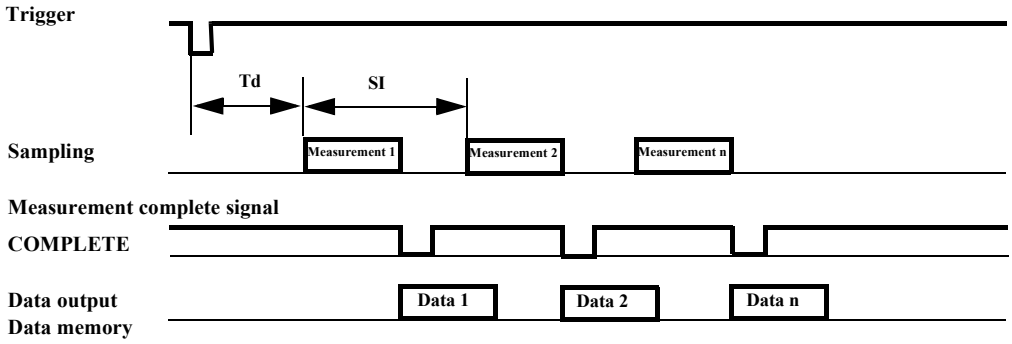
When each measurement is complete, the COMPLETE signal is output and the data output starts.

The measurement data is output to the display or interface.

If a sampling speed is faster than the transfer speed of an interface, the data cannot be output.

In this case, output the measurement data after acquiring all the measurement data in the data memory.

- When performing the data output in each measurement. (COMPLETE signal output: MULTI)



- When using a data memory. (COMPLETE signal output: MULTI)

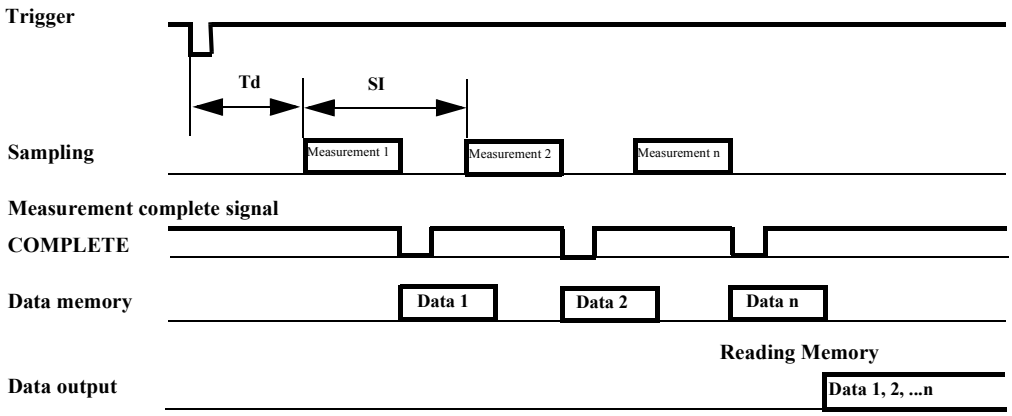


Figure 5-2 Measurement Sequence

When sampling over a precise interval or sampling at high speeds below 1 ms, use this instrument under the following conditions.

Function:	DCV, DCI, 2WΩ, LP2WΩ, DCV-Bch (Bch HI-COM)
Sampling count:	2 or more
Trigger source:	BUS, EXT, MAN
Complete signal output:	SINGLE
Auto-range:	OFF
Auto-zero:	OFF

5.7.1 Sampling Interval (SI) and Integration Time (IT) Settings

The integration time and sampling interval can be set by using MENU as well as the **RATE** key on the front panel.

If the integration time or sampling interval is set by using MENU, the FMS indicator in the display section is off. The display digit varies depending on the integration time, generating an additional error in the measurement accuracy. For more information, refer to Section 7.1, "Specifications."

The integration time can be set in units of PLC, which comes from the power supply frequency, and also seconds.

If the integration time is set in units of PLC, the sampling interval is rounded to the nearest integer.

Table 5-4 shows the setting range and resolution of the integration time.

Table 5-4 Setting Range and Resolution of the Integration Time (IT)

7461A/61P		7451A	
Setting range	Resolution	Setting range	Resolution
10 µs to 199 µs	1 µs	100 µs to 199 µs	1 µs
200 µs to 9.99 ms	10 µs	200 µs to 9.99 ms	10 µs
10 ms to 5999.9 ms	100 µs	10 ms to 5999.9 ms	100 µs
6 s to 10 s	200 µs	6 s to 10 s	200 µs
0.02 PLC to 100 PLC	0.01 PLC	0.02 PLC to 100 PLC	0.01 PLC

When setting the integration time in PLC unit, the integration time outside the range shown in the table can be set. However, the setting is converted to the units of second.

In this case, the setting range in seconds is applied and the value that is converted into the unit of PLC is displayed.

Set the integration time according to the following procedure.

Enter the integration time in 'IT SEC' or 'IT PLC' in **1 MEAS** in **MENU**.
Select the digit, decimal point, or unit prefix by using ◁ and ▷, and change it by using △ and ▽.
Press **ENTER** to apply the setting.
If a value that is less than the resolution is entered, the setting value is rounded to the resolution.

Table 5-5 shows the setting range of the sampling interval.

Table 5-5 Setting Range of Sampling Interval (SI)

7461A/61P	7451A
Setting range	Setting range
50 μs to 3600 s	200 μs to 3600 s

Although the sampling interval (SI) can be set arbitrarily, use the minimum measurement time or more shown in Table 5-6 as the sampling interval.

Although the sampling interval (SI) can be set to less than the minimum measurement time, the instrument operates in the minimum measurement time.

Table 5-6 Integration Time and Minimum Measurement Time

7461A/61P

Integration time (IT)	Minimum measurement time
10 μs ≤ IT < 200 μs	IT + IT + 70 μs
200 μs ≤ IT < 10 ms	IT + 80 μs
10 ms ≤ IT < 6 s	IT + 170 μs
6 s ≤ IT ≤ 10 s	IT + 270 μs

7451A

Integration time (IT)	Minimum measurement time
100 μs ≤ IT < 200 μs	IT + IT + 70 μs
200 μs ≤ IT < 10 ms	IT + 80 μs
10 ms ≤ IT < 6 s	IT + 170 μs
6 s ≤ IT ≤ 10 s	IT + 270 μs

Set the sampling interval according to the following procedure.

Enter the sampling interval in 'SI' in **2 TRIG** in **MENU**.
Select the digit, decimal point, or unit prefix by using ◁ and ▷, and change it by using △ and ▽.
Press **ENTER** to apply the setting.
If the value that is less than the resolution is entered, the setting value is rounded to the resolution.

If the sampling interval is set to less than 1 ms, Auto-range and Auto-zero are automatically set to OFF.

5.7.2 Sampling Count Setting

Although sampling is usually performed once when a trigger is detected, sampling can be performed more than once.

Enter the number of sampling in 'N.Smpl' in **2 TRIG** in **MENU**.

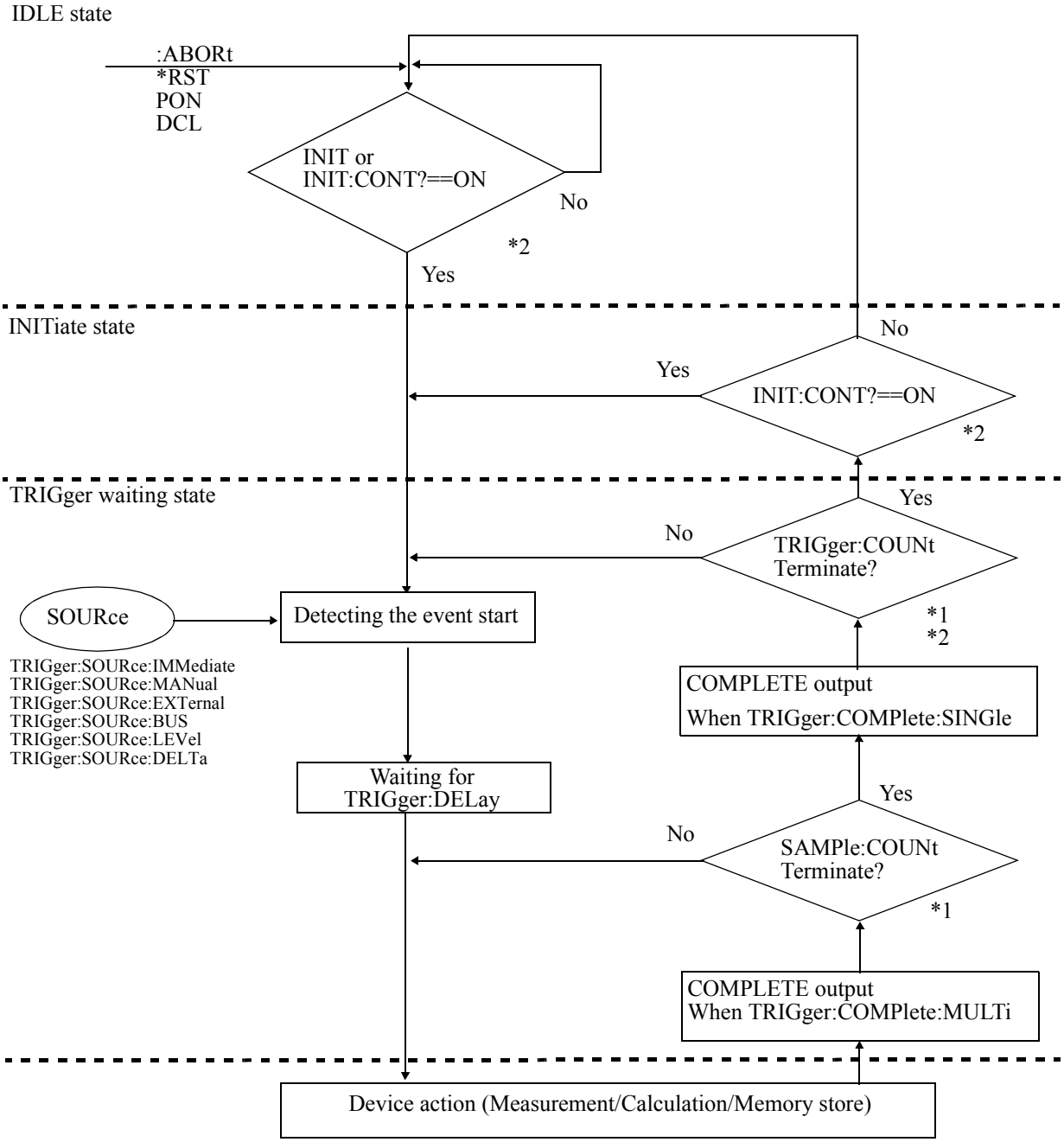
Sampling count 1 to 16000 samples/trigger

NOTE: ***When data is stored in the measurement data memory in conjunction with this setting on the 7461A or 7451A, the number of data to be stored is up to 10,000 even if the number of sampling is set to more than 10,000.***
For more information, refer to Section 5.11, "Storing and Recalling the Measurement Data."

5.8 Trigger Function

The trigger system of this instrument allows trigger detection, the sampling start to be delayed, and multiple samplings per trigger.

Figure 5-3 shows the trigger model.



*1: This judgment is disabled when IMMediate is used as the trigger source. (Always “Yes”)

*2: This judgment is disabled in the LOCAL mode. (Always “Yes”)

Figure 5-3 Trigger Model

5.8.1 Trigger System Operation

1. IDLE state

   The trigger system is in the IDLE state before the trigger sequence starts, after the trigger sequence is terminated, and while the processes of this system are not performed such as a reset state.

2. INITiate state

   The INITiate state means that the trigger system operation starts.

   The IDLE state is changed to the INITiate state by using either of the following trigger system commands: Starting the trigger system (INITiate), Setting CONTinuous to ON (INITiate:CONTinuous {ON})

   If CONTinuous is set to ON when the measurement is complete, the trigger system enters the INITiate state without returning to the IDLE state and waits for the next trigger.

3. TRIGger waiting state

   When the trigger event is detected from the selected trigger source, the device action starts after the trigger delay time.

   The trigger system performs the sampling twice or more by one trigger, outputs the complete signal, and returns to the INITiate state.

4. Device action

   The measurement and calculations are performed.

5.8.2 Trigger Source Selection

The trigger source is selected in 'SRC' in the **2 TRIG** category in **MENU**.

Trigger source

IMMED	Performs the continuous measurement at the timing of the sampling interval in this instrument.
MAN	Detects an event when the **TRIG** key is pressed or the trigger signal is input from the external trigger connector.
EXT	Detects an event when the trigger signal is input from the external trigger connector.
BUS	Detects an event when the *TRG command or GET command is received from the enabled USB or GPIB bus.
LEVEL	Detects an event when the measurement value exceeds the set threshold value. Detects an event at the rising edge (POSI) or the falling edge (NEGA) according to the SLOPE setting.
DELTA	Detects an event when the difference between delta 1 and delta 2 (refer to Figure 5-5) exceeds the set threshold value. Select SLOPE from the positive (POSI), absolute (BOTH), and negative (NEGA) polarities. Set PATRN to EMISS or CONV.

The following shows the parameters whose settings automatically change depending on the trigger source setting.

IMMED	HOLD: OFF
MAN	HOLD: ON
EXT	HOLD: ON
BUS	HOLD: ON
LEVEL	HOLD: ON, Auto-range: OFF, Measurement data memory store: ON
DELTA	HOLD: ON, Auto-range: OFF, Measurement data memory store: ON

Trigger key

If this instrument is in local state and HOLD is set to ON, the trigger event is generated when the **TRIG** key on the front panel is pressed. The trigger key can be used regardless of the trigger source selection.

Trigger event buffer

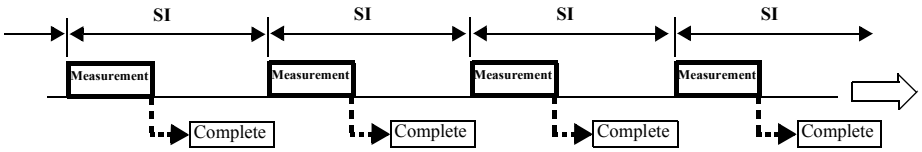
If triggers are input while this instrument is in the measurement state, one trigger event is buffered. In this case, the operation is the same as when the measurement is complete at the same time a trigger is generated.

5.8.2 Trigger Source Selection

[Measurement timing according to the trigger source]

A: HOLD: OFF

1. TRIGger:Source:IMMediate

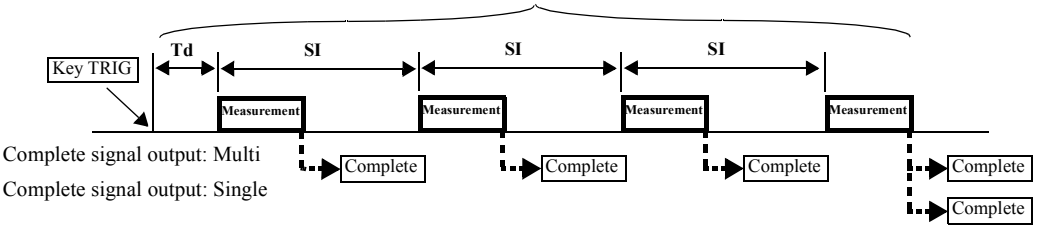


SI=TRIGger:TIMer<n>

TRIGger:DELay is invalid.

B: HOLD: ON

2. TRIGger:Source:MANual



Complete signal output: Multi

Complete signal output: Single

Td=TRIGger:DELay <n>

Cs=SAMPle:COUNt <n>

When TRIGger:COMPlete:SINGle	The Complete signal is output when the measurements of the set number of sampling (Cs) are complete.
When TRIGger:COMPlete:MULTi	The Complete signal is output each time when the measurement is complete.

3. TRIGger:Source:BUS

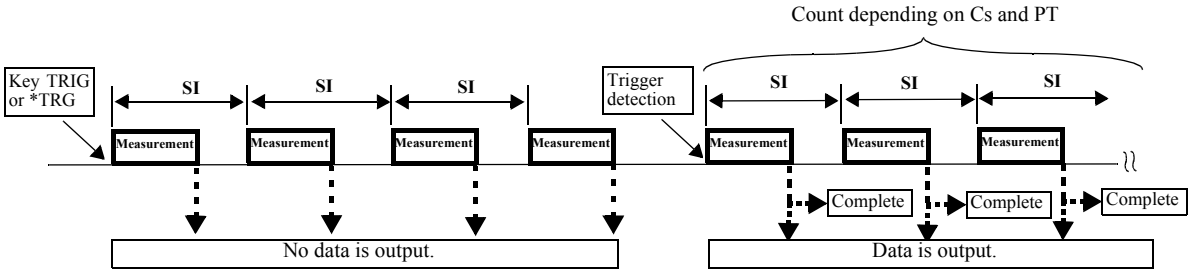
The same as TRIGger:Source:MANual However, the measurement starts when *TRG command is received.

4. TRIGger:Source:EXTernal

The same as TRIGger:Source:MANual However, the measurement starts when the external trigger signal is input.

C: Level trigger/Delta trigger

5. TRIGger:Source:LEVel (Memory store: ON, Auto-range: OFF, automatically set)



SI=TRIGger:TIMer<n>

Td=TRIGger:DELay <n>

Cs=SAMPle:COUNt <n>

PT=TSYStem:PRETrigger <n>

When TRIGger:COMPlete:SINGle	The Complete signal is output when the measurements of the set number of sampling (Cs) are complete.
When TRIGger:COMPlete:MULTi	The Complete signal is output each time when the measurement is complete.

(The figure shows a case of MULTI)

6. TRIGger:Source:DELTa

The same as TRIGger:Source:LEVel

5.8.3 Trigger Delay (Td)

Time from a trigger input to sampling start can be delayed by setting the trigger delay.

The measurement can start after the input becomes stable.

Table 5-7 shows the trigger delay time and setting resolution.

The trigger delay is set in 'DELAY' in the **2 TRIG** in **MENU**.

Table 5-7 Trigger Delay Time, Setting Resolution, and Accuracy

Trigger delay setting	Resolution *1	Delay time accuracy	
		Ext trigger	Bus trigger
0 to 600 ms	10 µs	0.9 ms to 1.6 ms + resolution (Software revision A01 and earlier) 0.1 ms to 0.5 ms + resolution (Software revision B00 and later)	3.2 ms to 3.7 ms + resolution (Software revision A01 and earlier) 2.6 ms to 3.1 ms + resolution (Software revision B00 and later)
600.1 m to 6 s	100 µs		
6001 m to 60 s	1 ms		
61 s to 3600 s	1 s	1 s	1 s

*1: If a value that is less than the resolution is set by using the panel or a remote command, the setting value is rounded to the resolution.

5.8.4 Trigger Count Setting

Although the trigger system usually returns to the Idle state when receiving a trigger, it can return to the trigger waiting state to receive triggers.

Trigger count can be set only by using a remote command.

Trigger count: 1 to 50000

5.8.5 Level Trigger and Delta Trigger

Data can be acquired into the measurement memory by detecting the change of an input signal and generating the trigger event.

The measurement data is always stored in the memory and the data before detecting the trigger is output according to the pre-trigger setting.

Procedure

1. Set the trigger source to LEVEL or DELTA.
2. Set the sampling count. (7451A/61A: up to 10000, 7461P: up to 16000)
3. Set the parameter for detecting a trigger level.(LEVEL, SLOPE, DELTA, PATRN)
4. The measurement in the set sampling interval starts when the **TRIG** key is pressed or *TRG command is received. The sampling interval can be set from 1 ms.
5. The level judgment is performed in each measurement and the trigger event is generated when the level exceeds the threshold value.
6. After the trigger event is generated, the number of measurements set in N.Smpl is performed and then the sampling stops.

   The number of measurements performed after a trigger is detected varies depending on the pre-trigger setting.

NOTE: ***If the trigger source is set to LEVEL or DELTA, HOLD: ON, Auto-range: OFF, Measurement data memory: ON, Scaling calculation: OFF, and dB/dBm calculation: OFF are set automatically.***
If the function is set to the continuity measurement, frequency measurement, diode measurement or temperature measurement, data acquisition starts without any trigger detection.

5.8.5.1 Level Trigger

The trigger is detected when the measurement data exceeds the set LEVEL (threshold value).

The trigger can be detected at the rising edge (POSI) or the falling edge (NEGA) according to the SLOPE setting.

LEVEL and SLOPE can be set in 'LEVEL' and 'SLOPE' in the **2 TRIG** category in **MENU**.

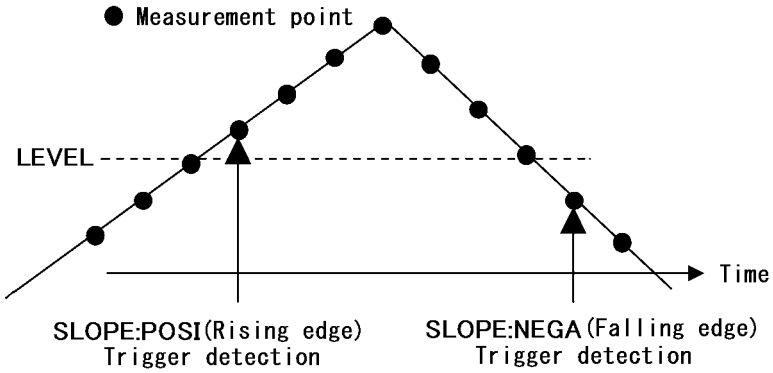


Figure 5-4 Level Trigger

Setting the level trigger

The LEVEL value is set by entering the display count up to the maximum number of display digits in the fixed measurement range.

Set the LEVEL value according to the following displayed number even if the displayed number becomes smaller than the maximum number of display digits because of the measurement function, sampling rate, or display digit settings.

7461A/61P: 1199999, 319999 (3 A range), 749999 (750 V range)

7451A: 319999, 119999 (DCV-Bch), 74999 (750 V range)

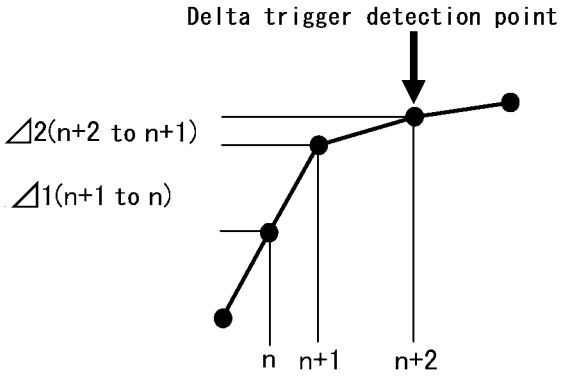
Example: Set LEVEL to 500000 when setting the threshold level to 5.00000 V in DCV 10V range of the 7461A.

If the setting exceeds the maximum display, the trigger cannot be detected.

Example: When LEVEL is set to 200000 in DCV-Bch of the 7451A.

5.8.5.2 Delta Trigger

An event is detected when the difference between delta 1 and delta 2 exceeds the set DELTA (threshold value).



Example: SLOPE:POSI, PATRN:CONV

The trigger is detected when Δ1 is larger than DELTA and Δ2 is smaller than DELTA.

Figure 5-5 Delta Trigger

The SLOPE setting can detect the polarity of a change that is positive (POSI), absolute (BOTH), or negative (NEGA).

The PATRN setting can detect either divergence (EMISS) or convergence (CONV).

DELTA, SLOPE, and PATRN can be set in 'DELTA', 'SLOPE', and 'PATRN' in the **2 TRIG** category in **MENU**.

The trigger is detected when the width of the change between the measurement values exceeds the set threshold value (DELTA).

One of the six detection patterns is selected by setting SLOPE and PATRN and the detection level is set in DELTA. The direction of the change is selected by setting PATRN.

Select one of the following in 'PATRN' in **2 TRIG** in **MENU**.

EMISS The trigger is detected when the width of the change becomes larger than DELTA.

CONV The trigger is detected when the width of the change becomes smaller than DELTA.

The polarity of the change is selected by setting SLOPE.

Select SLOPE from the positive (POSI), absolute (BOTH), and negative (NEGA) polarities in 'SLOPE' in **2 TRIG** in **MENU**.

The DELTA value is set by entering the display count up to the maximum number of display digits in the fixed measurement range.

Set the DELTA value according to the following displayed number even if the displayed number becomes smaller than the maximum number of display digits because the measurement function, sampling rate, or

display digit settings.

7461A/61P: 1199999, 319999 (3 A range), 749999 (750 V range)

7451A: 319999, 119999 (DCV-Bch), 74999 (750 V range)

Example: Set DELTA to 1000 when setting the threshold level to 10 mV in DCV 10V range of the 7461A.

If the setting exceeds the maximum display, the trigger cannot be detected.

Example: When DELTA is set to 200000 in DCV-Bch of the 7451A.

The trigger is detected when the polarity of change and direction of change judgment conditions are satisfied (refer to Table 5-8).

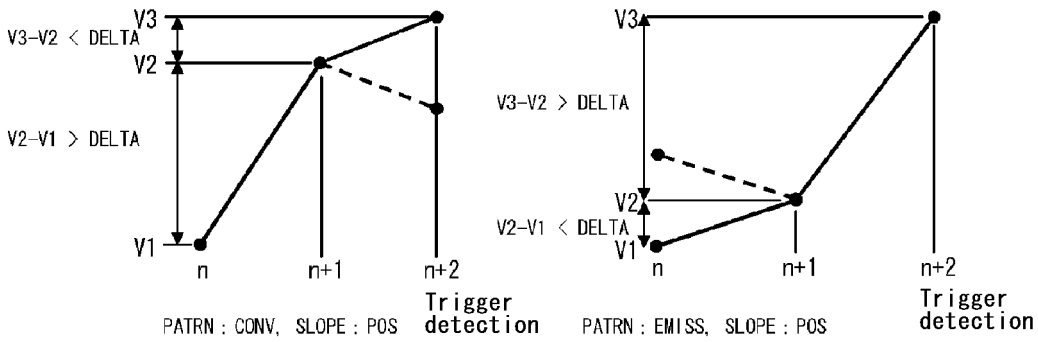
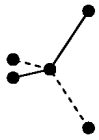


Figure 5-6 Delta Trigger

Table 5-8 Delta Trigger Detection Pattern

PATRN	CONV	CONV	CONV	EMISS	EMISS	EMISS
SLOPE	POSI	NEGA	BOTH	POSI	NEGA	BOTH
Trigger detection pattern						
Polarity of change judgment condition	V2-V1 > 0	V2-V1 < 0	V2-V1 ≠ 0	V3-V2 > 0	V3-V2 < 0	V3-V2 ≠ 0
Direction of change judgment condition	absolute value (V2-V1) > DELTA & absolute value(V3-V2) < DELTA			absolute value (V2-V1) < DELTA & absolute value (V3-V2) > DELTA		

For CONV: Both plus and minus polarities are allowed in V3-V2.
For EMISS: Both plus and minus polarities are allowed in V2-V1.

5.8.5.3 Pre-Trigger Operation

The measurement values before the trigger event can be acquired by using the pre-trigger function only when the trigger source is set to LEVEL or DELTA.

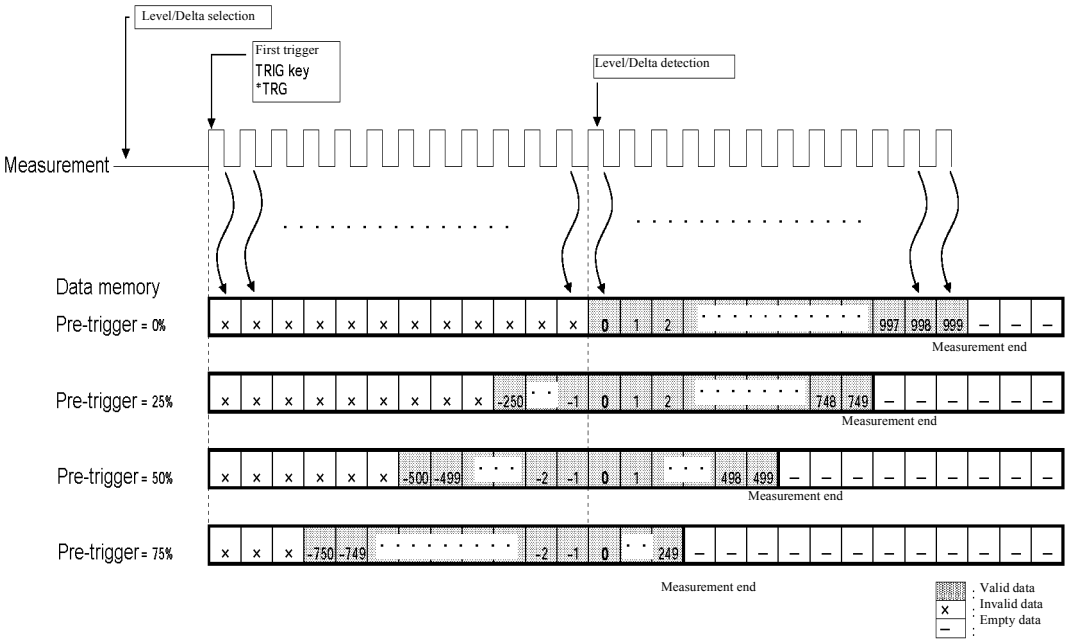
The pre-trigger setting can be selected from 0%, 25%, 50%, and 75%.

Although the data before the trigger event is stored in the measurement data memory by setting the pre-trigger, the total data amount cannot exceed the capacity of the data memory. The 7451A/61A stores up to 10000 data and the 7461P stores up to 20000 data.

Data memory in the pre-trigger operation

- Memory store

  Example: Sampling count = 1000



- Because the data memory operates as a ring-memory when the level or delta trigger is used, the measurement is repeated continuously until detecting the level or delta trigger.
- The number of sampling is up to 10000 when the level or delta trigger is used. Even if the sampling count is set to more than 10000, only 10000 measurements are performed. (7451A/61A)
  As for the 7461P, the number of sampling is up to 16000.
- If the trigger is detected before the pre-trigger count is reached, the number of valid measurement results in the memory is less than the set sampling count.

  Example: Pre-trigger = 75%
  The trigger is detected at the 100th measurement result after the first trigger.
  } The number of valid measurement results: 100 + 249 = 349

- Memory recall

Reads out the number of measurement results IRPO? → Outputs the number of valid measurement results. → Example of reply: IRPO01000

Reads out the data range IRNO? → Outputs the range of valid data. → Example of pre-trigger = 25%: IRNO-250, 749

Readout range setting IRD → Only current range of valid data can be set. → In pre-trigger = 25%: IRD-250, 749 OK

In pre-trigger = 75%: IRD-250, 749 NG

Reads out data IRO? → Outputs the data in the range set by IRD command.
If this command is executed before the trigger is detected, an error occurs.

5.9 Calculation Functions

This instrument supports the following eight calculation functions.

Item	Function
NULL calculation	Subtracts a fixed value from a measurement value.
Smoothing calculation	Calculates the moving average.
Scaling calculation	Calculates $(\frac{M-B}{A}) \times C$.
dB calculation	Calculates $20 \log_{10}(\frac{M}{D})$.
dBm calculation	Calculates $10 \log_{10}(\frac{M^2}{D} \times \frac{1}{10^{-3}})$.
Comparator calculation	Performs a comparative calculation (HI/GO/LO).
MAX/MIN calculation	Calculates the maximum, minimum and average values.
Statistical calculation	Calculates the maximum value, minimum value, average value, standard deviation, and dispersion.

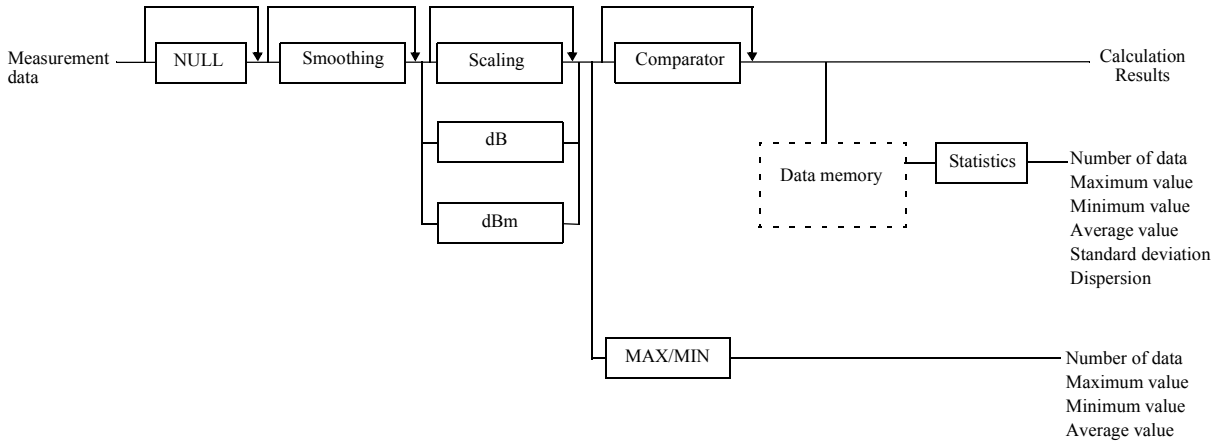


Figure 5-7 Calculation System Diagram

NOTE:

1. ***The following results may be obtained from calculating the data. This is because the internal resolutions of measurement values and setting values for the calculation are higher than the resolutions of the displayed data and remote output data.***
   - ***If the buzzer is set to ON in the MAX/MIN calculation, the buzzer sounds when the MAX or MIN value is renewed. However the buzzer may sound even when the display and remote output data are not changed.***
2. ***If the calculation is set to ON, the display of the calculation result is different from the ordinary measurement values and the decimal point and unit do not depend on the current measurement range.***
   <Example> DC voltage measurement, measurement range: 3000 mV, input voltage: 1 V
   Measurement value:1000.00 mV
   Calculation Result:1.00000 V

5.9.1 NULL Calculation

1. Function

   The NULL calculation subtracts the NULL constant from the measurement value and outputs the result.

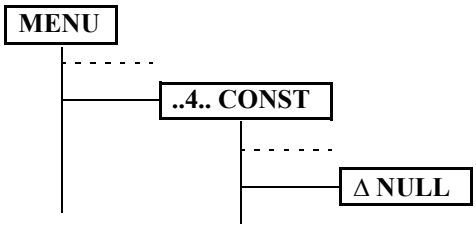
Measurement data output = Measurement value - NULL constant

   Pressing the **NULL** key performs the calculation and turns on the NULL lamp in the display section. When the **NULL** key is re-pressed while the NULL calculation is performed, the calculation is canceled and the NULL lamp turns off.

   CAUTION: ***The actual measurement value is not displayed while the calculation is performed. Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.***

2. NULL constant

   The measurement value when the **NULL** key is pressed is determined as the NULL constant. The NULL calculation cannot be performed when the overload (OL) occurs. The NULL constant can be checked in **MENU** during the calculation.

   The setting ranges are shown below.

	Setting range	Minimum resolution
7461A/61P	-9999999.E +6 to +9999999.E + 6	0.000000E - 9
7451A	-999999.E +6 to +999999.E + 6	

   The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

3. Changing the NULL constant

   1. Display the NULL constant according to item 2 (previous description).
   2. After pressing ▽, the blinking position of the following items, which can be changed, is moved by using ◁ and ▷.
      - From the most significant digit to the least significant digit in the numeric value
      - Unit prefix (exponential part setting)

- Decimal point

3. Change the setting value by using △ and ▽ at the blinking position.
   The unit prefix is switched according to the following order.
   p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T
4. Press **ENTER** to apply the setting.

4. Canceling the NULL calculation
   The NULL calculation is canceled when any of the following conditions occur.
   - The **NULL** key is pressed while the NULL calculation is performed.
   - The *RST command is executed.

5.9.2 Smoothing Calculation

1. Function
   The smoothing function is used when the noise is superimposed on the measurement signal. Because the measurement values in the specified number of times (smoothing count) are calculated by using the moving averaging method, the dispersion of the measurement values becomes small.
   The smoothing calculation equation is shown below.

Display = (Measurement value 1 + , , , , , + Measurement value N) / N

When the **SM** key is pressed, the calculation is performed and the SM lamp in the display section turns on or blinks.

When the **SM** key is re-pressed while the smoothing calculation is performed, the calculation is canceled and the SM lamp turns off.

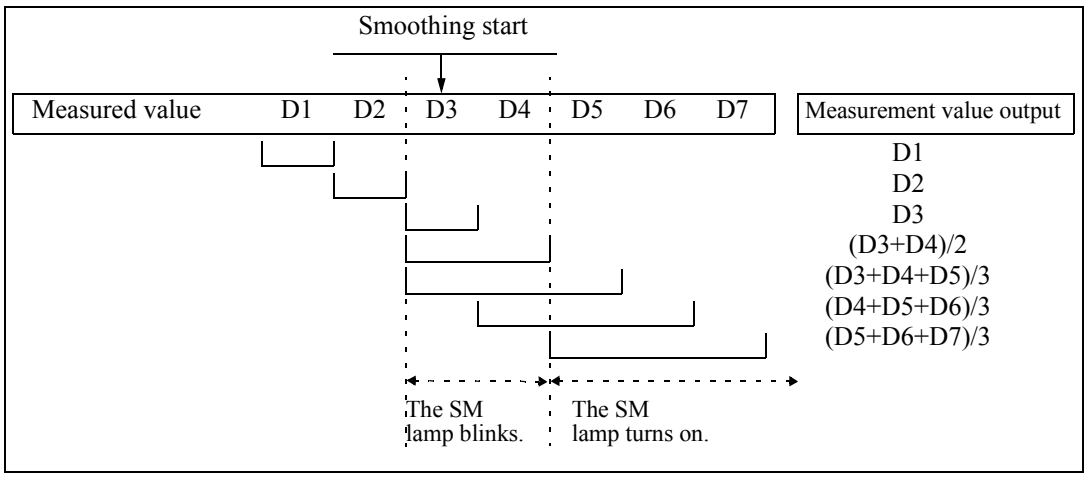
The SM lamp blinks from the smoothing start to the (N - 1)th measurement and turns on after the Nth measurement.

The (N + 1)th data is shown below.

Display = (Measurement value 2 + , , , , , + Measurement value (N + 1)) / N

The average of the measurement values, which have been acquired, is displayed from the smoothing start to the (N - 1)th measurement.

When the smoothing count, N, is set to 3, the operation is as follows.



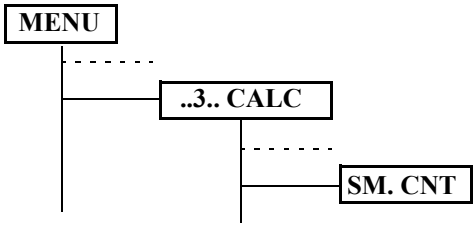
The smoothing calculation result, D (sm), at the n-th measurement, which exceeds the smoothing count, is shown below.

$$\text{Smoothing calculation result D (sm)} = \frac{1}{T}\sum_{i=n-T+1}^{n} Di$$

D (sm) :	Smoothing calculation result at the n-th measurement
Di:	Measurement value (before the smoothing calculation)
T:	Smoothing count (2 to 100 can be set.)

2. Setting the smoothing count

   The smoothing count is set in **MENU**.



   The smoothing count can be set between 2 and 100.

3. Restarting the smoothing calculation

   The smoothing calculation restarts from N=1 when any of the following conditions occur.

   - The power is turned on.
   - The smoothing count is changed.
   - The NULL calculation is performed.

4. Canceling the smoothing calculation
   The smoothing calculation is canceled when any of the following conditions occur.
   - The **SM** key is pressed while the smoothing calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.
5. Smoothing calculation and overload (OL)
   If the overload occurs during the smoothing process, this measurement value is ignored.
   (All measurement data except for OL data can be used in the smoothing calculation.)

5.9.3 Scaling Calculation

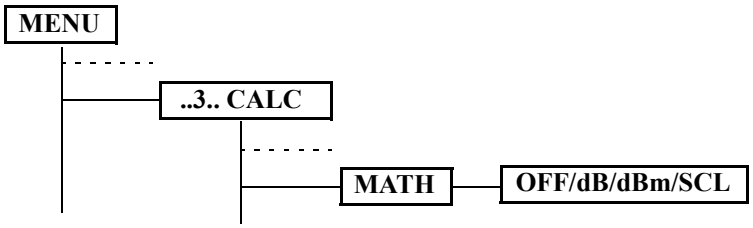
1. Function

   The scaling calculation equation is shown below.

$$\text{Display} = \frac{\text{Measurement value M - Constant B}}{\text{Constant A}} \times \text{Constant C}$$

   When SCL is selected from the MATH item in **MENU**, the MATH lamp in the display section turns on and the scaling calculation is performed.

   If OFF is selected from the MATH item in **MENU** while the scaling calculation is performed, the calculation is canceled and the MATH lamp turns off.

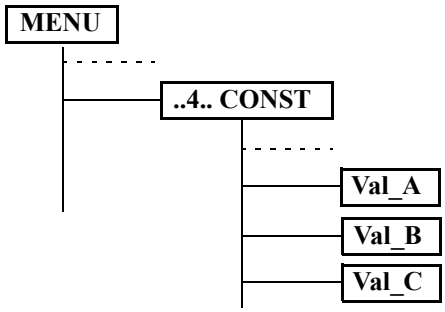


CAUTION: ***The actual measurement value is not displayed while the scaling calculation is performed. Remember that any hazardous voltage present on the input connector or test lead may not detected at this time.***

2. Setting scaling constants

NOTE: ***Press the*** **EXIT/LOCAL** ***key if quitting the setting procedure.***

   1. Set Constants A, B, and C in **MENU**.



   2. After selecting a constant to be set, the constant can be set by pressing ▽.

3. Setting the numerical value of the constant

   The setting ranges of Constants A, B, and C are shown below.

	Constant	Setting range	Minimum resolution
7461A/61P	A	-9999999.E +6 to +9999999.E + 6	0.000001E - 9
	B		0.000000E - 9
	C		
7451A	A	-999999.E +6 to +999999.E + 6	0.000001E - 9
	B		0.000000E - 9
	C		

   The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

   1. In the constant setting mode, the following items, which blink, can be selected by using ◁ and ▷.
      - From the most significant digit to the least significant digit in the numeric value
      - Unit prefix
      - Decimal point

   2. Change the numeric value and unit prefix by using △ and ▽.

      The unit prefix cycles through the following order.

      p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T

      Press the **TRIG** key in this mode if setting the measurement value as the scaling constant.

   3. Press **ENTER** to apply the setting.

4. S OL (scaling over)

   "±S OL" is displayed If the scaling calculation result exceeds the values of ±999.9999E + 6 (±999.999E + 6 for 7451A)

   In this time, the measurement range is not changed to the upper range even if the measurement range is set to Auto-range.

   (Auto-range is determined according to the measurement value acquired before the calculation is performed.)

5. Canceling the scaling calculation

   The scaling calculation is canceled when any of the following conditions occur.

   - OFF is selected from the MATH item in **MENU** while the scaling calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.
   - The dB or dBm calculation is performed.
   - The trigger source is set to LEVEL or DELTA.

5.9.4 dB and dBm calculation

1. Function

The dB calculation, in which the decibel conversion is performed, can be performed only in the voltage and current measurements.

The dBm calculation, which is useful to calculate the power gain per 1 mV, can be performed only in the voltage measurement.

If the zero is measured while the dB and dBm calculations are performed, the calculation error occurs and error message is displayed.

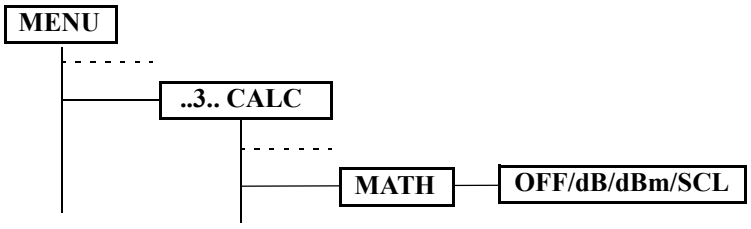
Only one calculation can be selected from the dB, dBm, and scaling calculations at a time.

The dB and dBm calculation equations are shown below.

$$\text{dB:}\quad \text{Display} = 20 \log_{10} \frac{|\ \text{Measurement value}\ |}{\text{Constant D}}$$

$$\text{dBm:}\quad \text{Display} = 10 \log_{10} \frac{(\text{Measurement value})^2 / \text{Constant D}}{10^{-3}}$$

When dB or dBm is selected from the MATH item in **MENU**, the MATH lamp in the display section turns on.



CAUTION: ***The actual measurement value is not displayed while the dB or dBm calculation is performed. Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.***

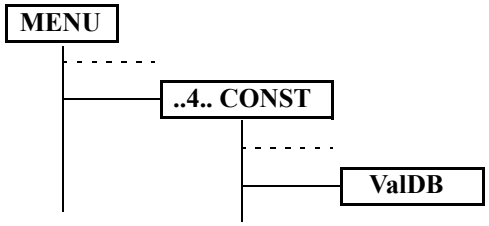
2. Setting Constant D

   The setting range of Constant D is shown below.

	Setting range	Minimum resolution
7461A/61P	0.000001E -9 to 9999999.E + 6	0.000001E - 9
7451A	0.000001E -9 to 999999.E + 6	

The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

1. Set Constant D in **MENU**.



2. After the constant setting mode is enabled by pressing ▽, the following items, which blink, can be selected by pressing ◁ and ▷.
   - From the most significant digit to the least significant digit in the numeric value
   - Unit prefix
   - Decimal point

3. Change the numeric value and unit prefix by using △ and ▽.

   The unit prefix cycles through the following order.

   p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T

   Press the **TRIG** key in this mode if setting the measurement value as Constant D.

4. Press **ENTER** to apply the setting.

3. Canceling the dB and dBm calculations

   The dB and dBm calculations are canceled when any of the following conditions occur.
   - The scaling calculation is set while the dB or dBm calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.
   - The trigger source is set to LEVEL or DELTA.

5.9.5 Comparator Calculation

1. Function

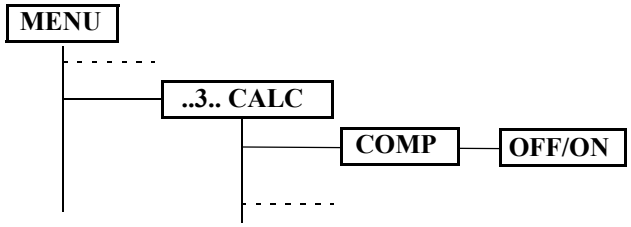
   The comparator calculation equation is shown below.

   HI = (Measurement value > HI setting value)
   LO = (Measurement value < LOW setting value)
   GO = (LOW setting value ≤ Measurement value ≤ HI setting value)

   (Processing the special data)

   - The overload (OL) of plus (+) data is judged as HI.
   - The overload (OL) of minus (-) data is judged as LO.

   The result whether HI, GO, or LO is displayed by setting COMP in **MENU** to ON.



   The judgment result can be output to the display, buzzer, and status register.

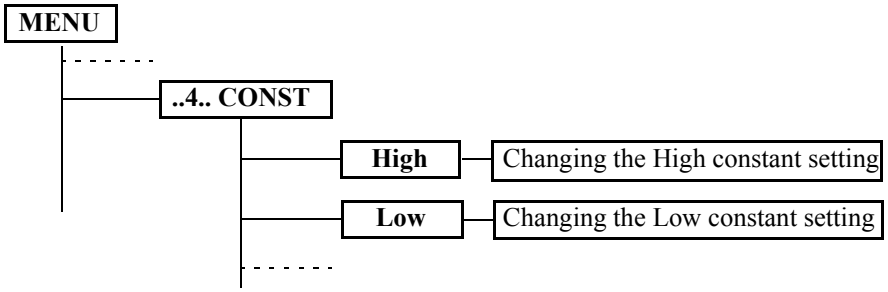
   (The buzzer sounds only when BEEP is set to ON.)

   The comparator calculation can be applied to all measurement functions.

   However, when MAX/MIN is set to ON, the comparator calculation is applied to the value before the MAX/MIN calculation.

2. Setting the judgment conditions

   The setting examples of the High constant, Low constant, and buzzer are shown below.



3. Setting the numerical value of the constant

   The setting ranges of the High and Low constants are shown below.

	Constant	Setting range	Minimum resolution
7461A/61P	High	-9999999.E +6 to +9999999.E + 6	0.0000001E - 9
	Low		
7451A	High	-999999.E +6 to +999999.E + 6	0.000001E - 9
	Low		

The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

1. After the constant setting mode is enabled by pressing ▽, the following items, which blink, can be selected by pressing ◁ and ▷.
   - From the most significant digit to the least significant digit in the numeric value
   - Unit prefix
   - Decimal point
2. Change the numeric value and unit prefix by using △ and ▽ at the blinking position.

   The unit prefix cycles through the following order.

   p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T

   Press the **TRIG** key in this mode if setting the measurement value as the High or Low constant.
3. Press **ENTER** to apply the setting.

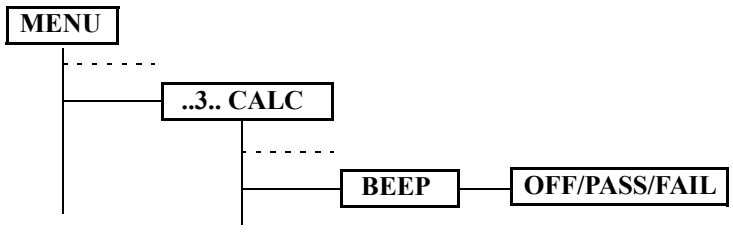
If the constants are set as “High constant < Low constant”, the calculation results may be judged as “HI and LO”. In this case, HI and LO are displayed at the same time.

4. Buzzer setting

   The buzzer output according to the PASS/FAIL judgments can be set.

   However, the buzzer output cannot be performed if •)) turns off.

   The buzzer setting is performed in BEEP in **MENU**.



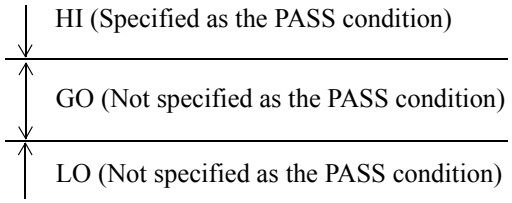
1. Press ▽ on the current setting display (BEEP) to enable the buzzer setting mode.
2. Select OFF, PASS, or FAIL by using ◁ and ▷.

5. PASS condition setting

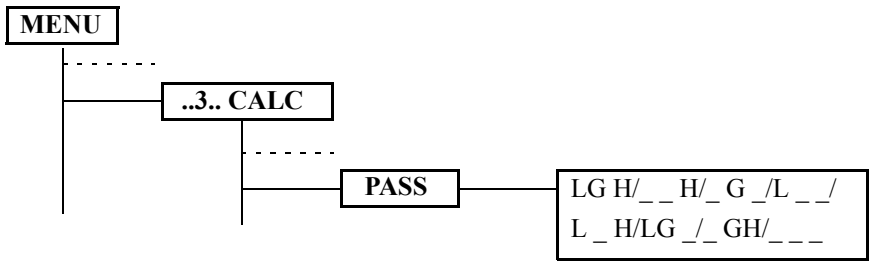
   In the PASS condition setting, each comparator calculation result (HI, GO, and LO) is specified as either PASS or not.

   For example, if HI is specified as the PASS condition and LO and GO are not specified as the PASS conditions, the comparator measurement result of HI is judged as PASS.

   The comparator calculation results of LO and GO are judged as FAIL.



   The PASS conditions are set in **MENU**.



   1. Press ▽ on the current setting display (PASS) to enable the PASS setting mode.
   2. Select the PASS condition by using ◁ and ▷.

6. Canceling the comparator calculation

   The comparator calculation is canceled when any of the following conditions occur.

   - The comparator calculation is set to OFF while the comparator calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.

7. Changing the measurement range and comparator operation

   The comparator function operates continuously even if the measurement range is changed.

   The reference values for the judgment are not changed because the values and the unit are set.

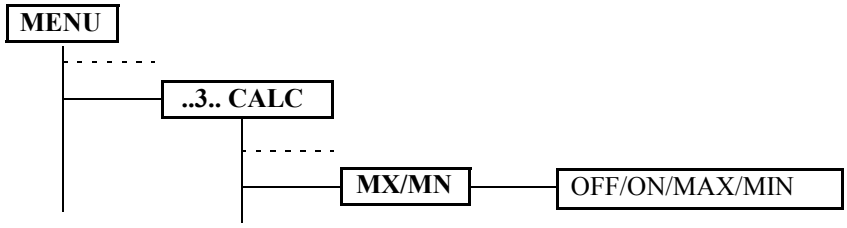
5.9.6 MAX/MIN Calculation

1. Function

    1. This function calculates the maximum, minimum, and average values while the MAX/MIN calculation is performed.
    2. The following calculations are performed at the same time in the MAX/MIN calculation.
        - Maximum value (MAX)
        - Minimum value (MIN)
        - Average value (AVE)
        - Measurement count
    3. All measurement data except for overload (OL) and calculation error data can be used for the calculation.
    4. If BEEP is set to ON, the buzzer sounds when the MAX or MIN value is renewed. However, although the displayed value is not changed, the buzzer may sound. This is because the resolution of measurement value is higher than the resolution of the displayed value.

2. Setting the calculation

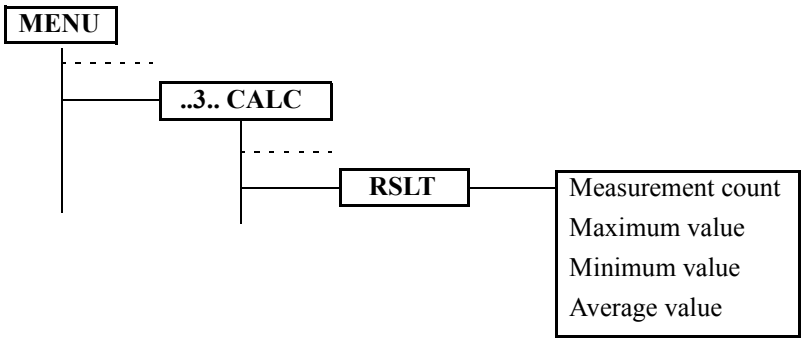
    The operation is selected as shown in below by selecting MX/MN in **MENU**.



OFF: Canceling the calculation [The MAX and MIN lamps turn off.]
ON: Performing the calculation (Measurement value display) [The MAX and MIN lamps turn on.]
MAX: Performing the calculation (Maximum value display) [The MAX lamp blinks.]
MIN: Performing the calculation (Minimum value display) [The MIN lamp blinks.]

3. How to call the MAX/MIN calculation result

    1. Select RSLT in **MENU** to see the calculation results display.



    2. Select a display item by pressing ◁ and ▷.

       The measurement count, maximum value, minimum value, and average value are displayed in this order.

4. MAX/MIN calculation value display function during the measurement

    1. The value while the MAX/MIN calculation is performed can be selected and displayed from the following three items.
       - Measurement value display (MAX and MIN lamps turn on.)
       - Maximum value display (MAX lamp blinks.)
       - Minimum value display (MIN lamp blinks.)

    2. How to set

       Select a display item according to "2. Setting the calculation" in this section.

       The selected value is displayed by pressing **ENTER**.

5. Canceling the MAX/MIN calculation

   The MAX/MIN calculation is canceled when any of the following conditions occur.
   - The MAX/MIN calculation is set to OFF while the MAX/MIN calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.

6. Restarting the MAX/MIN calculation

   The MAX/MIN calculation values are cleared and the MAX/MIN calculation restarts when any of the following conditions occur.
   - The power is turned on.
   - The MAX/MIN calculation is set to ON after it is set to OFF.
   - Any calculation except for the comparator calculation is changed to ON or OFF.
   - The NULL constant or the smoothing count is changed.

5.9.7 Statistical Calculation

1. Function

   1. This function calculates the maximum value, minimum value, average value, standard deviation, and range of the measurement data that is stored in the measurement memory.

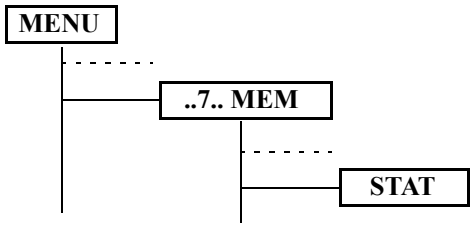
   2. The following calculations are performed at the same time in the statistical calculation.

      - Number of samples S_{CNT} =(Number of data) - (Number of overload), in the specified range
      - Maximum value S_{MAX} =Maximum value of the data to be calculated
      - Minimum value S_{MIN} =Minimum value of the data to be calculated
      - Average value $S_{AVE} = \frac{1}{S_{CNT}} \sum_{k=1}^{S_{CNT}} Dk$
      - Dispersion $S_{PTP} = | S_{MAX} - S_{MIN} |$
      - Standard deviation $S\sigma = \sqrt{\frac{1}{S_{CNT}-1} \sum_{k=1}^{S_{CNT}} (Dk-S_{AVE})^2}$

   3. All measurement data except for overload (OL) and calculation error data can be used for the calculation.

2. Performing the calculation

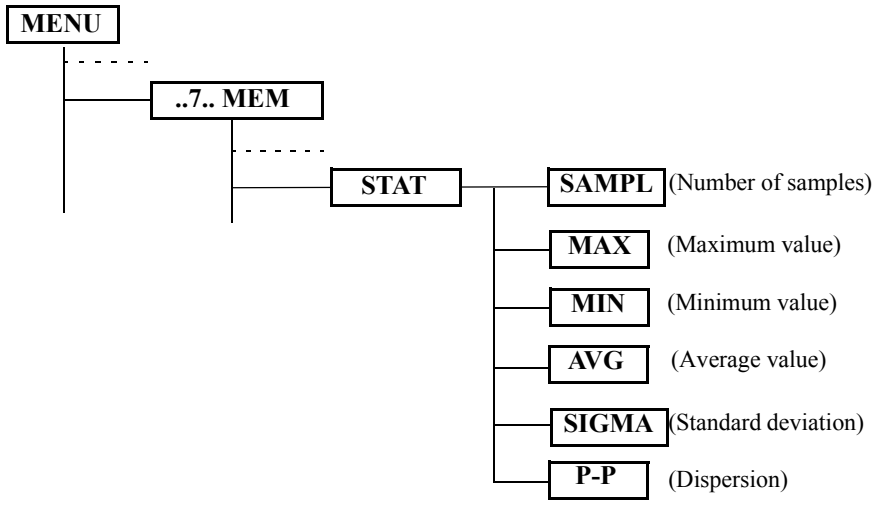
   1. Select STAT in **MENU**.



   2. Press **ENTER** to perform the calculation.
      'LAST' is displayed if no data is stored in the measurement data memory.

3. How to call the calculation result

   1. The display item of the calculation results can be selected in STAT in **MENU**.



   2. Select a display item by pressing ◁ and ▷ after the statistical calculation is performed.

5.10 Saving and Loading the Measurement Condition Settings

In this instrument, four files of measurement condition setting parameters can be saved in non-volatile memory USER0 to USER3.

For information on setting parameters, refer to Section 8.9, “System Recovery Procedure” and “Table 8-6 Initial Values of Setting Parameters.”

5.10.1 Auto-Loading when Turning on the Power

This instrument starts under the stored conditions when the power is turned off.

If the same startup conditions are always used, the measurement conditions stored in USER0 can be loaded when the power is turned on.

Select from the followings in ‘PON.LD’ in the **5 PARAM** category in **MENU**.

P.OFF	Starts this instrument under the measurement conditions stored when the power is turned off.
USER0	Loads the measurement conditions in USER0.

Press **ENTER** to apply the setting.

5.10.2 Saving the Setting Parameters

Select a memory from USER0 to USER3 in ‘SAVE’ in the **5 PARAM** category in **MENU**.

Press **ENTER** to save the setting parameters into the selected memory.

5.10.3 Loading the Setting Parameters

Select a memory from USER0 to USER3 and DFLT in ‘LOAD’ in the **5 PARAM** category in **MENU**.

Press **ENTER** to load the setting parameters from the selected memory.

If DFLT is selected, the initial values are loaded.

For more information on initial values, refer to Section 8.9, “System Recovery Procedure”.

5.11 Storing and Recalling the Measurement Data

This instrument includes the measurement data memory, in which the following number of measurement results can be stored.

- 7451A/61A: 10000 data
- 7461P: 20000 data

5.11.1 Memory Store

Select ON or OFF in 'Store' in **7 MEM** in **MENU**.

Pressing **ENTER** determines the setting and turns on the 'ST' indicator.

The measurement data is stored into the measurement data memory by closing MENU and starting the measurement.

When the number of measurement results reaches 10000 (20000 for 7461P), the buzzer sounds and the 'ST' indicator blinks.

5.11.2 Memory Recall

Enter the data number to be recalled in 'RCL' in **7 MEM** in **MENU**.

Pressing **ENTER** determines the setting and turns on the 'RCL' indicator.

The data number can be increased and decreased by using ◁ and ▷ before recalling the data number.

5.11.3 Memory Clear

The measurement data memory is cleared when any of the following conditions occur.

- The power is turned on.
- The device clear command is executed from the GPIB interface.
- The *RST or ICL command is executed.
- The parameter initialization is performed in **6 INIT** in **MENU**.
- The memory store setting of the measurement data is switched from OFF to ON.

5.12 System Settings

5.12.1 Buzzer Setting

Select ON or OFF in 'BEEP' in **9 SYS** in **MENU**.

ON	The buzzer sounds.
OFF	The buzzer does not sound. The buzzer does not sound regardless of the result of comparator calculation.

Press **ENTER** to apply the setting.

In the continuity measurement or if an error occurs, the buzzer sounds even when BEEP is set to OFF.

5.12.2 Selecting the Input Terminal

The following input terminals are on the front and rear panels respectively.

Front panel	DC voltage measurement, AC voltage measurement, Resistance measurement, Diode measurement, Continuity measurement, Frequency measurement, DC current measurement, and AC current measurement
Rear panel	DC voltage measurement, AC voltage measurement, Resistance measurement, Diode measurement, Continuity measurement, and Frequency measurement

Select the front or rear input terminals from 'TERM' in **9 SYS** in **MENU**.

As for the 7461P, the input terminals can be also selected by using ◁ and ▷ while pressing **SHIFT**.

While the rear input terminal is selected, the rear indicator in the display section turns on.

CAUTION: ***Do not switch between the input terminals on the front and rear panels while a voltage of 200 Vpeak or a current is applied at the input terminal.***
Do not apply a voltage of 200 V or more between the COM terminals on the front and rear panels.

5.12.3 Disabling Functions

To prevent incorrect functions from being selected because of malfunctions and to use this instrument safely, some functions can be disabled.

Select each function to be set to ENA or DIS in 'FInhI' in **9 SYS** in **MENU**.

Press ▽ to enter in the entry layer and select ENA (Enabling) or DIS (Disabling) by

using ◁ and ▷.

Press **ENTER** to apply the setting.

- The relationship according to disabling the function setting

  Although the ENA or DIS can be set for each function, some functions are automatically disabled depending on the settings of other functions.

  If a higher function setting in the dependency relation is disabled, no lower function setting can be changed.

Table 5-9 Relationship according to Disabling the Function

Higher function	Function depending on the higher function setting							
	DCV-Bch	Diode	ACV (AC+DC)	ACI (AC+DC)	Continuity	ACI	FREQ	TEMP
DCV	◎		○					
ACV			○	○		●	●	
DCI				○				
ACI				○			●	
2WΩ/LP-2WΩ		◎			○			○
4WΩ/LP-4WΩ					○			○
FREQ								

◎: Automatically set to ENA or DIS according to a higher function.

○: Disabled if any higher functions are set to DIS, and enabled if all higher functions are set to ENA.

●: Disabled if any of higher functions is set to DIS, and not enabled even if all higher functions are set to ENA.

- Canceling the setting

  To cancel the disabling setting and enable all functions, initialize in **6 INIT** in **MENU** or re-set all functions in 'F.Inhi' in **9 SYS** in **MENU**.

5.12.4 Front Panel Disabling Function

This function disables all panel keys on the front panel in this instrument.
The following shows how to set the front panel disabling function.

Password setting

The password is required to enable or disable the front panel keys.

A four-digit number from 0000 to 9999 can be used for the password.

Set the password according to the following procedure.

	Operation	Character display section
1.	Press **MENU** and select **9 SYS** by using ◁ and ▷.	MENU SYS
2.	Press ▽.	SYS BEEP
3.	Select 'PWD' by using ◁ and ▷.	SYS PWD
4.	Press ▽.	PWD NEW
5.	Select the digit in the numerical value entry section by using ◁ and ▷. Change the numerical value by using △ and ▽ to enter the password.	PWD NEW
6.	Press **ENTER**.	PWD OLD
7.	Enter the current password according to the above procedure.	PWD OLD
8.	Press **ENTER**. If the correct password is entered, PASS is displayed and the password is set.	PWD PASS
	If an incorrect password is entered, FAIL is displayed and the procedure returns to step 3. Re-set by using the correct password.	PWD FAIL
9.	Press **EXIT** to exit from MENU.	

Panel lock setting

Set the panel lock according to the following procedure.

	Operation	Character display section
1.	Press **MENU** and select **9 SYS** by using ◁ and ▷.	MENU SYS
2.	Press ▽.	SYS BEEP
3.	Select 'P.Lock' by using ◁ and ▷.	SYS P.Lock
4.	Press ▽.	P.Lock OFF
5.	Select ON by using ◁ and ▷. and press **ENTER**.	P.Lock ON
6.	Select the digit in the numerical value entry section by using ◁ and ▷. Change the numerical value by using △ and ▽ to enter the password.	PWD
7.	Press **ENTER**. If a correct password is entered, LOCK is displayed and exit from MENU. If an incorrect password is entered, the procedure returns to step 3. Re-set by using the correct password.	PWD LOCK

Canceling the panel lock

Cancel the panel lock according to the following procedure.

	Operation	Character display section
1.	If pressing any key while the panel lock is set, the entry layer of P.Lock is set.	PWD LOCK
2.	Select the digit in the numerical value entry section by using ◁ and ▷. Change the numerical value by using △ and ▽ to enter the password.	PWD LOCK
3.	Press **ENTER**.	
	If a correct password is entered, PASS is displayed and the password is set.	PWD PASS
	If an incorrect password is entered, FAIL is displayed and the procedure returns to step 3.	PWD FAIL

NOTE: ***The panel lock cannot be released if the password is forgotten. Do not forget the password. Copying the password onto paper is recommended.***

5.12.5 Error Queue

If any error occurs in this instrument, the error number is recorded and the 'ERR' indicator turns on.

Only the error number can be read from the front panel menu.

The error number and error message can be read from the GPIB or USB by using the query command.

Up to 20 errors can be recorded.

For more information on error numbers and solutions, refer to Section 8.11, “Error Message List.”

Reading the error queue

Errors are displayed in order by pressing ▽ in ‘QUEUE’ in **9 SYS** in **MENU**.

Clearing the error queue

The error queue is cleared by reading the errors.

6. HOW TO USE THE INTERFACE

The GPIB (IEEE-488) and USB interfaces are included as standard with this instrument.

However both interfaces cannot be used at the same time. Select either one.

6.1 Interface Selection

The interface can be selected and set only from MENU on the front panel.

1. Because the setting of the interface selection is saved in the nonvolatile memory, the setting does not change even if the power is turned off or the interface is reset.
2. The address specific to the instrument is set in the interface. To identify each DMM on the USB interface, the address of the USB interface (USB.ID) is set.
   The address is displayed by turning off the power or using the address setting in MENU.

The interface setting item and the initial setting when shipping are shown below.

Setting item	Initial setting when shipping
Header ON/OFF	ON
GPIB address/USB.ID	1
GPIB talker function	Addressable

Select the interface from GPIB and USB in 'BUS' in **8 I/F** in **MENU**.

Press **ENTER** to apply the setting.

Header ON/OFF

Select ON or OFF in 'HEADR' in **8 I/F** in **MENU**.

Press **ENTER** to apply the setting.

6.1.1 Response to the *IDN? Command (7451A/61A Only)

This instrument has changed its model name from "AD7451A/AD7461A" to "7451A/7461A." Accordingly, the response to the *IDN? command has also been changed. Note that an old model name response can also be set to maintain the compatibility with the application software.

The new model name is set when the instrument is shipped.

A new or old model name response setting does not change by the initialization command or the factory shipment initialization operation.

The new model name response to the *IDN? command is applied to software revision C00 or later.

6.1.1 Response to the *IDN? Command (7451A/61A Only)

Specification of response to *IDN?

New model name response	ADC Corp.,xxxxx,nnnnnnnnn,mmm xxxxx: 7451A or 7461A (Product model name) nnnnnnnnn: Serial No. mmm: Revision No.
Old model name response	ADC,xxxxxxx,nnnnnnnnn,mmm xxxxxxx: AD7451A or AD7461A (Product model name) nnnnnnnnn: Serial No. mmm: Revision No.

Changing the response to *IDN? by key operation

1. To change to the old model name response, turn on the power while pressing the **SHIFT** key.
2. After all displays are lit, “OLD *IDN?” is displayed and the response is changed to the old model name.
3. When the response is set to the old model name, the ‘ ▼ ’ indicator is activated at the lower right of the display section.
4. To change to the new model name response, turn on the power while pressing the **SHIFT** key again.
5. After all displays are lit, “NEW *IDN?” is displayed and the response is changed to the new model name.
6. When the response is set to the new model name, the ‘ ▼ ’ indicator at the lower right of the display section is turned off.

Changing the response to *IDN? by remote command

	Command	Description	Initial value
ADC command	OID0	The new model name is the response to *IDN?.	●
	OID1	The old model name is the response to *IDN?.	
	OID?	Response: OID0 or OID1	
SCPI command	:OIDentity {ON,1|OFF,0}	OFF,0: The new model name is the response to *IDN?.	●
		ON,1: The old model name is the response to *IDN?.	
	:OIDentity?	Response: 1 or 0	

When the response is set to the old model name, the ‘ ▼ ’ indicator is activated at the lower right of the display section.

6.2 GPIB

6.2.1 Overview

The automatic measurement system can be set up easily by using GPIB (General Purpose Interface Bus) because measurement functions and parameters can be set and measurement data can be read by using the external control.

The measurement values are not affected by any external devices connected to this instrument, because the GPIB signals from this instrument are electrically isolated from the measurement signal system.

The remote commands are common to the USB commands.

- General Specifications

  Standard: IEEE-488

  Code: ASCII code

  Logical level: Logic 0 “High” state +2.4 V or more

  Logic 1 “Low” state +0.4 V or less

Table 6-1 Interface Functions

Code	Function
SH1	Source handshake function
AH1	Acceptor handshake function
T5	Basic talker function, Listener-specified talker cancel function, Talk-Only mode function, Serial polling function
L4	Basic listener function, Talker-specified listener cancel function
SR1	Service request function
RL1	Remote/Local switching function
PP0	No parallel polling function
DC1	Device clear function (The SDC and DCL commands can be used.)
DT1	Device trigger function (The GET command can be used.)
C0	No controller function
E2	Using the tri-state bus driver

6.2.2 Precautions in Use of GPIB

1. Do not use too long a bus cable to connect any instrument or controller. Ensure that the total cable length does not exceed 20 m. ADC CORPORATION uses the following cables as the standard bus cable.

Table 6-2 Standard Bus Cable

Length	Name
0.5 m	408JE-1P5
1 m	408JE-101
2 m	408JE-102
4 m	408JE-104

2. Bus cables have piggyback connectors. A piggyback connector has both a male connector and a female connector by itself. The male and female connectors can be stacked on top of each other.

   When connecting bus cables, do not stack three or more connectors. Secure the connectors with the clamping screws.

3. Check the power requirements, grounding conditions, and setting conditions of each device before turning on the power of each device.

   Ensure that all devices on the bus are turned on. If any device on the bus is not turned on, the system may not operate correctly.

4. Connecting or disconnecting cables

   Turn off all devices on the bus before connecting or disconnecting GPIB cables. Check that the chassis of the devices on the bus are connected to each other and to the ground before connecting or disconnecting any GPIB cables.

5. ATN interruption while transmitting a message

   If an ATN interruption request is received on the bus while the messages are transmitted among devices, the request is allowed and previous status is cleared.

6. Do not connect the controller when using the Talk-Only mode.

7. Up to 255 characters in one transmission can be recognized as a program command. If the program command exceeds 255 characters, an error occurs.

8. Retain LOW on the REN line for 5 ms or more after the program command has been sent.

9. The next command can be received before the execution of the INI(ADC),INITiate(SCPI), or *TRG command is complete.
   To delay the execution of the next command until the current command is complete, use the *OPC, *OPC?, or *WAI command. Describe the *OPC, *OPC?, or *WAI command at the end of a line in the program.

   Example ":INIT;*OPC"
   ":INIT;*OPC?"
   ":INIT;*WAI"

6.2.3 GPIB Setting

The following setting menu can be used when the GPIB is selected as the interface.

Address setting

	Operations	Display
1.	Press **MENU** and select **8 I/F** by using ◁ and ▷.	MENU I/F
2.	Press **ENTER** to move to the I/F category layer.	I/F BUS
3.	Select BUS by using the ◁ and ▷ keys.	
4.	Press **ENTER** to move to the BUS selection layer.	BUS GPIB
5.	Select GPIB by using the ◁ and ▷ keys.	
6.	Press **ENTER** to set BUS to GPIB and move to the I/F category layer.	I/F BUS
7.	Select GP.Adr by using the ◁ and ▷ keys. 01 (Current address)	I/F GP.Adr
8.	Press **ENTER** to move to the address setting layer. 01 (Current address)	GP.Adr
9.	Select a digit to be changed by using the ◁ and ▷ keys. To set an address, increase and decrease a numeric value by using the △ and ▽ keys. 02	I/F GP.Adr
10.	Press **ENTER** to set the GPIB address. (New address)	
11.	Press **EXIT** to exit from MENU.	

6.2.3 GPIB Setting

Talk-Only setting

	Operations	Display
1.	Press **MENU** and select **8 I/F** by using the ◁ and ▷ keys.	MENU I/F
2.	Press **ENTER** to move to the I/F category layer.	I/F BUS
3.	Select BUS by using the ◁ and ▷ keys.	
4.	Press **ENTER** to move to the BUS selection layer.	BUS GPIB
5.	Select GPIB by using the ◁ and ▷ keys.	
6.	Press **ENTER** to set BUS to GPIB and move to the I/F category layer.	I/F BUS
7.	Select T.ONLY by using the ◁ and ▷ keys.	I/F T.ONLY
8.	Press **ENTER** to move to the Talk-Only setting layer.	T.ONLY ON
9.	Select ON or OFF by using the ◁ and ▷ keys.	
10.	Press **ENTER** to set Talk-Only to ON or OFF.	I/F T.ONLY
11.	Press **EXIT** to exit from MENU.	

6.3 USB

6.3.1 Overview

A USB (Universal Serial Bus), which is compliant with the USB2.0 Full Speed, is included in this instrument as standard.

By connecting multiple instruments to the USB ports on a personal computer, the functions of the instruments can be set, measurement data can be read from the personal computer, and the automatic measurement system can be set up easily.

NOTE: This instrument may not operate with all personal computers, hubs, etc.

6.3.2 USB Specifications

- Command language: ADC commands only
- Standard: Compliance with USB2.0 Full Speed
- Connector: USB B type (female)
- ID: Any number from 1 to 127 can be set as USBid.
- Remote/Local: Function is available.
- Input command: ASCII strings command which sets functions and the query command
- Output format ASCII strings which describe measurement data and query reply output
- Driver: The USB driver for ADC instruments is used.

6.3.3 Setting Up the USB Port

6.3.3.1 Connecting to a Personal Computer

Connect the connection cable to the USB connector (B type) on the rear panel of this instrument and the USB connector on the personal computer.

Insert the connector completely when connecting.

Use the USB hub when connecting several instruments to a personal computer.

When the 7461P is connected to a personal computer, the ADC Instruments USB Driver recognizes it as the 7461A. It does not affect the operation.

6.3.3.2 USBid Setting

The USBid setting menu can be used when USB is selected for the interface.

1. Enter any address from 1 to 127 in 'USBid' in **8 I/F** in **MENU**.
2. Press **ENTER** to apply the setting.

6.4 Control Signal

6.4.1 External Trigger Terminal (TRIGGER IN)

This instrument can be triggered by inputting a negative logic pulse from the external trigger terminal (TRIGGER IN) on the rear panel.

Select EXTernal as the trigger source when using this terminal.

The instrument starts measurement at the falling edge of TTL or the closed contact.

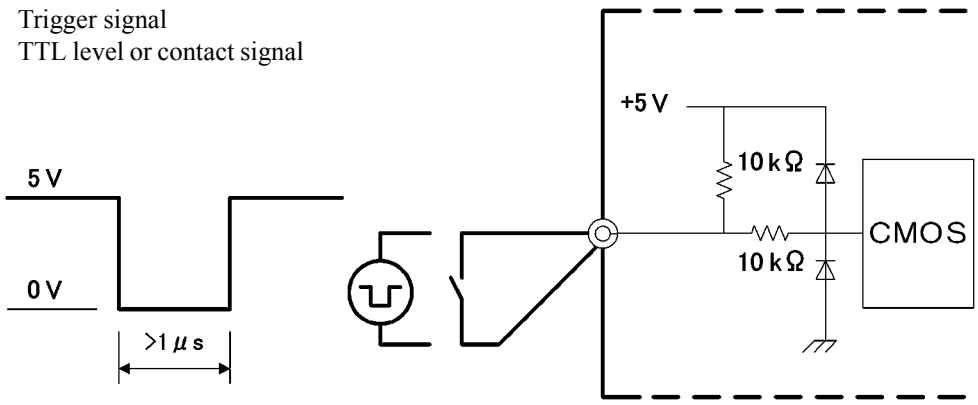


Figure 6-1 Simple Equivalent Circuit of Trigger Input Terminal

6.4.2 Measurement Complete Signal Terminal (COMPLETE OUT)

A negative logic pulse is output from the measurement complete signal terminal (COMPLETE OUT) on the rear panel when the measurement is complete.

The complete signal can be selected from the multi output and single output according to the output timing.

Multi output The signal is output for each sampling.

Single output The signal is output when the measurements set in sampling count are complete.

For more information, refer to "Measurement Timing according to the Trigger Source" in Section 5.8.2, "Trigger Source Selection."

Select Single (SGL) or Multi (MULTI) in 'C.Sig' in **9 SYS** in **MENU**.

Press **ENTER** to apply the setting.

The pulse width can be selected according to the connected device.

Select from the following in 'C.Widt' in **9 SYS** in **MENU**.

100 µs Pulse width: 100 µsec

5 µs Pulse width: 5 µsec

Press **ENTER** to apply the setting.

The measurement complete signal can be used as the TTL level and input signals that can be directly connected to the programmable controller.

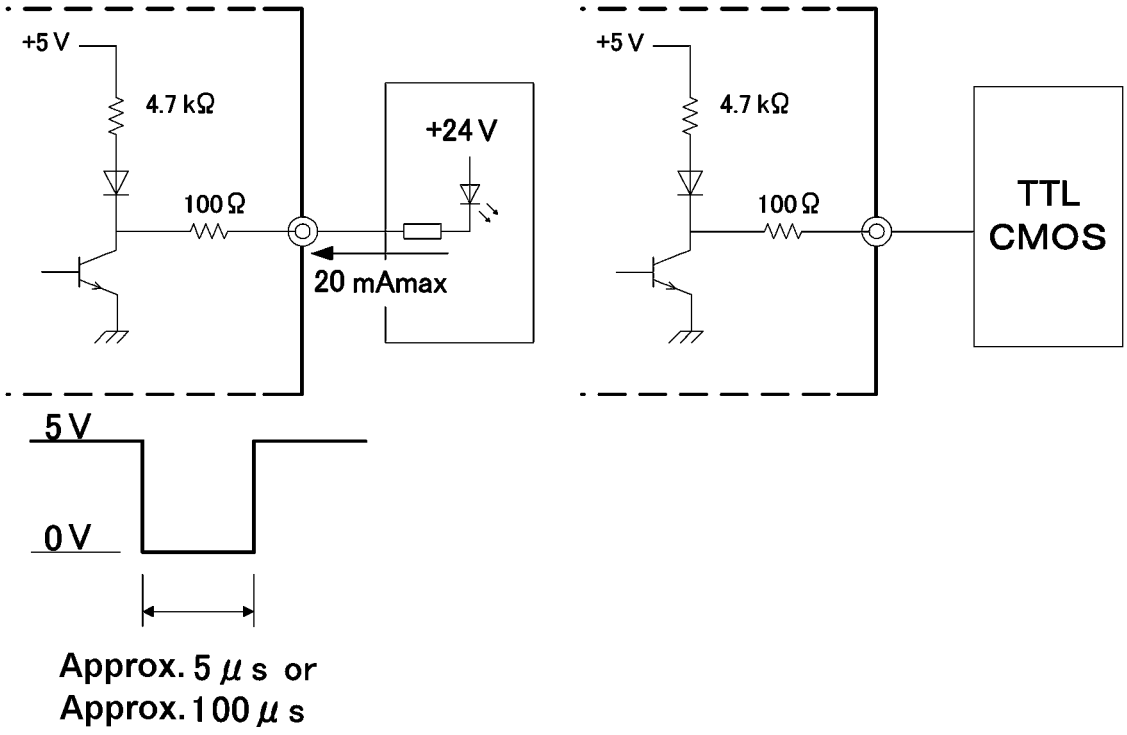


Figure 6-2 Simple Equivalent Circuit of Complete Signal Output Terminal

Complete signal output timing

The typical value of the complete signal output timing under the following setting conditions is shown.

Function: DCV, Auto-zero: OFF, Auto-range: OFF

1. Single output

   Time from the trigger signal input at the external trigger terminal to the complete signal output: Tc

   Tc=Tin × Sampling count + 2.5 msec

   Tin: The longest time in the followings.

   Sampling interval (SI)

   Integration time (IT) + 30 μsec

   310 μsec: Display OFF, Measurement data memory ON

   850 μsec: Display OFF, Measurement data memory OFF

   1.29 msec: Display ON

2. Multi output

   Period of the complete signal output: Tcp

   Tcp=Tin+1.4 msec

   Tin: The longest time in the followings.

   Sampling interval (SI)

   Integration time (IT) + 30 μsec

   310 μsec: Display OFF, Measurement data memory ON

   850 μsec: Display OFF, Measurement data memory OFF

   1.29 msec: Display ON

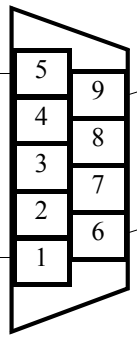
6.5 Comparator Output

The comparator calculation result is output from the comparator output terminal (COMPARATOR) on the rear panel as the TTL signal and the optical semiconductor relay contact signal.

Specifications

Connector	D-Sub9 pin
a. Optical semiconductor relay contact	
Allowable contact voltage (for break)	DC30 V
Allowable contact current	DC120 mA
Withstand voltage between contact and GND	30 V
Contact operating time	1 ms or less
Contact output	LO: 4 pin and 8 pin
	HI: 3 pin and 7 pin
	PASS: 2 pin and 6 pin
b. TTL output	
Output level	TTL
Maximum allowable applied voltage	12 V peak
Overload characteristics	Short-circuit (continuous)

Pin name	Pin number
Signal GND	5
RELAY LO	4
RELAY HI	3
RELAY PASS	2
TTL PASS	1



Pin number	Pin name
9	TTL FAIL
8	RELAY LO
7	RELAY HI
6	RELAY PASS

Figure 6-3 Signal Output Pin

The calculation result signals of Hi, Go, and Lo are output to the signal output pins of HI, PASS, and LO. The output to the PASS pin depends on the PASS conditions as shown in Table 6-3 and Table 6-4.

Table 6-3 Optical Semiconductor Relay Contact and TTL Logical Output (a)

Judgment result	Contact/TTL output pin			
	RELAY HI	RELAY PASS/TTL PASS	RELAY LO	TTL FAIL
Hi	make	See Table 6-4	break	See Table 6-4
Go	break		break	
Lo	break		make	

Table 6-4 Optical Semiconductor Relay Contact and TTL Logical Output (b)

PASS conditions				RELAY PASS TTL PASS			TTL FAIL		
				Judgment result			Judgment result		
Panel setting	Remote Command (ADC command)			Lo	Go	Hi	Lo	Go	Hi
LGH	LOP1	MIP1	HIP1	L	L	L	H	H	H
--H	LOP0	MIP0	HIP1	H	H	L	L	L	H
-G-	LOP0	MIP1	HIP0	H	L	H	L	H	L
L--	LOP1	MIP0	HIP0	L	H	H	H	L	L
L-H	LOP1	MIP0	HIP1	L	H	L	H	L	H
LG-	LOP1	MIP1	HIP0	L	L	H	H	H	L
-GH	LOP0	MIP1	HIP1	H	L	L	L	H	H
---	LOP0	MIP0	HIP0	H	H	H	L	L	L

H: break or Hi, L: make or Low

6.6 Digital Output (7461P Only)

The TTL signal or the optical semiconductor relay contact signal can be output arbitrarily from the comparator output terminal (COMPARATOR) on the rear panel by using the remote commands.

Only digital output enabled or disabled can be set by the panel keys. (Refer to Section 5.1.2, “Menu List.”) Other output settings are only possible by using the remote commands.

When the digital output is enabled, the comparator calculation results are not output from the output terminal. The comparator output is disabled.

Table 6-5 ADC Commands

<table>
<tr><th>Command</th><th>Description</th></tr>
<tr><td>DO0</td><td>Digital output disabled (Comparator output enabled)</td></tr>
<tr><td>DO1</td><td>Digital output enabled (Comparator output disabled)</td></tr>
<tr><td>DO?</td><td>Query of digital output
Reply: DO0 or DO1</td></tr>
<tr><td>DOSn</td><td>Controls arbitrarily the TTL signal or relay contact signal corresponding to 5 bits.
n (0 to 31) indicates a number that is converted from 5-bit binary to decimal.
<table>
<tr><th>Pin name</th><th>TTL FAIL</th><th>TTL PASS</th><th>RELAY PASS</th><th>RELAY HI</th><th>RELAY LO</th></tr>
<tr><td>Output signal</td><td>TTL output</td><td>TTL output</td><td>Relay contact</td><td>Relay contact</td><td>Relay contact</td></tr>
<tr><td>Pin number</td><td>9</td><td>1</td><td>2-6</td><td>3-7</td><td>4-8</td></tr>
<tr><td>Bit</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td>Setting</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td></tr>
</table>
H: break or Hi, L: make or Low

Example: DOS0
<table>
<tr><th>Pin name</th><th>TTL FAIL</th><th>TTL PASS</th><th>RELAY PASS</th><th>RELAY HI</th><th>RELAY LO</th></tr>
<tr><td>Output</td><td>Hi</td><td>Hi</td><td>break</td><td>break</td><td>break</td></tr>
</table></td></tr>
<tr><td>DOS?</td><td>Query of digital output setting
Replay: DOS0 to 31</td></tr>
</table>

Table 6-6 SCPI Commands

<table>
<tr><th>Command</th><th>Description</th></tr>
<tr><td>:DOUTput:STATe {OFF|ON}</td><td>Digital output disabled/enabled</td></tr>
<tr><td>:DOUTput:STATe?</td><td>Query of digital output</td></tr>
<tr><td>:DOUTput:BITS {0-31}</td><td>Digital output setting
Controls arbitrarily the TTL signal or relay contact signal corresponding to 5 bits. n (0 to 31) indicates a number that is converted from 5-bit binary to decimal.

<table>
<tr><td>Pin name</td><td>TTL FAIL</td><td>TTL PASS</td><td>RELAY PASS</td><td>RELAY HI</td><td>RELAY LO</td></tr>
<tr><td>Output signal</td><td>TTL output</td><td>TTL output</td><td>Relay contact</td><td>Relay contact</td><td>Relay contact</td></tr>
<tr><td>Pin number</td><td>9</td><td>1</td><td>2-6</td><td>3-7</td><td>4-8</td></tr>
<tr><td>Bit</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td>Setting</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td></tr>
</table>
H: break or Hi, L: make or Low

Example: DOUTput:BITS 0

<table>
<tr><td>Pin name</td><td>TTL FAIL</td><td>TTL PASS</td><td>RELAY PASS</td><td>RELAY HI</td><td>RELAY LO</td></tr>
<tr><td>Output</td><td>Hi</td><td>Hi</td><td>break</td><td>break</td><td>break</td></tr>
</table></td></tr>
<tr><td>:DOUTput:BITS?</td><td>Query reply: 0 to 31</td></tr>
</table>

Electrical specifications: The same as those described in Section 6.5, “Comparator Output”

Signal output pin: See Figure 6-3.

6.7 Command Reference

This section describes the command reference of this instrument.

6.7.1 Selecting the Command Language

This instrument can be operated remotely by using the following commands.

The command selection procedure is shown below.

1. Select the **8 I/F** category from the **MENU** mode and press ▽ to enter the selection layer.
2. Select the LANG parameter by pressing ◁ and ▷, and press ▽ to enter in the entry layer.
3. Select from SCPI, ADC, and R6552 by pressing ◁ and ▷, and press **ENTER** to apply the setting.

   SCPI: SCPI Command (GPIB only)

   ADC: Command used in ADC CORPORATION (GPIB/USB only)

   R6552: Command used in ADC CORPORATION's R6552 (only 7451A, GPIB only) For more information on operations, refer to “R6552 Series Digital Multimeter Operation Manual”.

6.7.1.1 Compatibility

If the R6552 command is selected, this instrument receives the R6552 command and executes it.

Command enabled when R6552 is selected	Command disabled when R6552 is selected
E	TRSn
M0 or M1	PTLn
M?	MSR?
LIT str	MSEn
LIT?	MSE?
DSR?	QSR?
DSE str	QSEn
DSE?	QSE?
SL n	*PSCn
SL?	

Command enabled when R6552 is selected	Command disabled when R6552 is selected
Z STP n RCLP n BZ n BZ?	

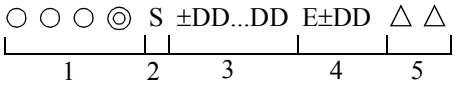
Note that the descriptions of the sampling rate setting commands vary depending on the selected language as follows:

Command	Description	
	ADC language selected (Integration time)	R6552 language selected (Integration time)
PR1	FAST1 (0.02 PLC)	FAST2 (0.1 PLC)
PR2	FAST2 (0.1 PLC)	MED (1 PLC)
PR3	MED (1 PLC)	SLOW2 (10 PLC)
PR4	SLOW1 (5 PLC)	Unacceptable
PR5	SLOW2 (10 PLC)	Unacceptable

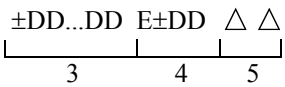
6.7.2 Output Data Format

1. Measurement data output Format (1 measurement result)

   With header

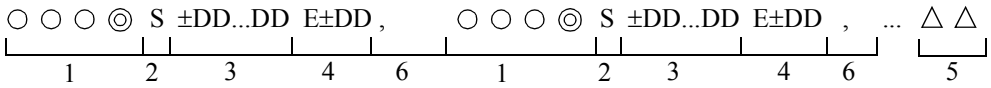
   Without header

2. Output format of the measurement data memory

   When two or more measurement results are output from the measurement data memory, the comma "," is output as a data delimiter.

With header



Without header

In the above case, header (1) and space (2) are not output.

1	○ ○ ○ ◎	Main header of 3 characters (○ ○ ○) + Sub header of 1 character (◎)
2	S	Space
3	Mantissa	Polarity + Decimal point + 6 or 7 digits numerical value
4	Exponent	E + Polarity + 2 digits numerical value
5	Block delimiter	

Main header	Description	
DCV	Measurement function	DC voltage measurement (DCV)
ACV		AC voltage measurement (ACV)
R2W		Resistance measurement (2WΩ)
R4W		Resistance measurement (4WΩ)
DCI		DC current measurement (DCI)
ACI		AC current measurement (ACI)
ADV		AC voltage (AC+DC coupling) measurement (ACV(AC+DC))
ADI		AC current (AC+DC coupling) measurement (ACI(AC+DC))
DOD		Diode measurement (Diode)
R2L		Low power resistance measurement (2WΩ)
R4L		Low power resistance measurement (4WΩ)
RCT		Continuity test (Cont)
FRQ		Frequency measurement (Freq)
BDV		DCV-Bch
TC2		Temperature measurement (TEMP-2W)°C
TC3		Temperature measurement (TEMP-3W)°C
TC4		Temperature measurement (TEMP-4W)°C
TF2		Temperature measurement (TEMP-2W)°F
TF3		Temperature measurement (TEMP-3W)°F
TF4		Temperature measurement (TEMP-4W)°F
TK2		Temperature measurement (TEMP-2W)K
TK3		Temperature measurement (TEMP-3W)K
TK4		Temperature measurement (TEMP-4W)K

Sub header	Description			Priority
O	Error information	Range over		1
O		Calculation	Scaling error	1
E			Err D (dB/dBm calculation error)	2
H	Calculation result information	Comparator result	HIGH	3
P			GO	3
L			LOW	3
M		MAX data		4
I		MIN data		4
B		dB data		5
W		dBm data		5
S		Scaling calculation data		6
N		NULL calculation data		7
_	None of the above			8

3. Measurement data
   - 7461A/61P

     ±○○○○○○○○E±○○ (6 1/2-digit display)

Overload (OL)	±9.999999E+37
OVER of the scaling calculation result	±9.999999E+36
dB/dBm calculation error	±9.999999E+35

   - 7451A

     ±○○○○○○○○E±○○ (5 1/2-digit display)

Overload (OL)	±9.99999E+37
OVER of the scaling calculation result	±9.99999E+36
dB/dBm calculation error	±9.99999E+35

7461A/61P			7451A		
DCV	100 mV	±ddd.ddddE-03	DCV	300 mV	±ddd.dddE-03
	1000 mV	±dddd.dddE-03		3000 mV	±dddd.ddE-03
	10 V	±dd.ddddE+00		30 V	±dd.ddddE+00
	100 V	±ddd.ddddE+00		300 V	±ddd.dddE+00
	1000 V	±dddd.dddE+00		1000 V	±dddd.ddE+00
ACV	100 mV	±ddd.ddddE-03	ACV	300 mV	±ddd.dddE-03
	1000 mV	±dddd.dddE-03		3000 mV	±dddd.ddE-03
	10 V	±dd.ddddE+00		30 V	±dd.ddddE+00
	100 V	±ddd.ddddE+00		300 V	±ddd.dddE+00
	700 V	±dddd.dddE+00		700 V	±dddd.ddE+00
ACV(AC+DC)	100 mV	±ddd.ddddE-03	ACV(AC+DC)	300 mV	±ddd.dddE-03
	1000 mV	±dddd.dddE-03		3000 mV	±dddd.ddE-03
	10 V	±dd.ddddE+00		30 V	±dd.ddddE+00
	100 V	±ddd.ddddE+00		300 V	±ddd.dddE+00
	700 V	±0ddd.dddE+00		700 V	±0ddd.ddE+00

6.7.2 Output Data Format

7461A/61P			7451A		
2WΩ			2WΩ	30 Ω	±dd.ddddE+00
	100 Ω	±ddd.ddddE+00		300 Ω	±ddd.dddE+00
	1000 Ω	±dddd.dddE+00		3000 Ω	±dddd.ddE+00
	10 kΩ	±dd.dddddE+03		30 kΩ	±dd.ddddE+03
	100 kΩ	±ddd.ddddE+03		300 kΩ	±ddd.dddE+03
	1000 kΩ	±dddd.dddE+03		3000 kΩ	±dddd.ddE+03
	10 MΩ	±dd.dddddE+06		30 MΩ	±dd.ddddE+06
	100 MΩ	±ddd.ddddE+06		300 MΩ	±ddd.dddE+06
4WΩ			4WΩ	30 Ω	±dd.ddddE+00
	100 Ω	±ddd.ddddE+00		300 Ω	±ddd.dddE+00
	1000 Ω	±dddd.dddE+00		3000 Ω	±dddd.ddE+00
	10 kΩ	±dd.dddddE+03		30 kΩ	±dd.ddddE+03
	100 kΩ	±ddd.ddddE+03		300 kΩ	±ddd.dddE+03
	1000 kΩ	±dddd.dddE+03		3000 kΩ	±dddd.ddE+03
	10 MΩ	±dd.dddddE+06		30 MΩ	±dd.ddddE+06
	100 MΩ	±ddd.ddddE+06		300 MΩ	±ddd.dddE+06
DCI	1000 µA	±dddd.dddE-06	DCI	3000 µA	±dddd.ddE-06
	10 mA	±dd.dddddE-03		30 mA	±dd.ddddE-03
	100 mA	±ddd.ddddE-03		300 mA	±ddd.dddE-03
	1000 mA	±dddd.dddE-03		3000 mA	±dddd.ddE-03
	3 A	±0d.dddddE+00			
ACI	1000 µA	±dddd.dddE-06	ACI	3000 µA	±dddd.ddE-06
	10 mA	±dd.dddddE-03		30 mA	±dd.ddddE-03
	100 mA	±ddd.ddddE-03		300 mA	±ddd.dddE-03
	1000 mA	±dddd.dddE-03		3000 mA	±dddd.ddE-03
	3 A	±0d.dddddE+00			
ACI(AC+DC)	1000 µA	±dddd.dddE-06	ACI(AC+DC)	3000 µA	±dddd.ddE-06
	10 mA	±dd.dddddE-03		30 mA	±dd.ddddE-03
	100 mA	±ddd.ddddE-03		300 mA	±ddd.dddE-03
	1000 mA	±dddd.dddE-03		3000 mA	±dddd.ddE-03
	3 A	±0d.dddddE+00			

7461A/61P			7451A		
FREQ	(<10 Hz)	(±d.dddddE+00)	FREQ	(<10 Hz)	(±d.dddddE+00)
	(<100 Hz)	(±dd.ddddE+00)		(<100 Hz)	(±dd.ddddE+00)
	(<1000 Hz)	(±ddd.dddE+00)		(<1000 Hz)	(±ddd.dddE+00)
	(<10 kHz)	(±d.dddddE+03)		(<10 kHz)	(±d.dddddE+03)
	(<100 kHz)	(±dd.ddddE+03)		(<100 kHz)	(±dd.ddddE+03)
	(<1000 kHz)	(±ddd.dddE+03)		(<1000 kHz)	(±ddd.dddE+03)
LP-2WΩ	100 Ω	±ddd.ddddE+00	LP-2WΩ	300 Ω	±ddd.dddE+00
	1000 Ω	±dddd.dddE+00		3000 Ω	±dddd.ddE+00
	10 kΩ	±dd.dddddE+03		30 kΩ	±dd.ddddE+03
	100 kΩ	±ddd.ddddE+03		300 kΩ	±ddd.dddE+03
	1000 kΩ	±dddd.dddE+03		3000 kΩ	±dddd.ddE+03
	10 MΩ	±dd.dddddE+06		30 MΩ	±dd.ddddE+06
LP-4WΩ	100 Ω	±ddd.ddddE+00	LP-4WΩ	300 Ω	±ddd.dddE+00
	1000 Ω	±dddd.dddE+00		3000 Ω	±dddd.ddE+00
	10 kΩ	±dd.dddddE+03		30 kΩ	±dd.ddddE+03
	100 kΩ	±ddd.ddddE+03		300 kΩ	±ddd.dddE+03
	1000 kΩ	±dddd.dddE+03		3000 kΩ	±dddd.ddE+03
	10 MΩ	±dd.dddddE+06		30 MΩ	±dd.ddddE+06
DIODE	1 range	±dddd.dddE-03	DIODE	1 range	±dddd.ddE-03
CONT	1 range	±dddd.dddE+00			
TEMP	1 range	±dddd.dddE+00	CONT	1 range	±dddd.ddE+00
DCV-Bch HI-COM measurement		±dd.dddddE+00	DCV-Bch HI-COM measurement		±dd.ddddE+00
DCV-Bch HI-LO measurement		±dd.dddddE+00	DCV-Bch HI-LO measurement		±dd.ddddE+00
DCV-Bch RATIO measurement		±d.ddddddE±dd	DCV-Bch RATIO measurement		±d.dddddE±dd
		±dd.dddddE±dd			±dd.ddddE±dd
		±ddd.ddddE±dd			±ddd.dddE±dd

( ): The Talker format in the frequency measurement function depends on the measurement range.
The Talker format of TEMP is only for the 7461P.

6.7.3 ADC Command Reference

This section describes the ADC command reference of this instrument.

"Initial value" shows the status when the *RST command is executed.

CAUTION: ***Ensure the following when using the USB interface.***
The device setting conditions are output by reading the data after executing the query command.
When executing the query command, set a wait time of 20 ms after the last command.

Item		Command	Description	Initial value
Measurement	Function	F1	DC voltage measurement (DCV)	●
		F2	AC voltage measurement (ACV)	
		F3	Resistance measurement (2WΩ)	
		F4	Resistance measurement (4WΩ)	
		F5	DC current measurement (DCI)	
		F6	AC current measurement (ACI)	
		F7	AC voltage (AC+DC coupling) measurement (ACV(AC+DC))	
		F8	AC current (AC+DC coupling) measurement (ACI(AC+DC))	
		F12	DC voltage Measurement DCV-Bch	
		F13	Diode measurement (Diode)	
		F20	Low power resistance measurement 2WΩ	
		F21	Low power resistance measurement 4WΩ	
		F22	Continuity test (Cont)	
		F42	Temperature measurement (TEMP, 2W) *7461P only	
		F43	Temperature measurement (TEMP, 3W) *7461P only	
		F44	Temperature measurement (TEMP, 4W) *7461P only	
		F50	Frequency measurement (Freq)	
		F?	Reply: One of F01 toF08, F12 to F13, F20 to F22, F42 to F44 or F50	
	Temperature unit *7461P only	TPU0	Temperature unit °C	●
		TPU1	Temperature unit °F	
		TPU2	Temperature unit K	
		TPU?	Reply: TPU0 to TPU2	
	Temperature sensor *7461P only	PT0	Platinum resistance bulb Pt100	●
		PT1	Platinum resistance bulb JPt100	
		PT?	Reply: PT0 or PT1	

Item		Command	Description	Initial value
Measurement	DCV-Bch measurement mode * Valid only in the DCV-Bch function (F12)	SDV0	HI-COM measurement	●
		SDV1	HI-LO measurement	
		SDV2	RATIO measurement	
		SDV?	Reply: One of SDV0 to SDV2	
	Invalid function	INH?n	Reads the invalid state of the function. n: 1 (DC voltage measurement) to 50 (Frequency measurement) Reply 0: Valid state of the function selection 1: Invalid state of the function selection	
	Trigger	*TRG	Trigger command	
	Measurement data memory	ST0	Store: OFF	●
		ST1	Store: ON	
		ST?	Reply: ST0 or ST1	
		IRDn, m	Readout range setting n, m: 0 to 9999 (7451A/61A) 0 to 19999 (7461P) • Parameters cannot be omitted.	(0, 0)
		IRO?	Data reading Reply: See Section 6.7.2, “Output Data Format.”	
		IRPO?	Number of measurement results readings Reply: IRPOdddd (7451A/61A) IRPOddddd (7461P)	
		IRNO?	Data range reading Reply: IRNOdddd, dddd (7451A/61A) IRNO±ddddd, ±ddddd (7461P) No data: IRNO0000, -001 (7451A/61A) IRNO+00000, -00001 (7461P)	
		ICL	Measurement data memory initialization	
Trigger system	Start	INI	Leaves the IDLE state.	
	Continue	INIC0	CONTINUOUS OFF	
		INIC1	CONTINUOUS ON	●
		INIC?	Reply: INIC0 or INIC1	
	Trigger abort	ABO	Moves to the IDLE state.	
	Trigger source selection	TRS0	IMMEDIATE	●
		TRS1	MANUAL	
		TRS2	EXTERNAL	
		TRS3	BUS	
		TRS4	LEVEL	
		TRS5	DELTA	
		TRS?	Reply: One of TRS0 to TRS5	

Item		Command	Description	Initial value
Trigger system	LEVEL trigger	TLVn	Level count value setting n:-1199999 to +1199999 (7461A/61P) n:-319999 to +319999 (7451A) Setting resolution: 1	(0)
		TLV?	Reply: TLV±ddddddd (7461A/61P) Reply: TLV±dddddd (7451A)	
		TLS0	NEGATIVE (Falling slope)	
		TLS1	POSITIVE (Rising slope)	●
		TLS?	Reply: TLS0 or TLS1	
	DELTA trigger	TDLn	Delta count value setting n: +1 to +600000 (7461A/61P) n: +1 to +160000 (7451A) Setting resolution:1	(1)
		TDL?	Reply: TDLddddddd (7461A/61P) Reply: TDLdddddd (7451A)	
		TDS0	NEGATIVE (Negative slope)	
		TDS1	POSITIVE (Positive slope)	●
		TDS2	BOTH (absolute value)	
		TDS?	Reply: One of TDS0 to TDS2	
		TDC0	CONVERGE (Convergence)	
		TDC1	EMISSION (Divergence)	●
		TDC?	Reply: TDC0 or TDC1	
	Sampling interval	TRTn	n:50 µs to 3600 s (7461A/61P) n:200 µs to 3600 s (7451A)	(500 ms)
		TRT?	Reply: TRT+d.ddddddE±dd (The position of the decimal point depends on the setting.)	
	Trigger delay	TRDn	n: 0 to 3600 (seconds)	(0)
		TRD?	Reply: TRD+d.ddddE±dd (The position of the decimal point depends on the setting.)	
	Trigger count	TRNn	n:1 to 50000 (counts)	(1)
		TRN?	Reply: TRNddddd	
	Sampling count (by one trigger)	SPNn	n:1 to 16000 (counts)	(1)
		SPN?	Reply: SPNddddd	

Item		Command	Description	Initial value
Trigger system	Pre-trigger value (LEVEL/DELTA trigger)	PTL0	0% (OFF)	●
		PTL1	25%	
		PTL2	50%	
		PTL3	75%	
		PTL?	Reply: PTLd	
	Complete signal output mode	TRCM0	SINGLE	●
		TRCM1	MULTI	
		TRCM?	Reply: TRCM0 or TRCM1	
	Complete signal width	CW0	Specifying the output signal width of the complete signal: 5 μs	
		CW1	Specifying the output signal width of the complete signal: 100 μs	●
		CW?	Reply: CW0 or CW1	

6.7.3 ADC Command Reference

Item		Command	Description										Initial value
Measurement condition	Measurement range		7461A/61P										
			DCV	ACV	ACV (AC+DC)	2WΩ	4WΩ	DCI	ACI	ACI (AC+DC)	LP-2WΩ	LP-4WΩ	Freq	
		R0	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	●
		R3	100 mV	100 mV	100 mV	100 Ω	100 Ω	-	-	-	100 Ω	100 Ω	(100 mV)	
		R4	1000 mV	1000 mV	1000 mV	1000 Ω	1000 Ω	1000 µA	1000 µA	1000 µA	1000 Ω	1000 Ω	(1000 mV)	
		R5	10 V	10 V	10 V	10 kΩ	10 kΩ	10 mA	10 mA	10 mA	10 kΩ	10 kΩ	(10 V)	
		R6	100 V	100 V	100 V	100 kΩ	100 kΩ	100 mA	100 mA	100 mA	100 kΩ	100 kΩ	(100 V)	
		R7	1000V	700 V	700 V	1000 kΩ	1000 kΩ	1000 mA	1000 mA	1000 mA	1000 kΩ	1000 kΩ	(700 V)	
		R8	-	-	-	10 MΩ	10 MΩ	3 A	3 A	3 A	10 MΩ	10 MΩ	-	
		R9	-	-	-	100 MΩ	100 MΩ	-	-	-	-	-	-	
		R?	Reply: One of R0, R3 to R9											
			7451A											
			DCV	ACV	ACV (AC+DC)	2WΩ	4WΩ	DCI	ACI	ACI (AC+DC)	LP-2WΩ	LP-4WΩ	Freq	
		R0	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	●
		R2	-	-	-	30 Ω	30 Ω	-	-	-	-	-	-	
		R3	300 mV	300 mV	300 mV	300 Ω	300 Ω	-	-	-	300 Ω	300 Ω	(300 mV)	
		R4	3000 mV	3000 mV	3000 mV	3000 Ω	3000 Ω	3000 µA	3000 µA	3000 µA	3000 Ω	3000 Ω	(3000 mV)	
		R5	30 V	30 V	30 V	30 kΩ	30 kΩ	30 mA	30 mA	30 mA	30 kΩ	30 kΩ	(30 V)	
		R6	300 V	300 V	300 V	300 kΩ	300 kΩ	300 mA	300 mA	300 mA	300 kΩ	300 kΩ	(300 V)	
		R7	1000 V	700 V	700 V	3000 kΩ	3000 kΩ	3000 mA	3000 mA	3000 mA	3000 kΩ	3000 kΩ	(700 V)	
		R8	-	-	-	30 MΩ	30 MΩ	-	-	-	30 MΩ	30 MΩ	-	
		R9	-	-	-	300 MΩ	300 MΩ	-	-	-	-	-	-	
		R?	Reply: One of R0, R2 to R9											
	Range Fix	RX	Range switching from AUTO to MANUAL											
	Sampling rate	PR0	FREE (PR0 is returned when the sampling interval or integration time is changed.)											
		PR1	FAST1											
		PR2	FAST2 (The same as PR1 in the frequency measurement)											
		PR3	MED											
		PR4	SLOW1											
		PR5	SLOW2 (The same as PR4 in the frequency measurement)											●
		PR?	Reply: One of PR0 to PR5											

Item		Command	Description	Initial value
Measurement condition	Integration time *3	ITPn	Sets the integration time in units of PLC. 0.02 PLC to 100 PLC *1	(10)
		ITP?	Reply: ITP+d.dddddE±dd (The position of the decimal point depends on the setting.)	
		ITSn	Sets the integration time in unit of second. • 7461A/61P n:0.00001 to 10 • 7451A n:0.0001 to 10	(0.2)
		ITS?	Reply: ITS+d.dddddE±dd (The position of the decimal point depends on the setting.)	
	Number of display digits	RE3	3 1/2-digit display	
		RE4	4 1/2-digit display	
		RE5	5 1/2-digit display	●
		RE6	6 1/2-digit display (7461A/61P only)	●
		RE?	Reply : One of RE3 to RE6 (7461A/61P) : One of RE3 to RE5 (7451A)	
	AC filter*2	FL1	FAST	
		FL2	MED	●
		FL3	SLOW	
		FL?	Reply: One of FL1 to FL3	
	Auto-zero	AZ0	OFF	
		AZ1	ON	●
		AZ2	ONCE (Auto-zero is set to OFF after executing once.)	
		AZ?	Reply: AZ0 or AZ1	
	Continuity threshold constant	KOMn	n:1 to 1000 (Ω) setting resolution: 1 Ω	(10)
		KOM?	Reply: KOMdddd	
	DCV input impedance	IMP0	HiZ	●
		IMP1	10 MΩ	
		IMP?	Reply: IMP0 or IMP1	

*1: When the integration time is outside this range, it is set in seconds. Refer to Section 5.7.1, “Sampling Interval (SI) and Integration Time (IT) Settings.”
*2: The AC filter is enabled in the current set function that is ACV, ACV(AC+DC), ACI, or ACI(AC+DC).
*3: The integration time is enabled in all functions except for the frequency measurement.

Item		Command	Description	Initial value
Calculation	NULL calculation	NL0	OFF	●
		NL1	ON	
		NL?	Reply: NL0 or NL1	
		KNLn	NULL constant setting n:-9999999.E+6 to +9999999.E+6 (7461A/61P) n:-999999.E+6 to +999999.E+6 (7451A) Setting resolution: 0.000001E-9 Note: The setting is disabled when the NULL calculation is set to OFF.	(0)
		KNL?	Reply: KNL±d.dddddd E±dd (7461A/61P)*3 Reply: KNL±d.ddddd E±dd (7451A)	
	Smoothing calcula-tion	SM0	OFF	●
		SM1	ON	
		SM?	Reply: SM0 or SM1	
		TIn	Smoothing count n: 2 to 100 (counts)	(10)
		TI?	Reply: TIddd	
	Scaling calculation	SC0	OFF	●
		SC1	ON	
		SC?	Reply: SC0 or SC1	
		KAn	Constant A (Zero cannot be set). n:-9999999.E+6 to +9999999.E+6 (7461A/61P) n:-999999.E+6 to +999999.E+6 (7451A) Setting resolution: 0.000001E-9	(1)
		KBn	Constant B n:-9999999.E+6 to +9999999.E+6 (7461A/61P) n:-999999.E+6 to +999999.E+6 (7451A) Setting resolution: 0.000001E-9	(0)
		KCn	Constant C n:-9999999.E+6 to +9999999.E+6 (7461A/61P) n:-999999.E+6 to +999999.E+6 (7451A) Setting resolution: 0.000001E-9	(1)
		KAM	Sets Constant A to the measurement value.	
		KBM	Sets Constant B to the measurement value.	
		KCM	Sets Constant C to the measurement value.	
		KA?	Reply: KA±d.dddddd E±dd (7461A/61P)*3 Reply: KA±d.ddddd E±dd (7451A)	
		KB?	Reply: KB±d.dddddd E±dd (7461A/61P) Reply: KB±d.ddddd E±dd (7451A)	

*3: The position of the decimal point in the reply is fixed.

Item		Command	Description	Initial value
Calculation	Scaling calculation	KC?	Reply: KC±d.dddddd E±dd (7461A/61P) Reply: KC±d.ddddd E±dd (7451A)	
	dB/dBm calculation	DB0	dB calculation OFF	●
		DB1	dB calculation ON	
		DB2	dBm calculation ON	
		DB?	Reply: One of DB0 to DB2	
		KDn	Constant D n:0.000001E-9 to 9999999.E+6 (7461A/61P) n:0.00001E-9 to 999999.E+6 (7451A)	(1)
		KDM	Sets Constant D to the measurement value.	
		KD?	Reply: KD±d.dddddd E±dd (7461A/61P)*3 Reply: KD±d.ddddd E±dd (7451A)	
	MAX•MIN calculation ○○○ : Header (Refer to Section 6.7.2, "Output Data Format.")	MN0	MAX•MIN calculation OFF	●
		MN1	MAX•MIN calculation ON	
		MN?	Reply: One of MN0 to MN1	
		MAX?	MAX value reading*3 Reply: ○○○ M ±d.dddddd E±dd (7461A/61P) Reply: ○○○ M ±d.ddddd E±dd (7451A)	
		MIN?	MIN value reading*3 Reply: ○○○ I ±d.dddddd E±dd (7461A/61P) Reply: ○○○ I ±d.ddddd E±dd (7451A)	
		AVE?	AVE value reading*3 Reply: ○○○ A ±d.dddddd E±dd (7461A/61P) Reply: ○○○ A ±d.ddddd E±dd (7451A)	
		AVN?	Measurement count reading*3 Reply: AVN±d.dddddd E±dd (7461A/61P) Reply: AVN±d.ddddd E±dd (7451A)	
	Comparator calculation	CO0	OFF	●
		CO1	ON	
		CO?	Reply: CO0 or CO1	
		HIn	HIGH constant n:-9999999.E+6 to +9999999.E+6 (7461A/61P) n:-999999.E+6 to +999999.E+6 (7451A) Setting resolution: 0.000001E-9	(0)

*3: The position of the decimal point in the reply is fixed.

Item		Command	Description	Initial value
Calculation	Comparator calculation	LOn	LOW constant n:-9999999.E+6 to +9999999.E+6 (7461A/61P) n:-999999.E+6 to +999999.E+6 (7451A) Setting resolution:0.000001E-9	(0)
		HIM	Sets the HIGH constant to the measurement value.	
		LOM	Sets the LOW constant to the measurement value.	
		HI?	Reply: HI±d.dddddE±dd (7461A/61P) *3 Reply: HI±d.ddddE±dd (7451A)	
		LO?	Reply: LO±d.dddddE±dd (7461A/61P) *3 Reply: LO±d.ddddE±dd (7451A)	
	PASS condition range setting	LOP0	Calculation result LO is not specified as the PASS condition.	●
		LOP1	Calculation result LO is specified as the PASS condition.	
		LOP?	Reply: LOP0 or LOP1	
		MIP0	Calculation result GO is not specified as the PASS condition.	
		MIP1	Calculation result GO is specified as the PASS condition.	●
		MIP?	Reply: MIP0 or MIP1	
		HIP0	Calculation result HI is not specified as the PASS condition.	●
		HIP1	Calculation result HI is specified as the PASS condition.	
		HIP?	Reply: HIP0 or HIP1	
	Statistical calculation	SIRDn, m	Statistical calculation range setting and performing the calculation n, m: 0 to 9999 (7451A/61A) n, m: 0 to 19999 (7461P) Note: 1. An error occurs if no data exists in the range set by the measurement data memory. 2. No parameters can be omitted. (0, -1)	(0, -1)
		SIRD?	Statistical calculation range reading Reply: SIRDdddd,dddd (7451A/61A) Reply: ±SIRDddddd, ±ddddd (7461P) Initial state: SIRD+0000, -0001 (7451A/61A) Initial state: SIRD+00000, -00001 (7461P)	
			Statistical calculation result reading	
		SCNT?	Number of samples reading Reply: SCNT+d.dddddE+dd (7461A/61P) *3 Reply: SCNT+d.ddddE+dd (7451A)	
		SMAX?	Maximum value reading Reply: SMAX±d.dddddE±dd (7461A/61P) *3 Reply: SMAX±d.ddddE±dd (7451A)	
		SMIN?	Minimum value reading Reply: SMIN±d.dddddE±dd (7461A/61P) *3 Reply: SMIN±d.ddddE±dd (7451A)	

*3: The position of the decimal point in the reply is fixed.

<table>
<tr><th colspan="2">Item</th><th>Command</th><th>Description</th><th>Initial value</th></tr>
<tr><td rowspan="11">Calculation</td><td rowspan="3">Statistical calculation</td><td>SAVE?</td><td>Average value reading
Reply: SAVE±d.dddddd E±dd (7461A/61P) *3
Reply: SAVE±d.ddddd E±dd (7451A)</td><td></td></tr>
<tr><td>SSIG?</td><td>Standard deviation value reading
Reply: SSIG+d.dddddd E±dd (7461A/61P) *3
Reply: SSIG+d.ddddd E±dd (7451A)
However, if the number of samples is 1 or less:
Outputs SSIG+9.999999E+30 (7461A/61P)
Outputs SSIG+9.99999E+30 (7451A)</td><td></td></tr>
<tr><td>SPTP?</td><td>MAX-MIN reading
Reply: SPTP±d.dddddd E±dd (7461A/61P) *3
Reply: SPTP±d.ddddd E±dd (7451A)</td><td></td></tr>
<tr><td rowspan="8">Digital output
*7461P only</td><td>DO0</td><td>Digital output disabled (comparator output enabled)</td><td>●</td></tr>
<tr><td>DO1</td><td>Digital output enabled (comparator output disabled)</td><td></td></tr>
<tr><td>DO?</td><td>Query of digital output
Reply: DO0 or DO1</td><td></td></tr>
<tr><td>DOSn</td><td>Controls arbitrarily the TTL signal or relay contact signal corresponding to 5 bits.
n (0 to 31) indicates a number that is converted from 5-bit binary to decimal.
<table>
<tr><th>Pin name</th><th>TTL FAIL</th><th>TTL PASS</th><th>RELAY PASS</th><th>RELAY HI</th><th>RELAY LO</th></tr>
<tr><td>Output signal</td><td>TTL output</td><td>TTL output</td><td>Relay contact</td><td>Relay contact</td><td>Relay contact</td></tr>
<tr><td>Pin number</td><td>9</td><td>1</td><td>2-6</td><td>3-7</td><td>4-8</td></tr>
<tr><td>Bit</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td>Setting</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td><td>0: H
1: L</td></tr>
</table>
H: break or Hi, L: make or Low

Example: DOS0
<table>
<tr><th>Pin name</th><th>TTL FAIL</th><th>TTL PASS</th><th>RELAY PASS</th><th>RELAY HI</th><th>RELAY LO</th></tr>
<tr><td>Output</td><td>Hi</td><td>Hi</td><td>break</td><td>break</td><td>break</td></tr>
</table></td><td>0</td></tr>
<tr><td>DOS?</td><td>Query of digital output setting
Replay: DOS0 to 31</td><td></td></tr>
</table>

*3: The position of the decimal point in the reply is fixed.

Item		Command	Description	Initial value
System	Buzzer	BP0	OFF	●
		BP1	The buzzer sounds when the comparator calculation result is FAIL.	
		BP2	The buzzer sounds when the comparator calculation result is PASS.	
		BP?	Reply: One of BP0 to BP2	
	Measurement data display	DS0	OFF	
		DS1	ON	●
		DS?	Reply: DS0 or DS1	
	Power supply frequency	LF?	Reply LF0: 50 Hz LF1: 60 Hz	
	Initialization	*RST	Parameter initialization	
		C	Device clear	
	Changing the response to *IDN? *5	OID0	The new model name is the response to *IDN?.	●
		OID1	The old model name is the response to *IDN?.	
		OID?	Response: OID0 or OID1	
	Instrument information *For 7461P, new model name response only	*IDN?	New model name response ADC Corp.,xxxxx,nnnnnnnnn,mmm xxxxx: 7451A, 7461A or 7461P (Product model name) nnnnnnnnn: Serial No. mmm: Revision No. The new model name response is applied to software revision C00 or later. Old model name response ADC,xxxxxxx,nnnnnnnnn,mmm xxxxxxx: AD7451A or AD7461A (Product model name) nnnnnnnnn: Serial No. mmm: Revision No.	
	Header	H0	Header OFF	
		H1	Header ON	
		H?	Reply: H0 or H1	
	Block delimiter *4	DL0	CR/LF+EOI	●
		DL1	LF	
		DL2	EOI	
		DL?	Reply: One of DL0 to DL2	
	SRQ	S1	Prohibition of SRQ transmitting	
		S0	Permission of SRQ transmitting	●
		S?	Reply: S0 or S1	

*4: EOI is a GPIB function. EOI is not output in USB.
*5: This command is applied to software revision C00 or later, and is not available on the 7461P.

Item		Command	Description	Initial value
System	Status	*CLS	Clears each status byte.	
		*STB?	Reads the status byte register Reply: ddd	
		*SREn	Sets the service request enable register (SRER). n: 0 to 255 (However, bit6 cannot be set.)	
		*SRE?	Reply: ddd	
		*ESR?	Reads the standard event status register (SESR). Reply: ddd	
		*ESEn	Sets the standard event status enable register (SESER). n:0 to 255	
		*ESE?	Reply: ddd	
		MSR?	Reads the measurement event register (MER). Reply: ddddd	
		MSEn	Sets the measurement event enable register (MEER). n:0 to 65535	
		MSE?	Reply: ddddd	
		QSR?	Reads the questionable event register (QER). Reply: ddddd	
		QSEn	Sets the questionable event enable register (QEER). n:0 to 65535	
		QSE?	Reply: ddddd	
		OSR?	Reads the operation event register (OER). Reply: ddddd	
		OSEn	Sets the operation event enable register (OEER). n:0 to 65535	
		OSE?	Reply: ddddd	
		*PSCn	n:-32767 to +32767 If "n" is set to any value except for zero, SRER and SESER are cleared when the power is turned on. If "n" is set to zero, SRER and SESER are not cleared when the power is turned on.	
		*PSC?	Reply: 0 or 1 (When any value except for zero is set.)	
	Operation complete	*OPC	Sets the operation complete bit (bit0) of the Standard Event Status Register after all operations are complete.	
		*OPC?	Reply: 1 (After all operations are complete.)	
		*WAI	Waits for the completion of all operations. (GPIB only)	
	Self-Test	*TST?	Performs the self- test and reads the result. (It takes time to perform the self-test. Read the result after the self-test is complete.) Reply 0 : Pass 1 : Fail	

Item		Command	Description	Initial value
System	Calibration	CAL0	Sets the calibration mode to OFF. (Writes the calibration factor when leaving the calibration mode.)	●
		CAL1	Sets the calibration mode to ON.	
		CAL?	Reply: CAL0 or CAL1	
		XOUT	Cancels (sets to OFF) the calibration mode. (Does not write the calibration factor.)	
		PCn	Enters the STD value. (Displayed count value) n:0 to ±999999	
		XDTn	Enters the STD value. (Displayed value) n: Displayed value on STD	
		XTERM?	Performs the zero calibration at the rear input terminal. Reply: 0 (Normal) or 1 (Abnormal)	
		CMNT "str"	Sets the calibration information. Enclose the character strings by using (") or ('). Up to 50 characters (one byte alphanumeric characters) can be stored.	
		CMNT?	Reply: CMNT"xxxx・・・xxx" Up to 50 characters	
	Setting parameter	*SAVn	n:0 to 3 Saves the setting parameters in area [n] of the non-volatile memory.	
		SINI	Sets the values when shipping to all area from [0] to [3].	
		*RCLn	n:0 to 3 Loads the setting parameters from area [n] of the non-volatile memory.	
		RINI	Loads the values when shipping as the setting parameters.	
	Error log reading	ERR?	Reads the error details. Reply: ±ddd,"xxxxxxxxxx" Error character strings (Up to 80 characters) Error code • Up to 20 error queues can be stored. The error is output according to the FIFO system. • If more than 20 errors occur, the last error is replaced with -350, "Queue overflow". Any errors after this are not saved. • If no error occurs, +000, "No error" is returned.	
	Input terminal	IN0	Front	●
		IN1	Rear	
		IN?	Reply: IN0 or IN1	

6.7.4 SCPI Command Reference

6.7.4.1 SCPI Command Reference Syntax

This section describes the syntax to explain each SCPI command.

Descriptions of each command include the following items:

Function description

SCPI command

Parameter

Query reply

- [Function description]

  How to use commands and the operation of this instrument when commands are executed are described.

- [SCPI command]

  The syntax, which is used when the command is sent from the external controller to this instrument, is described. The syntax consists of a command section and a parameter section. The command and parameter sections are delimited by a space.

  Commas (,) are used to delimit two or more parameters. The ellipsis (...) displayed between commas shows that the parameters are omitted at that position.

  For example, the description "<numeric value 1>",...,"<numeric value 4>" shows "<numeric value 1>, <numeric value 2>, <numeric value 3>, <numeric value 4>".

  The part, which is written by lowercase alphabetical characters in the syntax, shows that it can be omitted.

  For example, ":CALibration:CABLe" can be abbreviated to ":CAL:CABL".

  The symbols used in the syntax are defined as follows:

< >	Shows a parameter required for sending a command
[ ]	Shows that the command is optional. It can be omitted.
{}	Shows that only one item need be selected from multiple items.
|	Written in curly brackets {..} and used as a delimiter for multiple items.

- [Parameter]

  Describes a parameter required for sending a command.

  When the parameter is a numeric type or a character (string) type, it is enclosed in angle brackets (< >).

  If the parameter is optional, it is enclosed in braces { }.

  Optional items are delimited by (|) and the multiple descriptions in one optional item are delimited by (,).

  Example: {ON,1|OFF,0}

  This example shows that “ON” has the same meaning of “1”, “OFF” has the same meaning of “0”, and the following four options: ON, 1, OFF, and 0 can be selected.

  If the range of the value to be entered is restricted, the range is shown as [Minimum value| Default value| Maximum value] or [Minimum value| Maximum value].

  In this manual, parameter types are described in the following formats:

<int_d>	Shows a numeric value that can be entered in each format of NR1, NR2, and NR3 and is rounded to an integer in this instrument. “d” shows the number of digits.
<real_d>	Shows a numeric value that can be entered in each format of NR1, NR2, and NR3 and is rounded to a real number in this instrument. “d” shows the number of decimals.
<str>	Shows a character string or alphanumeric symbols enclosed in quotation (') or double quotation (") marks.
<func>	Shows the same character string as the query reply to the [:SENSe]:FUNCtion command.

- [Query reply]

  Describes the data format used when a query, which relates to the command, is read.

  Each parameter that is read is enclosed in braces { }. If multiple items in braces { } are delimited by a vertical bar (|), only one item is read. If multiple parameters are read, they are delimited by commas (,) and displayed. An ellipsis (...) displayed between commas indicates that data is omitted at that position. For example, the description "{numeric value 1},..., {numeric value 4}" shows that the four parameters, {numeric value 1}, {numeric value 2}, {numeric value 3}, and {numeric value 4} are read.

MEMO: ***The initial value of Continuous is set to OFF in software revision B00 and later. The initial value of Continuous can be set to ON in the MENU setting (I/F-CONT).***

NOTE: ***When using the SCPI commands via USB, note the following point.***
After sending a setting command, send a query command and receive its response.
Sending more than one setting command in succession may cause unintended behavior or an error to occur.
As query command for setting commands, error check by ":SYSTem:ERRor?" and completion check by "*OPC?" are generally used.

6.7.4.2 Remote Command

1. Measurement command

Function description	SCPI command	Parameter	Query reply
Setting the DCV measurement function and sending the read out request at the same time	:MEASure?		Refer to Section 6.7.2, “Output Data Format.”
1Performing the measurement once and reading the measure-ment value	:MEASure:<Function>? <Function> can be selected from the followings:		Refer to Section 6.7.2, “Output Data Format.”
DCV	VOLTage:DC		
ACV	VOLTage:AC		
2WΩ	RESistance		
4WΩ	FRESistance		
DCI	CURRent:DC		
ACI	CURRent:AC		
ACV(AC+DC)	VOLTage:ACDC		
ACI(AC+DC)	CURRent:ACDC		
Diode	DIODe		
LP-2WΩ	RESistance:LPOWer		
LP-4WΩ	FRESistance:LPOWer		
Continuity	CONTinuity		
Freq	FREQuency		
TEMP *7461P only	TEMPerature		
DCV-Bch HI-COM	VOLTage:BDC:BCOM		
DCV-Bch HI-LO	VOLTage:BDC:B		
DCV-Bch RATIO	VOLTage:BDC:RATio		
Presetting the specified function	:CONFigure:<Function>		
Reading the measurement function setting	:CONFigure?		"<func>,<real>" (String, Range)
Changing the measurement func-tion			
DCV	[:SENSe]:FUNCtion	"VOLTage:DC"	"VOLT:DC"
ACV		"VOLTage:AC"	"VOLT:AC"
2WΩ		"RESistance"	"RES"
4WΩ		"FRESistance"	"FRES"
DCI		"CURRent:DC"	"CURR:DC"
ACI		"CURRent:AC"	"CURR:AC"
ACV(AC+DC)		"VOLTage:ACDC"	"VOLT:ACDC"
ACI(AC+DC)		"CURRent:ACDC"	"CURR:ACDC"
Diode		"DIODe"	"DIOD"

Function description	SCPI command	Parameter	Query reply
LP-2WΩ		"RESistance:LPOWer"	"RES:LPOW"
LP-4WΩ		"FRESistance: LPOWer"	"FRES:LPOW"
Continuity		"CONTinuity"	"CONT"
Freq		"FREQuency"	"FREQ"
TEMP *7461P only		"TEMPerature"	"TEMP"
DCV-Bch HI-COM		"VOLTage:BDC: BCOM"	"VOLT:BDC:BCOM"
DCV-Bch HI-LO		"VOLTage:BDC:B"	"VOLT:BDC:B"
DCV-Bch RATIO		"VOLTage:BDC: RATio"	"VOLT:BDC:RAT"
Auto-range setting The setting to Continuity, Diode, and DCV-Bch is disabled.	[:SENSe]:<Function>:RANGe:AUTO	{ON,1|OFF,0}	{1|0}
Temperature unit setting *7461P only	:UNIT:TEMPerature	{C|F|K}	{C|F|K}
Temperature measurement method setting *7461P only	[:SENSe]:TEMPerature:TRANsducer:TYPE	{RTD|TRTD|FRTD} RTD:2W TRTD:3W FRTD:4W	{RTD|TRTD|FRTD}
Temperature sensor standard setting *7461P only	[:SENSe]:TEMPerature:TRANsducer:STANdard	{PT100|JPT100}	{PT100|JPT100}
Measurement range setting DCV ACV, ACV(AC+DC), Freq 2WΩ, 4WΩ DCI, ACI, ACI(AC+DC) LP-2WΩ, LP-4WΩ The setting to Continuity, Diode, and DCV-Bch is disabled.	[:SENSe]:<Function>:RANGe	0 ≤ Setting value ≤ Maximum value in the measurement range (750 V in the frequency measurement)	<real_6> (7461A/61P) <real_5> (7451A)
Sampling rate setting	[:SENSe]:<Function>:SRATe	{FREE|SFASt|FAST|MED|SLOW|SSLow}	{FREE|SFAS|FAST|MED|SLOW|SSL} *1
Display digit setting	[:SENSe]:<Function>:DIGits *3	[3|6|6] (7461A/61P) [3|5|5] (7451A)	<int_1>
Integration time setting (PLC) The setting is disabled in the frequency measurement.	[:SENSe]:<Function>:NPLCycles	[0.02|10|100] *2	<real_6>
Integration time setting (second) The setting is disabled in the frequency measurement.	[:SENSe]:<Function>:APERture	[0.00001|10] (7461A/61P) [0.0001|10] (7451A)	<real_6>

*1: 'FREE' is returned when the sampling interval or integration time is set.
*2: When the integration time is outside this range, it is set in seconds. (Refer to Section 5.7.1, "Sampling Interval (SI) and Integration Time (IT) Settings".)
*3: The setting is applied to all functions.

Function description	SCPI command	Parameter	Query reply
AC filter setting The setting is enabled in the ACV, ACV(AC+DC), ACI, and ACI(AC+DC) functions.	[:SENSe]:<Function>:FILTer	{FAST|MED|SLOW}	{FAST|MED|SLOW}
Auto-zero setting	[:SENSe]:ZERO:AUTO	{ON,1|OFF,0|ONCE}	{1|0}
Continuity threshold constant	[:SENSe]:CONTinuity:THReshold	[1|10|1000]	<int_4>
Switching the DCV input impedance	:INPut:IMPedance:AUTO	{ON,1|OFF,0}	{1|0} (1:HiZ, 0:10 MΩ)

2. Calculation function command

Function description	SCPI Command	Parameter	Query reply
NULL			
Calculation setting	[:SENSe]:<Function>:REFerence:STATe	{ON,1|OFF,0}	{1|0}
NULL constant setting The setting is enabled when the NULL calculation is set to ON.	[:SENSe]:<Function>:REFerence	[-9999999.E+6|0|9999999.E+6] (7461A/61P) [-999999.E+6|0|999999.E+6] (7451A)	<real_6> (7461A/61P) <real_5> (7451A)
Smoothing			
Calculation setting	:CALCulate:SMOothing	{ON,1|OFF,0}	{1|0}
Smoothing count setting	:CALCulate:SMOothing:POINts	[2|10|100]	<int_3>
Scaling			
Calculation setting	:CALCulate:SCALing	{ON,1|OFF,0}	{1|0}
Constant A setting*4	:CALCulate:SCALing:A	{ [-9999999.E+6|0|9999999.E+6]| MEASurement} (7461A/61P) { [-999999.E+6|0|999999.E+6]| MEASurement} (7451A)	<real_6> (7461A/61P) <real_5> (7451A)
Constant B setting*4	:CALCulate:SCALing:B		
Constant C setting*4	:CALCulate:SCALing:C		
MAX/MIN			
MAX/MIN calculation setting	:CALCulate:AVERage	{ON,1|OFF,0}	{1|0}
MAX value reading	:CALCulate:AVERage:MAXimum?		Refer to Section 6.7.2, "Output Data Format."
MIN value reading	:CALCulate:AVERage:MINimum?		
AVE value reading	:CALCulate:AVERage:AVERage?		
Measurement count reading	:CALCulate:AVERage:COUNt?		<real_6> (7461A/61P) <real_5> (7451A)
Comparator calculation			
Calculation setting	:CALCulate:LIMit	{ON,1|OFF,0}	{1|0}
High constant setting*4	:CALCulate:LIMit:UPPer	{ [-9999999.E+6|0|9999999.E+6]| MEASurement} (7461A/61P) { [-999999.E+6|0|999999.E+6]| MEASurement} (7451A)	<real_6> (7461A/61P) <real_5> (7451A)
LOW constant setting*4	:CALCulate:LIMit:LOWer		
Whether to set the calculation result LOW to PASS.	:CALCulate:LIMit:PASS:LOWer	{ON,1|OFF,0}	{1|0}
Whether to set the calculation result GO to PASS.	:CALCulate:LIMit:PASS:MED	{ON,1|OFF,0}	{1|0}

*4: How to set the parameter can be selected from entering the value directly and substituting the measurement value.

Function description	SCPI Command	Parameter	Query reply
Whether to set the calculation result HI to PASS.	:CALCulate:LIMit:PASS:UPPer	{ON,1|OFF,0}	{1|0}
Buzzer setting	:CALCulate:LIMit:BEEPer	{OFF|FAIL|PASS}	{OFF|FAIL|PASS}
Statistical calculation			
Calculation range specifying and performing	:CALCulate:STATistics	[0|0|9999], [0|-1|9999]	<int_4>,<int_4>
Number of samples reading	:CALCulate:STATistics:COUNt?		<real_6> (7461A/61P) <real_5> (7451A)
Maximum value reading	:CALCulate:STATistics:MAXimum?		
Minimum value reading	:CALCulate:STATistics:MINimum?		
Average value reading	:CALCulate:STATistics:AVERage?		
Standard deviation value reading	CALCulate:STATistics:DEViation?		
MAX-MIN reading	CALCulate:STATistics:MAXMin?		
dB/dBm			
Calculation setting	:CALCulate:DB	{DB|DBM|OFF}	{DB|DBM|OFF}
Constant D setting	:CALCulate:DB:D	{[0.000001E-9|1| 9999999.E+6]| MEASurement} (7461A/61P) {[0.000001E-9|1| 999999.E+6]| MEASurement} (7451A)	<real_6> (7461A/61P) <real_5> (7451A)
Digital output *7461P only			
Enable/Disable setting	:DOUTput:STATe	{ON,1|OFF,0}	{1|0}
Output setting	:DOUTput:BITS	{0-31} Refer to Section 6.6, "Digital Output (7461P Only)."	{0-31}

3. Trigger system command

Function description	SCPI Command	Parameter	Query reply
Trigger system starting	:INITiate		
Continuous setting	:INITiate:CONTinuous	{ON,1|OFF,0}	{1|0}
Trigger system initialization and measurement data reading	:READ?		Refer to Section 6.7.2, “Output Data Format.”
Only measurement data reading	:FETCh?		Refer to Section 6.7.2, “Output Data Format.”
Trigger system abort	:ABORt		
Sampling count (1 trigger)	:SAMPle:COUNt	[1|1|16000]	<int_4>
Trigger			
Trigger source selection	:TRIGger:SOURce	IMMediate	IMM
		MANual	MAN
		EXTernal	EXT
		BUS	BUS
		LEVel	LEV
		DELTa	DELT
Trigger count	:TRIGger:COUNt	[1|1|50000]	<int_5>
Trigger delay	:TRIGger:DELay	[0|0|3600]	<real_4>
Sampling interval	:TRIGger:TIMer	[0.00005|0.5|3600] (7461A/61P) [0.0002|0.5|3600] (7451A)	<real_6>
LEVEL trigger			
Level count value setting	:TSYStem:LEVel	[-1199999|0|1199999] (7461A/61P) [-319999|0|319999] (7451A)	<int_7> (7461A/61P) <int_6> (7451A)
Slope setting	:TSYStem:LEVel:SLOPe	{POSitive|NEGative}	{POS|NEG}
DELTA trigger			
Delta count value setting	:TSYStem:DELTa	[1|1|600000] (7461A/61P) [1|1|160000] (7451A)	<int_7> (7461A/61P) <int_6> (7451A)
Slope setting	:TSYStem:DELTa:SLOPe	{POSitive|NEGative|BOTH}	{POS|NEG|BOTH}
Convergence/Divergence setting	:TSYStem:DELTa:DIRection	{CONVergence|EMISsion}	CONV|EMIS
Pre-trigger value setting in the LEVEL or DELTA trigger modes	:TSYStem:PRETrigger	{0|25|50|75}	{0|25|50|75}

4. System relevant command

Function description	SCPI Command	Parameter	Query reply
Buzzer Setting	:SYSTem:BEEPer:STATe	{ON,1|OFF,0}	{1|0}
Error queue reading	:SYSTem:ERRor?		<int_3>,<str>
SCPI version reading	:SYSTem:VERSion?		<str>
Block delimiter	:SYSTem:GPIB:DELimiter:BLOCk	{CRLF|LF|EOI}	{CRLF|LF|EOI}
Power supply frequency	[:SENSe]:LFRQency?		50|60
Invalid state of the function reading	:INHibit:FUNCtion:<Function>?		{0|1} (0: Selectable, 1: Non-selectable)
Input terminal setting (Front/Rear)	:ROUTe:TERMinals	{FRONt|REAR}	{FRON|REAR}
Measurement data display setting	:DISPlay	{ON,1|OFF,0}	{1|0}
Measurement data memory	:TRACe:STATe		
Store setting		{ON,1|OFF,0}	{1|0}
Initialization	:TRACe:CLEar		
Readout range setting	:TRACe:NUMBer	[0|0|9999], [0|0|9999] (7451A/61A) [0|0|19999], [0|0|19999] (7461P)	
Data reading	:TRACe:DATA?		Refer to Section 6.7.2, "Output Data Format."
Data range reading	:TRACe:DATA:NUMBer?		<int_4>,<int_4> (7451A/61A) <int_5>,<int_5> (7461P)
Number of measurement results reading	:TRACe:DATA:POINts?		<int_5>
Complete signal output mode	:TRIGger:COMPlete	SINGle|MULTi	SING|MULT
Complete signal pulse width selection	:TRIGger:COMPlete:WIDTh	[0.000005|0.0001|0.0001]	<real_6>
Selection of the old model name for the response to *IDN?	:OIDentity This command is applied to software revision C00 or later, and is not available on the 7461P.	{ON,1|OFF,0} (Initial value is OFF)	{1|0}

5. Status command

Function description	SCPI Command	Parameter	Query reply
Operation Event Enable Register setting	:STATus:OPERation:ENABle	[0|0|65535]	<int_5>
Operation Event Register reading	:STATus:OPERation:EVENt?		<int_5>
Questionable Event Enable Register setting	:STATus:QUEStionable:ENABle	[0|0|65535]	<int_5>
Questionable Event Register reading	:STATus:QUEStionable:EVENt?		<int_5>
Measurement Event Enable Register setting	:STATus:MEASurement:ENABle n	[0|0|65535]	<int_5>
Measurement Event Register reading	:STATus:MEASurement:EVENt?		<int_5>

6. Common command to IEEE-488.2 (Can be used in USB.)

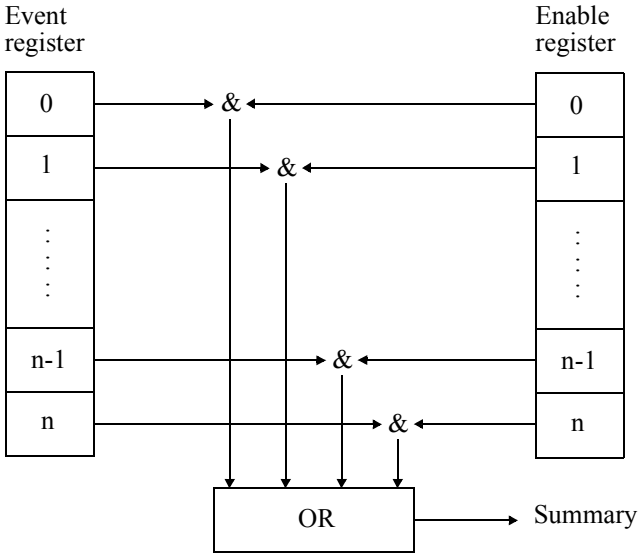
Function description	SCPI Command	Parameter	Query reply
Each status byte clear	*CLS		
Standard Event Status Enable Register setting	*ESE	{0|0|127]	<int_3>
Standard Event Status Register reading	*ESR?		<int_3>
Instrument information reading	*IDN?		For more information on a query response, refer to page 6-32.
Setting the operation complete bit (bit0) of the Standard Event Status Register after all operations are complete	*OPC		
Processing state of the operarion is returned	*OPC?		{0|1} (0: completed, 1: uncompleted)
Instrument initialization	*RST		
Service Request Enable Register setting	*SRE	[64|64|127] (bit6 is always 1)	<int_3>
Status Byte Register reading	*STB?		<int_3>
Trigger command	*TRG		
Self-test performing and the result reading	*TST?		{0|1} (0: Normal, 1: Abnormal)
All operation completion waiting	*WAI (GPIB only)		

6.7.5 Status Register Structure

This instrument has a layered status register structure that is compliant with IEEE standard 488.2-1987 and can send various statuses of this instrument to the controller. This section describes the operation model of status structure and allocation of events.

1. Status register

   This instrument adopts the model of the status register defined in IEEE standard 488.2-1987. The status register consists of the event register and enable register.



- Event register

  The event register latches and retains statuses correspondent to each event. (Sometimes the changes are retained.)

  Once this register is set, the setting value is kept until it is read by a query or cleared by *CLS.

  Data cannot be written into the event register.

- Enable register

  The enable register specifies which bit in the event register is set as an effective status to generate a summary. The enable register is ANDed with the event register and the OR of the result is generated as a summary. The summary is written into the status byte register.

  Data can be written into the enable register.

This instrument uses the following five types of status registers:

- Status Byte Register (STB)
- Standard Event Status Register (SESR)
- Measurement Event Register (MER)
- Questionable Event Register (QER)
- Operation Event Register (OER)

The status register structure in this instrument is shown in Figure 6-4 and Figure 6-5.

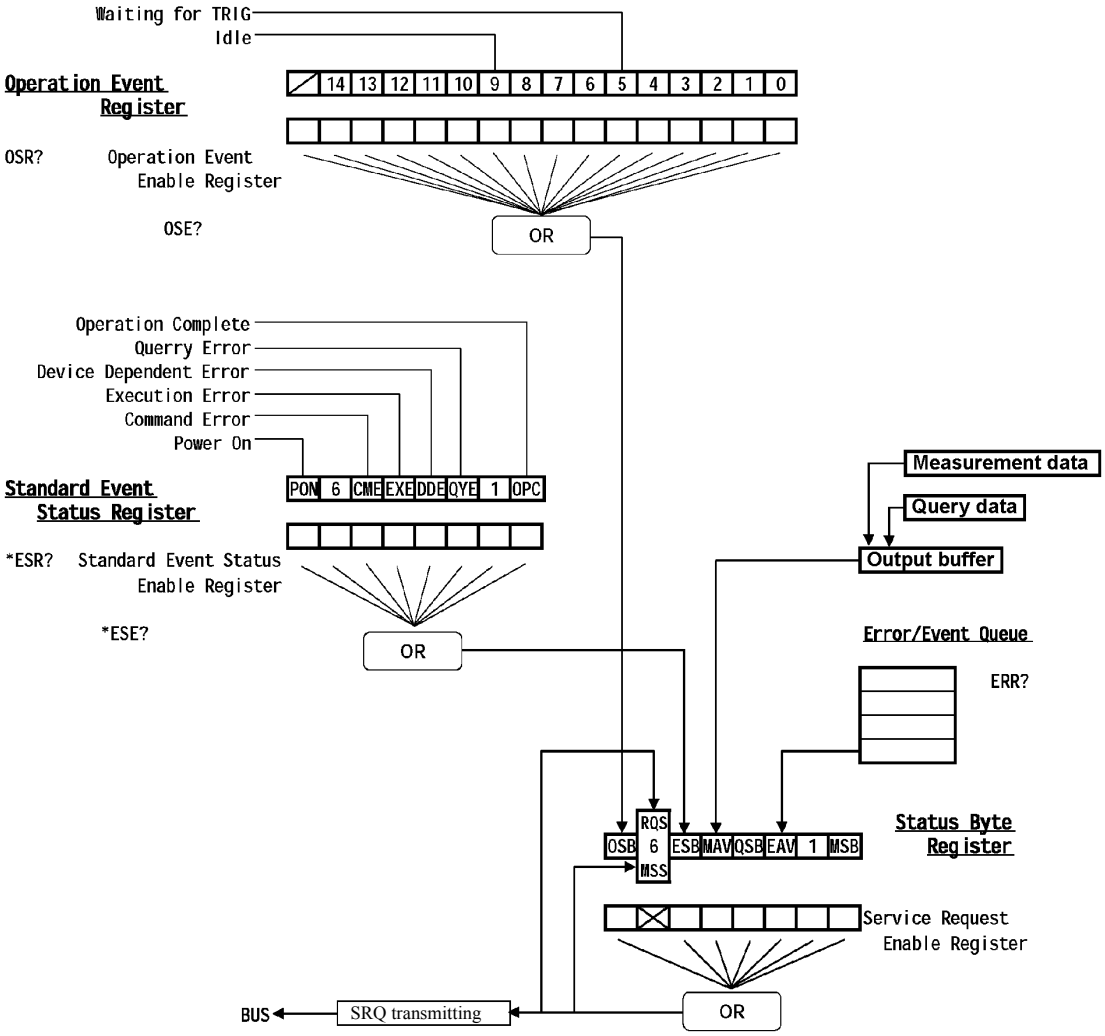


Figure 6-4 Status Register Structure (1/2)

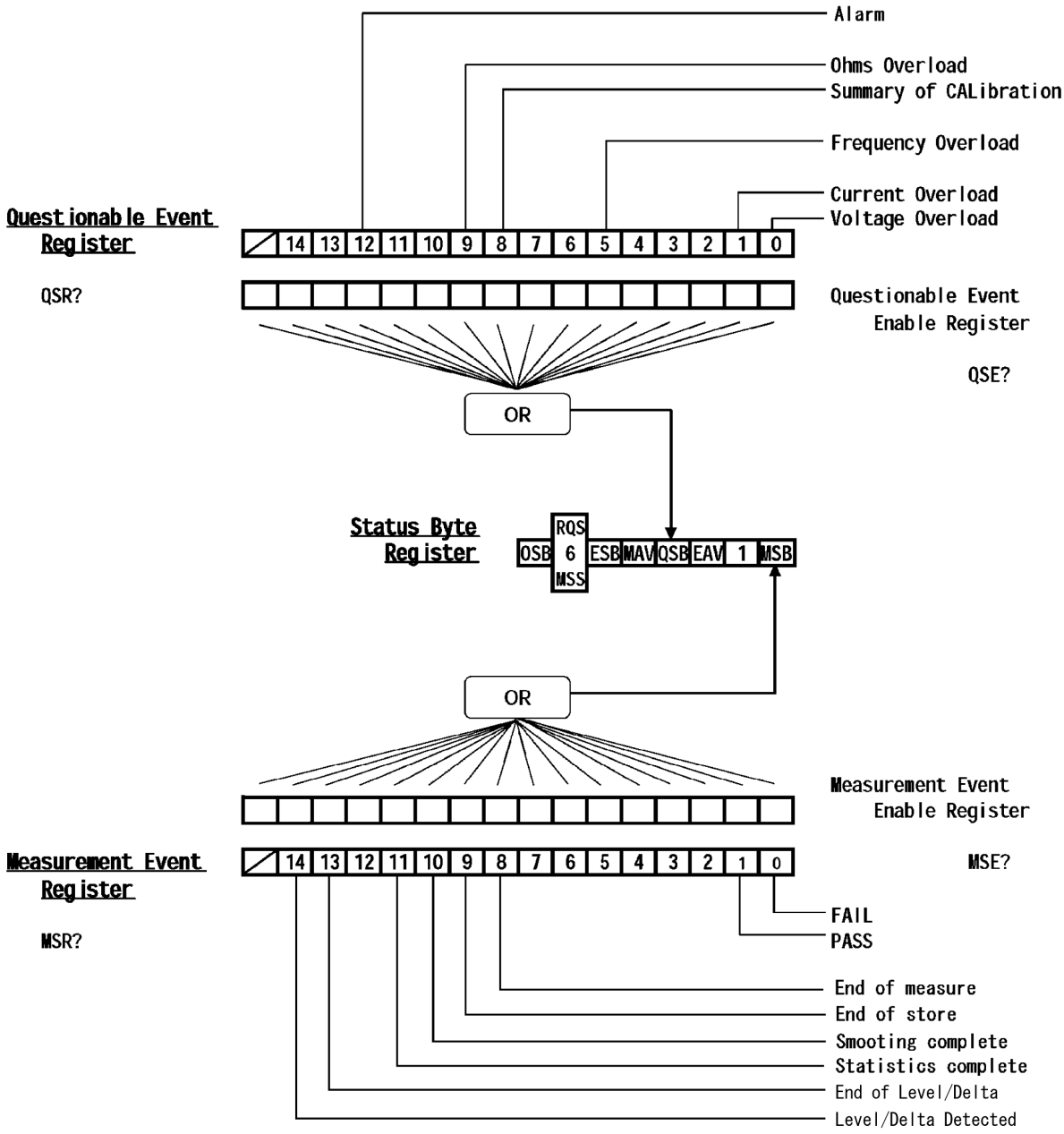


Figure 6-5 Status Register Structure (2/2)

2. Event enable register

   Each event register has an enable register that determines which bit is to be enabled. The enable register sets the corresponding bit in decimal.

   - Service Request Enable Register (SRER) setting: *SRE
   - Standard Event Status Enable Register (SESER) setting: *ESE
   - Measurement Event Enable Register (MEER): MSE

- Questionable Event Enable Register (QEER): QSE
- Operation Event Enable Register (OEER) OSE

3. Status byte register

   The status byte register summarizes the information from the status register. A summary of this status byte register is sent to the controller as a service request. Therefore, the status byte register operates slightly differently than the status register structure. This section describes the status byte register.

   The status byte register structure is shown in Figure 6-6.

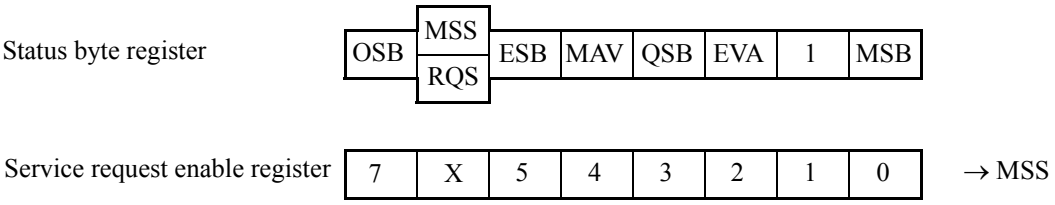


Figure 6-6 Status Byte Register Structure

This status byte register follows the status register except for the following three points:

- The summary of the status byte register is written to bit6 of the status byte register.
- Bit6 of the enable register is always enabled and cannot be changed.
- Bit6 (MSS) of the status byte register writes the RQS of the service request.

This register responds to the serial polling from the controller. When the register responds to the serial polling, bit0 to bit5, bit7, and the RQS of the status byte register are read, and then the RQS is reset to 0. No other bits are cleared until each factor is set to 0.

The status byte register, RQS, and MSS can be cleared by executing "*CLS". Consequently, the SRQ line is set to FALSE.

A description of each bit in the status byte register is shown in Table 6-7.

Table 6-7 Status Byte Register (STB)

bit	Name	Description
0	MSB Measurement Summary Bit	ON: When any event of Measurement Event Register occurs and Measurement Event Register is set to 1, this bit is set to 1 if the corresponding bit of the Measurement Event Enable Register is set to 1. OFF: This bit is set to 0 when Measurement Event Register is cleared by being read.
1	Not used	Always 0
2	EAV Error Available	ON: This bit is set to 1 when the error information is stored in Error Queue. OFF: This bit is set to 0 when the error information is read from Error Queue and no data has existed in Error Queue.
3	QSB Questionable Summary Bit	ON: When any event of Questionable Event Register occurs and Questionable Event Register is set to 1, this bit is set to 1 if the corresponding bit of the Questionable Event Enable Register is set to 1. OFF: This bit is set to 0 when Questionable Event Register is cleared by being read.
4	MAV Message Available	ON: This bit is set to 1 when the output data is entered in the output buffer. OFF: This bit is set to 0 when the data is read from the output buffer.
5	ESB Standard Event Status	ON: When any event of SESR occurs and SESR is set to 1, this bit is set to 1 if the corresponding bit of the SESER is set to 1. OFF: This bit is set to 0 when SESR is cleared by being read (*ESR?).
6	MSS Master Summary	ON: When any event of STB occurs, this bit is set to 1 if the corresponding bit of the SRER is set to 1.
	RQS Request Service	ON: If MSS is set to 1, SRQ is generated and RQS is set to 1. OFF: When STB is read in the serial polling, RQS is set to 0.
7	OSB Operation Summary Bit	ON: When any event of Operation Event Register occurs and Operation Event Register is set to 1, this bit is set to 1 if the corresponding bit of the Operation Event Enable Register is set to 1. OFF: This bit is set to 0 when Operation Event Register is cleared by being read.

Conditions which clear the status byte register

- When the power is turned on.
- When the *CLS command is executed. However, the MAV bit is not cleared if data exists in the output buffer.
- The status byte register is not cleared by executing *STB? command.

Conditions which clear the service request enable register

- When the power is turned on. (When the PSC flag is set to 1)
- When the *SRE0 command is executed.

4. Standard event status register

   Allocation in the standard event register is shown in Table 6-8.

Table 6-8 Standard Event Status Register (SESR)

bit	Name	Description
0	OPC Operation Complete	ON: This bit is set to 1 when all operations are complete after receiving the *OPC command.
1	Not used	Always 0
2	QYE Query Error	ON: This bit is set to 1 under any of the following conditions. • When the data is read, but no data exists in the output buffer and no data to be output exists. • When data in the output buffer is lost.
3	DDE Device Dependent Error	ON: This bit is set to 1 when the error, which depends on the device, occurs.
4	EXE Execution Error	ON: This bit is set to 1 when the received command cannot be executed. This bit is set to 1 when the command parameter contains an error.
5	CME Command Error	ON: This bit is set to 1 when the received command spelling is incorrect.
6	Not used	Always 0
7	PON Power On	ON: This bit is set to 1 when the power is turned on.

Conditions which clear the standard event status register

- When the power is turned on.
- When the *CLS command is executed.
- When the *ESR? command is executed..

Conditions which clear the standard event status enable register

- When the power is turned on. (When the PSC flag is set to 1)
- When the *ESE0 command is executed.

5. Measurement event register

Allocation in the measurement event register is shown in Table 6-9.

Table 6-9 Measurement Event Register (MER)

bit	Name	Description
0	FL FAIL	ON: This bit is set to 1 when the comparator calculation result satisfies the FAIL conditions.
1	PS PASS	ON: This bit is set to 1 when the comparator calculation result satisfies the PASS conditions.
2	Not used	Always 0
3	Not used	Always 0
4	Not used	Always 0
5	Not used	Always 0
6	Not used	Always 0
7	Not used	Always 0
8	EOM End of measure	ON: This bit is set to 1 when the measurement is complete.
9	EOS End of store	ON: This bit is set to 1 when no more measurement data can be stored in the measurement memory.
10	SM Smoothing complete	ON: This bit is set to 1 when the smoothing count reaches the specified number.
11	STAT Statistics complete	ON: This bit is set to 1 when the statistical processing is complete.
12	Not used	Always 0
13	ELD End of Level/Delta	ON: This bit is set to 1 when all measurements are complete after the LEVEL trigger or DELTA trigger is detected.
14	LDD Level/Delta Detected	ON: This bit is set to 1 when the LEVEL trigger or DELTA trigger is detected.
15	Not used	Always 0

Conditions which clear the measurement event register

- When the power is turned on.
- When the *CLS command is executed.
- When the :STATus:MEASurement[:EVENt]?" command is executed.
- When the MSR? command is executed.

Conditions which clear the measurement event enable register

- When the power is turned on.
- When the :STATus:MEASurement:ENABle 0 command is executed.
- When the MSE0 command is executed.

6. Questionable event register

Allocation in the questionable event register is shown in Table 6-10.

Table 6-10 Questionable Event Register (QER)

bit	Name	Description
0	Voltage Overload	ON: This bit is set to 1 when OL occurs in the voltage or diode measurement.
1	Current Overload	ON: This bit is set to 1 when OL occurs in the current measurement.
2	Not used	Always 0
3	Not used	Always 0
4	Not used	Always 0
5	Frequency Overload	ON: This bit is set to 1 when OL occurs in the frequency measurement.
6	Not used	Always 0
7	Not used	Always 0
8	Summary of Calibration	ON: This bit is set 1 when the default calibration values or the calibration values acquired in previous power ON is used due to the calibration data SUM failure in the power ON check.
9	Ohms Overload	ON: This bit is set to 1 when OL occurs in the resistance measurement.
10	Not used	Always 0
11	Not used	Always 0
12	Alarm	ON: This bit is set to 1 when an alarm occurs in the measurement.
13	Not used	Always 0
14	Not used	Always 0
15	Not used	Always 0

Conditions which clear the questionable event register

- When the power is turned on.
- When the *CLS command is executed.
- When the :STATus:QUEStionable[:EVENt]? command is executed.
- When the QSR? command is executed.

Conditions which clear the measurement event enable register

- When the power is turned on.
- When the :STATus:QUEStionable:ENABle 0 command is executed.
- When the QSE0 command is executed.

7. Operation event register

Allocation in the operation event register is shown in Table 6-11.

Table 6-11 Operation Event Register (OER)

bit	Name	Description
0	Not used	Always 0
1	Not used	Always 0
2	Not used	Always 0
3	Not used	Always 0
4	Not used	Always 0
5	Waiting for TRIG	ON: This bit is set to 1 when this instrument is in Trigger Layer.
6	Not used	Always 0
7	Not used	Always 0
8	Not used	Always 0
9	Idle	ON: This bit is sets to 1 when this instrument is in the Idle state.
10	Not used	Always 0
11	Not used	Always 0
12	Not used	Always 0
13	Not used	Always 0
14	Not used	Always 0
15	Not used	Always 0

Conditions which clear the operation event register

- When the power is turned on.
- When the *CLS command is executed.
- When the :STATus:OPERation[:EVENt]? command is executed.
- When the OSR? command is executed.

Conditions which clear the operation event enable register

- When the power is turned on.
- When the :STATus:OPERation:ENABle 0 command is executed.
- When the OSE0 command is executed.

6.8 Sample Programs (Command used in ADC CORPORATION)

This section describes examples of a program in which a computer controls this instrument through GPIB.

The USB driver includes examples of a program in which this instrument is controlled through USB.

For more information on the USB driver, refer to 3.3.2, “Caution when Connecting Peripherals”.

Computer: FMV-5350ML3 (OS:Windows98) made by Fujitsu Limited

GPIB hardware: PCI-GPIB made by NATIONAL INSTRUMENTS

Module: Niglobal.bas, Vbib-32.bas (Included software in PCI-GPIB)

Compatible language: Microsoft Excel Visual Basic for Application

---

NOTE: ***These samples are programmed assuming that the 7461A is used. If the 7451A is used, change the settings such as a range configuration which depends on the multimeter.***

Example 1 Measures a voltage in the DCV 10 V range and reads the measurement data from the 7461A. The GPIB address of the 7461A is set to 1.

```
Dim DMM_ADR As Integer          ' Declares the variable of the 7461A
                                ' GPIB address.
Dim dmm As Integer              ' Declares the variable of the device descriptor.
Dim dt As String * 100          ' Declares the variable of the buffer used for receiving the GPIB data.

DMM_ADR = 1                     ' The GPIB number of the 7461A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                                ' Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)
                                ' Sets the transmitting and receiving addresses individually.
Call ibwrt(dmm, "*RST" & Chr(10))
                                ' Initializes the 7461A.
Call ibwrt(dmm, "H1" & Chr(10))
                                ' Sets the header of the output data to ON.
Call ibwrt(dmm, "F1" & Chr(10))
                                ' Sets the measurement function to DCV.
Call ibwrt(dmm, "R5" & Chr(10))
                                ' Sets the measurement range to 10 V.
Call ibwrt(dmm, "PR5" & Chr(10))
                                ' Sets the sampling rate to SLOW2.

Call ibrd(dmm, dt)              ' Reads the measurement value data.

Cells(1, 1) = "'" & Left(dt, 18)
                                ' Substitutes the measurement value for a cell.

Call ibonl(dmm, 0)              ' Terminates.
```

Example 2 Sets the measurement function to the 2WΩ measurement, detects the measurement end by using the status byte, and reads the measurement data from the 7461A. The GPIB address of the 7461A is set to 1.

```
Dim DMM_ADR As Integer          ' Declares the variable of the 7461A GPIB address.
Dim dmm As Integer              ' Declares the variable of the device descriptor.
Dim dt As String * 100          ' Declares the variable of the buffer used for receiving the GPIB data.

DMM_ADR = 1                     ' The GPIB number of the 7461A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                                ' Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)
                                ' Sets the transmitting and receiving addresses individually.
Call ibwrt(dmm, "*RST" & Chr(10))
                                ' Initializes the 7461A.
Call ibwrt(dmm, "H1" & Chr(10))
                                ' Sets the header of the output data to ON.
Call ibwrt(dmm, "F3" & Chr(10))
                                ' Sets the measurement function to 2WΩ.
Call ibwrt(dmm, "R4" & Chr(10))
                                ' Sets the measurement range to 1kΩ.
Call ibwrt(dmm, "ITP1" & Chr(10))
                                ' Sets the integration time to 1 PLC.
Call ibwrt(dmm, "TRS3" & Chr(10))
                                ' Specifies "BUS" as the trigger.
Call ibwrt(dmm, "*CLS" & Chr(10))
                                ' Clears the status byte.

Call ibwrt(dmm, "*TRG" & Chr(10))
                                ' Triggers.

Do                              ' Reads the status byte of the measurement end.
  Call ibwrt(dmm, "*STB?" &Chr(10))
                                ' Requests the contents of the status byte.
  Call ibrd(dmm, dt)            ' Writes the contents of the status byte into the variable dt.
  dt = dt And 16                ' Performs the AND operation in bit4(MAV).
Loop While (dt <> 16)

Call ibrd(dmm, dt)              ' Reads the measurement data.

Cells(1, 1) = "'" & Left(dt, 18)
                                ' Substitutes the measurement value for a cell.

Call ibonl(dmm, 0)              ' Terminates.
```

Example 3 Sets the measurement function to the DCV 10 V range and performs the high-speed measurement (Sampling interval: 500 μs). In the high-speed measurement, the data is read after the measurements of the specified count are complete. The GPIB address of the 7461A is set to 1.

```
Dim DMM_ADR As Integer          ' Declares the variable of the 7461A GPIB address.
Dim dmm As Integer              ' Declares the variable of the device descriptor.
Dim dt As String * 20000        ' Declares the variable of the buffer used for receiving the GPIB data.
Dim s As Integer                ' The variable for storing the result of serial polling

DMM_ADR = 1                     ' The GPIB number of the 7461A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                                ' Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)
                                ' Sets the transmitting and receiving addresses individually.
```

```
Call ibconfig(0, IbcAUTOPOLL, 0)
                                    ' Disables the automatic serial polling. (Note)
Call ibwrt(dmm, "*RST" & Chr(10))
                                    ' Initializes the 7461A.
Call ibwrt(dmm, "H0" & Chr(10))
                                    ' Sets the header of the output data to OFF.
Call ibwrt(dmm, "S0" & Chr(10))
                                    ' Permits the SRQ transmitting.
Call ibwrt(dmm, "F1" & Chr(10))
                                    ' Sets the measurement function to DCV.
Call ibwrt(dmm, "R5" & Chr(10))
                                    ' Sets the measurement range to 10 V.

Call ibwrt(dmm, "TRS3" & Chr(10))
                                    ' Specifies "BUS" as the trigger.
Call ibwrt(dmm, "AZ0" & Chr(10))
                                    ' Disables the auto-zero.
Call ibwrt(dmm, "ITS0.0001" & Chr(10))
                                    ' Sets integration time to 100 ms.
Call ibwrt(dmm, "TRT0.0005" & Chr(10))
                                    ' Sets the sampling interval to 500 µs.
Call ibwrt(dmm, "DS0" & Chr(10))
                                    ' Sets the display to OFF.

Call ibwrt(dmm, "ST1" & Chr(10))
                                    ' Enables the measurement memory.
Call ibwrt(dmm, "ICL" & Chr(10))
                                    ' Clears the measurement memory.
Call ibwrt(dmm, "SPN1000" & Chr(10))
                                    ' Sets the sampling counts per one trigger. (1000 counts)
Call ibwrt(dmm, "IRD0,999" & Chr(10))
                                    ' Sets the range of the measurement memory recall. (0 to 999)
Call ibwrt(dmm, "*CLS" & Chr(10))
                                    ' Clears the status byte.
Call ibwrt(dmm, "MSE256" & Chr(10))
                                    ' Sets the measurement end flag of MSR.
Call ibwrt(dmm, "*SRE1" & Chr(10))
                                    ' Sets the measurement end flag of SRE.

Call ibwrt(dmm, "*TRG" & Chr(10))
                                    ' Triggers.

Call ibwait(dmm, RQS Or TIMO)   ' Waits for the SRQ transmitting.

If (ibsta And TIMO) Then        ' Judges whether a time-out occurs.
  Call MsgBox("SRQ Time Out", vbOKOnly, "Error")
                                    ' Sends a message if the time-out occurs.
  Call ibwrt(dmm, "DS1" & Chr(10))
                                    ' Sets the display to ON.
  Call ibonl(dmm, 0)            ' Terminates.
  Exit Sub
Else
  Call ibrsp(dmm, s)            ' Reads the status.
End If

Call ibwrt(dmm, "IRO?" & Chr(10))
                                    ' Requests the measurement value data.
Call ibrd(dmm, dt)              ' Substitutes the measurement value for a variable.

Cells(1, 1) = "'" & dt          ' Substitutes the measurement value for a cell.
```

```
Call ibwrt(dmm, "DS1" & Chr(10))
                                        ' Sets the display to ON.

Call ibonl(dmm, 0)                      ' Terminates.
```

(Note): The GPIB hardware setting is changed. Return to the original setting if other programs may case a failure.

6.9 Sample Programs (SCPI command)

This section describes examples of a program in which a computer controls this instrument through GPIB.

Computer: FMV-5350ML3 (OS:Windows98) made by Fujitsu Limited

GPIB hardware: PCI-GPIB made by NATIONAL INSTRUMENTS

Module: Niglobal.bas, Vbib-32.bas (Included software in PCI-GPIB)

Compatible language: Microsoft Excel Visual Basic for Application

---

NOTE: ***These samples are programmed assuming that the 7461A is used. If the 7451A is used, change the settings such as a range configuration which depends on the multimeter.***

Example 1 Measures the D.U.T in the DCV 10 V range and reads the measurement data from the 7461A. The GPIB address of the 7461A is set to 1.

```
Dim DMM_ADR As Integer          ' Declares the variable of the 7461A
                                ' GPIB address.
Dim dmm As Integer              ' Declares the variable of the device descriptor.
Dim dt As String * 100          ' Declares the variable of the buffer used for receiving the GPIB data.

DMM_ADR = 1                     ' The GPIB address of the 7461A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                                ' Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)
                                ' Sets the transmitting and receiving addresses individually.
Call ibwrt(dmm, "*RST" & Chr(10))
                                ' Initializes the 7461A.

Call ibwrt(dmm, ":SENSE:FUNCTION 'VOLTAGE:DC'" & Chr(10))
                                ' Sets the measurement function to DCV.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:RANGE 10" & Chr(10))
                                ' Sets the measurement range to 10 V.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:SRATE SSLOW" & Chr(10))
                                ' Sets the sampling rate to SLOW2.

Call ibwrt(dmm, ":FETCH?" & Chr(10))
                                ' Requests the measurement value data.
                                ' (Software revision A01 and earlier) *1
Call ibwrt(dmm, ":READ?" & Chr(10))
                                ' Requests the measurement value data.
                                ' (Software revision B00 and later) *1
Call ibrd(dmm, dt)              ' Substitutes the measurement value for a variable.

Cells(1, 1) = "'" & Left(dt, 13)
                                ' Substitutes the measurement value for a cell.

Call ibonl(dmm, 0)              ' Terminates.
```

NOTE:
*1: Select either one according to the software revision.

Example 2 Sets the measurement function to the 2WΩ measurement, detects the measurement end by using the status byte, and reads the measurement data from the 7461A. The GPIB address of the 7461A is set to 1.

```
Dim DMM_ADR As Integer          ' Declares the variable of the 7461A GPIB address.
Dim dmm As Integer              ' Declares the variable of the device descriptor.
Dim dt As String * 100          ' Declares the variable of the buffer used for receiving the GPIB data.

DMM_ADR = 1                     ' The GPIB address of the 7461A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                                ' Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)
                                ' Sets the transmitting and receiving addresses individually.
Call ibwrt(dmm, "*RST" & Chr(10))
                                ' Initializes the 7461A.

Call ibwrt(dmm, ":SENSE:FUNCTION 'RESISTANCE'" & Chr(10))
                                ' Sets the measurement function to 2WΩ.
Call ibwrt(dmm, ":SENSE:RESISTANCE:RANGE 1000" & Chr(10))
                                ' Sets the measurement range to 1kΩ.
Call ibwrt(dmm, ":SENSE:RESISTANCE:NPLCYCLES 1" & Chr(10))
                                ' Sets integration time to 1 PLC.
Call ibwrt(dmm, ":TRIGGER:SOURCE BUS" & Chr(10))
                                ' Specifies "BUS" as the trigger.
Call ibwrt(dmm, ":INITIATE:CONTINUOUS ON" & Chr(10))
                                ' Sets Continuous to ON.
Call ibwrt(dmm, "*CLS" & Chr(10))
                                ' Clears the status byte.

Call ibwrt(dmm, "*TRG" & Chr(10))
                                ' Triggers.

Do                              ' Reads the status byte of the measurement end.
  Call ibwrt(dmm, "*STB?" & Chr(10))
                                ' Requests the contents of the status byte.
  Call ibrd(dmm, dt)            ' Writes the contents of the status byte into the variable dt.
  dt = dt And 16                ' Performs the AND operation in bit4(MAV).
Loop While (dt <> 16)

Call ibwrt(dmm, ":FETCH?" & Chr(10))
                                ' Requests the measurement data.
Call ibrd(dmm, dt)              ' Reads the measurement data.

Cells(1, 1) = "'" & Left(dt, 13)
                                ' Substitutes the measurement value for a cell.

Call ibonl(dmm, 0)              ' Terminates.
```

Example 3 Sets the measurement function to the DCV 10 V range and performs the high-speed measurement (Sampling interval: 500 µs). In the high-speed measurement, the data is read after the measurements of the specified count are complete. The GPIB address of the 7461A is set to 1.

```
Dim DMM_ADR As Integer          ' Declares the variable of the 7461A GPIB address.
Dim dmm As Integer              ' Declares the variable of the device descriptor.
Dim dt As String * 20000        ' Declares the variable of the buffer used for receiving the GPIB data.
Dim s As Integer                ' The variable in which the serial polling result is stored

DMM_ADR = 1                     ' The GPIB address of the 7461A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                                ' Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)
                                ' Sets the transmitting and receiving addresses individually.
Call ibconfig(0, IbcAUTOPOLL, 0)
                                ' Disables the automatic serial polling.
Call ibwrt(dmm, "*RST" & Chr(10))
                                ' Initializes the 7461A.

Call ibwrt(dmm, ":SENSE:FUNCTION 'VOLTAGE:DC'" & Chr(10))
                                ' Sets the measurement function to DCV.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:RANGE 10" & Chr(10))
                                ' Sets the measurement range to 10 V.

Call ibwrt(dmm, ":TRIGGER:SOURCE BUS" & Chr(10))
                                ' Specifies "BUS" as the trigger.
Call ibwrt(dmm, ":INITIATE:CONTINUOUS ON" & Chr(10))
                                ' Sets Continuous to ON.
Call ibwrt(dmm, ":SENSE:ZERO:AUTO OFF" & Chr(10))
                                ' Disables the auto-zero.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:APERTURE 100E-6" & Chr(10))
                                ' Sets the integration time to 100 µs.
Call ibwrt(dmm, ":TRIGGER:TIMER 500E-6" & Chr(10))
                                ' Sets the sampling interval to 500 µs.
Call ibwrt(dmm, ":DIPLAY OFF" & Chr(10))
                                ' Sets the display to OFF.

Call ibwrt(dmm, ":TRACE:STATE ON" & Chr(10))
                                ' Enables the measurement memory.
Call ibwrt(dmm, ":TRACE:CLEAR" & Chr(10))
                                ' Clears the measurement memory.
Call ibwrt(dmm, ":SAMPLE:COUNT 1000" & Chr(10))
                                ' Sets the sampling counts per one trigger. (1000 counts)
Call ibwrt(dmm, ":TRACE:NUMBER 0,999" & Chr(10))
                                ' Sets the range of the recall. (0 to 999)
Call ibwrt(dmm, "*CLS" & Chr(10))
                                ' Clears the status byte.

Call ibwrt(dmm, ":STATUS:MEASUREMENT:ENABLE 256" & Chr(10))
                                ' Sets the measurement end flag of MSR.
Call ibwrt(dmm, "*SRE 1" & Chr(10))
                                ' Sets the measurement end flag of SRE.

Call ibwrt(dmm, "*TRG" & Chr(10))
                                ' Triggers.

Call ibwait(dmm, RQS Or TIMO)   ' Waits for the SRQ transmitting.
If (ibsta And TIMO) Then        ' Judges whether a time-out occurs.
  Call MsgBox("SRQ Time Out", vbOKOnly, "Error")
                                ' Sends a message if the time-out occurs.
```

```
  Call ibwrt(dmm, ":DIPLAY ON" & Chr(10))
                                    ' Sets the display to ON.
  Call ibonl(dmm, 0)                ' Terminates.
  Exit Sub
Else
  Call ibrsp(dmm, s)                ' Reads the status byte.
End If

Call ibwrt(dmm, ":TRACE:DATA?" & Chr(10))
                                    ' Requests the measurement value data.
Call ibrd(dmm, dt)                  ' Substitutes the measurement value for a variable.

Cells(1, 1) = "'" & dt              ' Substitutes the measurement value for a cell.

Call ibwrt(dmm, ":DISPLAY ON" & Chr(10))
                                    ' Sets the display to ON.

Call ibonl(dmm, 0)                  ' Terminates.
```

7. SPECIFICATIONS

7.1 Specifications

Unless otherwise specified, the measurement accuracy of this instrument is guaranteed for one year under the following conditions: Temperature; 23 ±5°C, relative humidity; 85% or less (75% or less in a resistance measurement of larger than 10 MΩ and low power resistance measurement of larger than 1 MΩ)

7.1.1 DC Voltage Measurement (DCV)

- **7461A/61P**

Input terminal	Range	Maximum display	Resolution	Input impedance *3	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
					24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
V-COM	100 mV	119.9999	100 nV	More than 1 GΩ/ 10 MΩ ±1%	0.0030+30.0	0.0040+35.0	0.0040+35.0	5.0+5.0	5.0+7.0
	1000 mV	1199.999	1 μV	More than 1 GΩ/ 10 MΩ ±1%	0.0020+4.0	0.0030+5.0	0.0040+5.0	5.0+1.0	5.0+5.0
	10 V	11.99999	10 μV	More than 1 GΩ/ 10 MΩ ±1%	0.002+3.0	0.0030+3.0	0.0035+3.0	5.0+1.0	5.0+5.0
	100 V	119.9999	100 μV	10 MΩ ±1%	0.0020+5.0	0.0035+5.0	0.0045+5.0	5.0+1.0	5.0+5.0
	1000 V	1099.999	1 mV	10 MΩ ±1%	0.0020+5.0	0.0035+5.0	0.0040+5.0	5.0+1.0	5.0+5.0
BchHi-COM	10 V	11.99999	10 μV	More than 1 GΩ	0.002+3.0	0.0030+3.0	0.0035+3.0	5.0+1.0	5.0+5.0

*1 Integration time: 10 PLC, Display digit: 6 1/2, Auto-zero: ON
*2 Relative value with respect to the calibration standard
*3 In the 100 mV, 1000 mV, and 10 V ranges, the input impedance can be selected from either 10 MΩ or more than 1 GΩ.

Additional error depending on the integration time

Integration time		Additional error ±(digits + μV)	
10 μs ≤ IT ≤ 200 μs	-	In a 3 1/2-digit display	2+20
200 μs < IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	2+20
2 ms < IT < 1 PLC	-	In a 5 1/2-digit display	2+20
1 PLC ≤ IT < 10 PLC	Integral multiple of 1 PLC	In a 6 1/2-digit display	1+0
10 PLC < IT ≤ 10 s	Integral multiple of 1 PLC	In a 6 1/2-digit display	2+0

Maximum allowable applied voltage

Between V and COM	1000 Vpeak
Between BchHi and BchLo	200 Vpeak
Between BchHi, BchLo and COM	200 Vpeak
Between COM and the chassis	500 V

Noise rejection ratio

Integration time	Effective CMRR *4		NMRR
	DC	50/60 Hz ±0.08%	50/60 Hz ±0.08%
Integral multiple of 1 PLC	130 dB	120 dB	60 dB
Others	130 dB	60 dB	0 dB

*4 Unbalanced impedance of 1 kΩ

- **7451A**

Input terminal	Range	Maximum display	Resolution	Input impedance *3	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
					24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
V-COM	300 mV	319.999	1 µV	More than 1 GΩ/ 10 MΩ ±1%	0.0020+5.0	0.0060+7.0	0.0140+7.0	8.0+1.0	8.0+5.0
	3000 mV	3199.99	10 µV	More than 1 GΩ/ 10 MΩ ±1%	0.0020+2.0	0.0060+3.0	0.0100+3.0	7.0+0.1	7.0+3.0
	30 V	31.9999	100 µV	10 MΩ±1%	0.0020+3.0	0.0070+6.0	0.0150+6.0	8.0+1.0	8.0+3.0
	300 V	319.999	1 mV	10 MΩ±1%	0.0020+2.0	0.0060+3.0	0.0140+3.0	8.0+0.1	8.0+3.0
	1000 V	1099.99	10 mV	10 MΩ±1%	0.0020+2.0	0.0060+3.0	0.0140+3.0	8.0+0.1	8.0+1.2
BchHi-COM	10 V	11.9999	100 µV	More than 1 GΩ	0.0020+2.0	0.0100+3.0	0.0100+3.0	8.0+0.1	8.0+1.0

*1 Integration time: 10 PLC, Display digit: 5 1/2, Auto-zero: ON
*2 Relative value with respect to the calibration standard
*3 In the 300 mV and 3000 mV ranges, the input impedance can be selected from either 10 MΩ or more than 1 GΩ.

Additional error depending on the integration time

Integration time		Additional error ±(digits + µV)	
100 µs ≤ IT ≤ 200 µs	-	In a 3 1/2-digit display	2+20
200 µs < IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	10+20
2.01 ms < IT < 9.99 ms	-	In a 5 1/2-digit display	20+20
10 ms < IT < 1 PLC	-	In a 5 1/2-digit display	2+20
1 PLC ≤ IT < 10 PLC	Integral multiple of 1 PLC	In a 5 1/2-digit display	1+0
10 PLC < IT ≤ 10 s	Integral multiple of 1 PLC	In a 5 1/2-digit display	2+0

Maximum allowable applied voltage

Between V and COM	1000 Vpeak
Between BchHi and BchLo	200 Vpeak
Between BchHi, BchLo and COM	200 Vpeak
Between COM and the chassis	500 V

Noise rejection ratio

Integration time	Effective CMRR *4		NMRR
	DC	50/60 Hz ±0.08%	50/60 Hz ±0.08%
Integral multiple of 1 PLC	130 dB	120 dB	60 dB
Others	130 dB	60 dB	0 dB

*4 Unbalanced impedance of 1 kΩ

7.1.2 AC Voltage Measurement (ACV, ACV(AC+DC))

- **7461A/61P**

  Measurement method True RMS measurement, RMS display

  Input range 5% or more of a full scale

  Crest factor 5:1 at a full scale

  Temperature coefficient (1/10 of measurement accuracy that includes the additional error)/°C in each range and frequency range

  ACV

Input terminal	Range	Maximum display	Resolution	Input impedance	Measurement accuracy *1 ±(% of reading + digits) 23°C±5°C					
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz	100 kHz - 300 kHz *2
V-COM	100 mV	119.9999	100 nV	1 MΩ ±2%, 140 pF or less	0.28+600	0.1+600	0.06+600	0.12+1800	0.6+2100	5+6000
	1000 mV	1199.999	1 µV	1 MΩ ±2%, 140 pF or less	0.28+500	0.1+500	0.06+500	0.12+500	0.6+800	4+5000
	10 V	11.99999	10 µV	1 MΩ ±2%, 140 pF or less	0.28+400	0.1+400	0.06+400	0.12+500	0.6+800	4+5000
	100 V	119.9999	100 µV	1 MΩ ±2%, 140 pF or less	0.28+400	0.1+400	0.06+400	0.12+500	0.6+800	4+5000
	700 V	749.999	1 mV	1 MΩ ±2%, 140 pF or less	0.28+280	0.1+280	0.06+280	0.12+350	0.6+580	4+3500

*1 AC filter: SLOW, Integration time: 10 PLC, Display digit: 6 1/2, Sinusoidal input voltage
The input voltage is restricted to 2.2×10^7 V•Hz in the 100 V and 700 V ranges.
If the input voltage is 10% or less of the full scale, 30 µV is added to the measurement accuracy as the additional error.

*2 100 kHz-200 kHz in the 100 mV and 1000 mV ranges

7.1.2 AC Voltage Measurement (ACV, ACV(AC+DC))

ACV(AC+DC)

Input terminal	Range	Maximum display	Resolution	Input impedance	Measurement accuracy *1 ±(% of reading + digits) 23°C±5°C					
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz	100 kHz - 300 kHz *2
V-COM	100 mV	119.99	10 µV	1 MΩ ±12%, 140 pF or less	0.28+6	0.1+6	0.06+6	0.12+18	0.6+21	5+60
	1000 mV	1199.9	100 µV	1 MΩ ±12%, 140 pF or less	0.28+5	0.1+5	0.06+5	0.12+5	0.6+8	4+50
	10 V	11.999	1 mV	1 MΩ ±12%, 140 pF or less	0.28+4	0.1+4	0.06+4	0.12+5	0.6+8	4+50
	100 V	119.99	10 mV	1 MΩ ±12%, 140 pF or less	0.28+4	0.1+4	0.06+4	0.12+5	0.6+8	4+50
	700 V	749.9	100 mV	1 MΩ ±12%, 140 pF or less	0.28+3	0.1+3	0.06+3	0.12+4	0.6+6	4+35

*1 AC filter: SLOW, Integration time: 10 PLC, Display digit: 4 1/2, Sinusoidal input voltage
The input voltage is restricted to 2.2×10^7 V•Hz in the 100 V and 700 V ranges.
If the input voltage is 10% or less of the full scale, 30 µV is added to the measurement accuracy as the additional error.

*2 100 kHz-200 kHz in the 100 mV and 1000 mV ranges

Additional error depending on the crest factor (For a non-sinusoidal input voltage)

1 - 2	0.05% of range
2 - 3	0.15% of range
3 - 5	0.4% of range

Additional error depending on the AC filter Integration time: 10 PLC

Frequency bandwidth		Additional error (% of reading)				Response time *1
		20 Hz - 45 Hz	45 Hz - 100Hz	100 Hz - 1 kHz	More than 1 kHz	
SLOW	20 Hz - 300kHz	0	0	0	0	3 s
MED	20 Hz - 300kHz	0.7	0.23	0.04	0	500 ms
FAST	300 Hz - 300kHz	-	-	0.1	0	100 ms

*1 Time until the measurement value reaches within 0.1% of the final value in the same range

Maximum allowable applied voltage

Between V and COM terminals	700 Vrms, 1000 Vpeak, 2.2×10^7 V・Hz
Between COM terminal and chassis	500 V

Additional error depending on the integration time AC filter: SLOW

Integration time		Additional error (±digit)	
10 µs ≤ IT < 200 µs	-	In a 3 1/2-digit display	2
200 µs ≤ IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	4
2 ms < IT < 1 PLC	-	In a 5 1/2-digit display	20
1 PLC ≤ IT < 10 PLC	-	In a 6 1/2-digit display	120
10 PLC < IT ≤ 10 s	-	In a 6 1/2-digit display	2

- **7451A**

Measurement method True RMS measurement, RMS display

Input range 5% or more of a full scale

Crest factor 3:1 at a full scale

Temperature coefficient (1/10 of measurement accuracy that includes the additional error)/°C in each range and frequency range

ACV

Input terminal	Range	Maximum display	Resolution	Input impedance	Measurement accuracy *1 ±(% of reading + digits)					
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz	100 kHz - 300 kHz *2
V-COM	300 mV	319.999	1 µV	1 MΩ ±2%, 140 pF or less	0.28+120	0.1+120	0.06+120	0.2+150	0.6+240	5+1500
	3000 mV	3199.99	10 µV	1 MΩ ±2%, 140 pF or less	0.28+120	0.1+120	0.06+120	0.2+150	0.6+240	4+1500
	30 V	31.9999	100 µV	1 MΩ ±2%, 140 pF or less	0.28+120	0.1+120	0.06+120	0.2+150	0.6+240	4+1500
	300 V	319.999	1 mV	1 MΩ ±2%, 140 pF or less	0.28+120	0.1+120	0.06+120	0.2+150	0.6+240	4+1500
	700 V	749.99	10 mV	1 MΩ ±2%, 140 pF or less	0.28+28	0.1+28	0.06+28	-	-	-

*1 AC filter: SLOW, Integration time: 10 PLC, Display digit: 5 1/2, Sinusoidal input voltage
The input voltage is restricted to 2.2×10^7 V•Hz in the 300 V and 700 V ranges.

*2 100 kHz-200 kHz in the 300 mV and 3000 mV ranges

7.1.2 AC Voltage Measurement (ACV, ACV(AC+DC))

ACV(AC+DC)

Input terminal	Range	Maximum display	Resolution	Input impedance	Measurement accuracy *1 ±(% of reading + digits)					
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz	100 kHz - 300 kHz *2
V-COM	300 mV	319.99	10 µV	1 MΩ ±12%, 140 pF or less	0.28+12	0.1+12	0.06+12	0.2+15	0.6+24	4+150
	3000 mV	3199.9	100 µV	1 MΩ ±12%, 140 pF or less	0.28+12	0.1+12	0.06+12	0.2+15	0.6+24	4+150
	30 V	31.999	1 mV	1 MΩ ±12%, 140 pF or less	0.28+12	0.1+12	0.06+12	0.2+15	0.6+24	4+150
	300 V	319.99	10 mV	1 MΩ ±12%, 140 pF or less	0.28+12	0.1+12	0.06+12	0.2+15	0.6+24	4+150
	700 V	749.9	100 mV	1 MΩ ±12%, 140 pF or less	0.28+3	0.1+3	0.06+3	-	-	-

*1 AC filter: SLOW, Integration time: 10 PLC, Display digit: 4 1/2, Sinusoidal input voltage
The input voltage is restricted to 2.2×10^7 V•Hz in the 300 V and 700 V ranges.

*2 100 kHz-200 kHz in the 300 mV and 3000 mV ranges

Additional error depending on the crest factor (For a non-sinusoidal input voltage)

1 - 2	0.05% of range
2 - 3	0.15% of range

Additional error depending on the AC filter Integration time: 10 PLC

Frequency bandwidth		Additional error (% of reading)				Response time *1
		20 Hz - 45 Hz	45 Hz - 100Hz	100 Hz - 1 kHz	More than 1 kHz	
SLOW	20 Hz - 300 kHz	0	0	0	0	3 s
MED	20 Hz - 300 kHz	0.7	0.23	0.04	0	500 ms
FAST	300 Hz - 300 kHz	-	-	0.1	0	100 ms

*1 Time until the measurement value reaches within 0.1% of the final value in the same range

Maximum allowable applied voltage

Between V and COM terminals	700 Vrms, 1000 Vpeak, 2.2×10^7 V・Hz
Between COM terminal and chassis	500 V

Additional error depending on the integration time AC filter: SLOW

Integration time		Additional error (±digit)	
100 µs ≤ IT < 200 µs	-	In a 3 1/2-digit display	2
200 µs ≤ IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	10
2.01 ms < IT < 9.99 ms	-	In a 5 1/2-digit display	20
10 ms < IT < 1 PLC	-	In a 5 1/2-digit display	12
1 PLC ≤ IT < 10 PLC	-	In a 5 1/2-digit display	10
10 PLC < IT ≤ 10 s	-	In a 5 1/2-digit display	2

7.1.3 Resistance Measurement (2WΩ, LP-2WΩ, 4WΩ, LP-4WΩ)

- **7461A/61P**

  2WΩ and 4WΩ measurements

Range	Maximum display	Resolution	Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient *4 ±(ppm of reading + digits)/°C	
				24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
100 Ω	119.9999	100 μΩ	10 mA	0.0030+30.0	0.0060+35.0	0.0080+35.0	6.0+5.0	6.0+5.0
1000 Ω	1199.999	1 mΩ	1 mA	0.0020+5.0	0.0060+5.0	0.0080+8.0	6.0+1.0	6.0+5.0
10 kΩ	11.99999	10 mΩ	100 μA	0.0020+5.0	0.0060+5.0	0.0080+8.0	6.0+1.0	6.0+5.0
100 kΩ	119.9999	100 mΩ	10 μA	0.0020+5.0	0.0060+5.0	0.0080+8.0	6.0+1.0	6.0+5.0
1000 kΩ	1199.999	1 Ω	5 μA	0.0020+10.0	0.0060+30.0	0.0080+30.0	10.0+5.0	10.0+7.0
10 MΩ	11.99999	10 Ω	500 nA	0.0150+10.0	0.0200+30.0	0.0300+30.0	30.0+5.0 *3	30.0+7.0 *3
100 MΩ	119.9999	100 Ω	50 nA	0.3000+100.0	0.5000+100.0	0.6000+100.0	700.0+5.0	700.0+7.0

*1 Integration time: 10 PLC, Auto-zero: ON for 2WΩ, Display digit: 6 1/2
The measurement cable resistance and 0.2 Ω (internal circuit resistance error) are added to the digit error in 2WΩ measurement.
*2 Relative value with respect to the calibration standard
*3 15 ppm/°C is added to the temperature coefficient in the temperature range of 30°C to 50°C.
*4 The temperature coefficient in Auto-zero ON is used in the 4WΩ measurement.

Low Power 2WΩ and 4WΩ measurements (LP-2WΩ/LP-4WΩ)

Range	Maximum display	Resolution	Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient *4 ±(ppm of reading + digits)/°C	
				24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
100 Ω	119.9999	100 μΩ	1 mA	0.0030+30.0	0.0070+50.0	0.0100+60.0	6.0+5.0	6.0+8.0
1000 Ω	1199.999	1 mΩ	100 μA	0.0020+30.0	0.0070+50.0	0.0100+60.0	6.0+5.0	6.0+8.0
10 kΩ	11.99999	10 mΩ	10 μA	0.0020+30.0	0.0070+50.0	0.0100+60.0	7.0+5.0	7.0+8.0
100 kΩ	119.9999	100 mΩ	5 μA	0.0020+15.0	0.0070+50.0	0.0100+60.0	10.0+5.0	10.0+25.0
1000 kΩ	1199.999	1 Ω	500 nA	0.0150+15.0	0.0450+50.0	0.0500+60.0	65.0+5.0 *3	65.0+8.0 *3
10 MΩ	11.99999	10 Ω	50 nA	0.3000+100.0	0.6000+100.0	0.8000+100.0	700.0+5.0	700.0+8.0 *3

*1 Integration time: 10 PLC, Auto zero: ON for 2WΩ, Display digit: 6 1/2
The measurement cable resistance and 0.2 Ω (internal circuit resistance error) are added to the digit error in 2WΩ measurement.
*2 Relative value with respect to the calibration standard
*3 15 ppm/°C is added to the temperature coefficient in the temperature range of 30°C to 50°C.
*4 The temperature coefficient in Auto-zero ON is used in the LP-4WΩ measurement.

Response time: 100 MΩ = 5 s
(Time until the measurement value reaches within 0.1% of the final value)
1 MΩ/10 MΩ = 0.5 s
(Time until the measurement value reaches within 0.1% of the final value)

7.1.3 Resistance Measurement (2WΩ, LP-2WΩ, 4WΩ, LP-4WΩ)

Additional error depending on the integration time

Integration time		Additional error ±(digit + mΩ)	
10 µs ≤ IT ≤ 200 µs	-	In a 3 1/2-digit display	2+20
200 µs < IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	2+20
2 ms < IT < 1 PLC	-	In a 5 1/2-digit display	2+20
1 PLC ≤ IT < 10 PLC	Integral multiple of 1 PLC	In a 6 1/2-digit display	1+0
10 PLC < IT ≤ 10 s	Integral multiple of 1 PLC	In a 6 1/2-digit display	2+0

Voltage between input terminals without connections.9.5 V or less

Maximum allowable applied voltage

Between Ω and COM	1000 Vpeak
Between COM and the chassis	500 V
Between 4WΩHi-4WΩLo	200 Vpeak
Between 4WΩHi, 4WΩLo and the chassis	500 V

- **7451A**

  2WΩ and 4WΩ measurements

Range	Maximum display	Resolution	Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient *5 ±(ppm of reading + digits)/°C	
				24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
30 Ω	31.9999	100 µΩ	10 mA	0.0030+30.0	0.0100+40.0	0.0150+40.0	10.0+5.0	10.0+50.0
300 Ω	319.999	1 mΩ	1 mA	0.0020+5.0	0.0080+11.0	0.0150+11.0	10.0+1.0	10.0+6.0
3000 Ω	3199.99	10 mΩ	1 mA	0.0020+3.0	0.0070+3.0	0.0120+3.0	10.0+0.2	10.0+3.0
30 kΩ	31.9999	100 mΩ	100 µA	0.0020+3.0	0.0070+3.0	0.0130+3.0	10.0+0.2	10.0+3.0
300 kΩ	319.999	1 Ω	10 µA	0.0020+3.0	0.0090+3.0	0.0140+3.0	8.0+0.2	8.0+3.0
3000 kΩ	3199.99	10 Ω	1 µA	0.0070+14.0	0.0300+19.0	0.0300+19.0	35.0+5.0	35.0+5.0
30 MΩ	31.9999	100 Ω	100 nA	0.0600+14.0	0.1800+19.0	0.2000+19.0	100.0+5.0 *3	100.0+5.0 *3
300 MΩ	319.999	1 kΩ	10 nA	0.6000+14.0	1.7000+19.0	2.0000+19.0	1000.0+5.0 *4	1000.0+5.0 *4

*1 Integration time: 10 PLC, Auto-zero: ON for 2WΩ, Display digit: 5 1/2
The measurement cable resistance and 0.2 Ω (internal circuit resistance error) are added to the digit error in 2WΩ measurement.
*2 Relative value with respect to the calibration standard
*3 115 ppm/°C is added to the temperature coefficient in the temperature range of 30°C to 50°C.
*4 1150 ppm/°C is added to the temperature coefficient in the temperature range of 30°C to 50°C.
*5 The temperature coefficient in Auto-zero ON is used in the 4WΩ measurement.

Low Power 2WΩ and 4WΩ measurements (LP-2WΩ/LP-4WΩ)

Range	Maximum display	Resolution	Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient *5 ±(ppm of reading + digits)/°C	
				24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
300 Ω	319.999	1 mΩ	100 μA	0.0030+30.0	0.0080+40.0	0.0150+40.0	10.0+5.0	10.0+50.0
3000 Ω	3199.99	10 mΩ	100 μA	0.0020+5.0	0.0080+11.0	0.0150+11.0	10.0+1.0	10.0+5.0
30 kΩ	31.9999	100 mΩ	10 μA	0.0020+5.0	0.0080+11.0	0.0150+11.0	10.0+1.0	10.0+5.0
300 kΩ	319.999	1 Ω	1 μA	0.0070+5.0	0.0300+11.0	0.0300+11.0	30.0+1.0	30.0+5.0
3000 kΩ	3199.99	10 Ω	100 nA	0.0600+20.0	0.1800+33.0	0.2000+33.0	100.0+5.0 *3	100.0+5.0 *3
30 MΩ	31.9999	100 Ω	10 nA	0.6000+20.0	1.7000+33.0	2.0000+33.0	1000.0+5.0 *4	1000.0+5.0 *4

*1 Integration time: 10 PLC, Auto zero: ON for 2WΩ, Display digit: 5 1/2
The measurement cable resistance and 0.2 Ω (internal circuit resistance error) are added to the digit error in 2WΩ measurement.
*2 Relative value with respect to the calibration standard
*3 115 ppm/°C is added to the temperature coefficient in the temperature range of 30°C to 50°C.
*4 1150 ppm/°C is added to the temperature coefficient in the temperature range of 30°C to 50°C.
*5 The temperature coefficient in Auto-zero ON is used in the 4WΩ measurement.

Response time: 300 MΩ = 5 s
(Time until the measurement value reaches within 0.1% of the final value)
3 MΩ/30 MΩ = 0.5 s
(Time until the measurement value reaches within 0.1% of the final value)

Additional error depending on the integration time

Integration time		Additional error ±(digit + mΩ)	
100 μs ≤ IT ≤ 200 μs	-	In a 3 1/2-digit display	2+20
200 μs < IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	10+20
2.01 ms < IT < 9.99 ms	-	In a 5 1/2-digit display	20+20
10 ms < IT < 1 PLC	-	In a 5 1/2-digit display	2+20
1 PLC ≤ IT < 10 PLC	Integral multiple of 1 PLC	In a 5 1/2-digit display	1+0
10 PLC < IT ≤ 10 s	Integral multiple of 1 PLC	In a 5 1/2-digit display	2+0

Voltage between input terminals without connections. 9.5 V or less

Maximum allowable applied voltage

Between Ω and COM	1000 Vpeak
Between COM and the chassis	500 V
Between 4WΩHi-4WΩLo	200 Vpeak
Between 4WΩHi, 4WΩLo and the chassis	500 V

7.1.4 DC Current Measurement (DCI)

- **7461A/61P**

Input terminal	Range	Maximum display	Resolution	Resistance between input terminals	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
					24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
mA-COM	1000 µA	1199.999	1 nA	103 Ω	0.005+15	0.0180+20.0	0.0200+20.0	30.0+5.0	30.0+6.0
	10 mA	11.99999	10 nA	10.5 Ω	0.005+15	0.0250+20.0	0.0250+30.0	30.0+5.0	30.0+6.0
	100 mA	119.9999	100 nA	1.5 Ω	0.008+15	0.0250+20.0	0.0250+30.0	35.0+5.0	35.0+6.0
	1000 mA	1199.999	1 µA	0.5 Ω	0.03+15	0.0300+20.0	0.0500+30.0	50.0+10.0	50.0+10.0
	3 A	3.19999	10 µA	0.5 Ω	0.1+15	0.1000+20.0	0.1000+30.0	50.0+20.0	50.0+30.0

*1 Integration time: 10 PLC, Display digit: 6 1/2, Auto-zero: ON
*2 Relative value with respect to the calibration standard

Additional error depending on the integration time

Integration time		Additional error (±digit)	
10 µs ≤ IT < 200 µs	-	In a 3 1/2-digit display	2
200 µs ≤ IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	2
2 ms < IT < 1 PLC	-	In a 5 1/2-digit display	2
1 PLC ≤ IT < 10 PLC	Integral multiple of 1 PLC	In a 6 1/2-digit display	1
10 PLC < IT ≤ 10 s	Integral multiple of 1 PLC	In a 6 1/2-digit display	2

Maximum allowable applied current

Between mA and COM terminals

3A (DC or ACrms)

Input protection By using the 3.15 A/250 V fast blow fuse which is compliant with IEC60127-2 sheet1
The fuse can be changed on the rear panel.

- **7451A/61P**

Input terminal	Range	Maximum display	Resolution	Resistance between input terminals	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
					24 hours 23°C ±1°C *2	90 days 23°C ±5°C	1 year 23°C ±5°C	Auto-zero ON	Auto-zero OFF
mA-COM	3000 µA	3199.99	10 nA	10.5 Ω	0.01+30	0.0300+40.0	0.0500+40.0	45.0+4.0	45.0+10.0
	30 mA	31.9999	100 nA	10.5 Ω	0.01+4.5	0.0300+6.0	0.0500+6.0	45.0+0.6	45.0+7.0
	300 mA	319.999	1 µA	1.5 Ω	0.03+4.5	0.0600+6.0	0.1000+6.0	50.0+0.6	50.0+7.0
	3000 mA	3199.99	10 µA	0.5 Ω	0.12+4.5	0.1200+6.0	0.1200+6.0	50.0+0.6	50.0+7.0

*1 Integration time: 10 PLC, Display digit: 5 1/2, Auto-zero: ON
*2 Relative value with respect to the calibration standard

Additional error depending on the integration time

Integration time		Additional error (±digit)	
100 µs ≤ IT < 200 µs	-	In a 3 1/2-digit display	2
200 µs ≤ IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	10
2.01 ms < IT < 9.99 ms	-	In a 5 1/2-digit display	20
10 ms < IT < 1 PLC	-	In a 5 1/2-digit display	2
1 PLC ≤ IT < 10 PLC	Integral multiple of 1 PLC	In a 5 1/2-digit display	1
10 PLC < IT ≤ 10 s	Integral multiple of 1 PLC	In a 5 1/2-digit display	2

Maximum allowable applied current

Between mA and COM terminals
3A (DC or ACrms)

Input protection By using the 3.15 A/250 V fast blow fuse which is compliant with IEC60127-2 sheet1
The fuse can be changed on the rear panel.

7.1.5 AC Current Measurement (ACI, ACI(AC+DC))

- **7461A/61P**

Measurement method True RMS measurement, RMS display

Input range 5% or more of a full scale

Crest factor 5:1 at a full scale (Except for the 3 A range)

Temperature coefficient (1/10 of measurement accuracy that includes the additional error)/°C in each range and frequency range

ACI

Input terminal	Range	Maximum display	Resolution	Resistance between input terminals	Measurement accuracy *1 ±(% of reading + digits) 23°C±5°C		
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 5 kHz
mA-COM	1000 µA	1199.999	1 nA	103 Ω	0.45+400	0.25+400	0.11+400
	10 mA	11.99999	10 nA	10.5 Ω	0.45+400	0.25+400	0.11+400
	100 mA	119.9999	100 nA	1.5 Ω	0.45+400	0.25+400	0.11+400
	1000 mA	1199.999	1 µA	0.5 Ω	0.45+400	0.25+400	0.11+400
	3 A	3.19999	10 µA	0.5 Ω	0.45+120	0.25+120	0.15+120

*1: AC filter: SLOW, Integration time: 10 PLC, Display digit: 6 1/2, Sinusoidal input current

7.1.5 AC Current Measurement (ACI, ACI(AC+DC))

ACI(AC+DC)

Input terminal	Range	Maximum display	Resolution	Resistance between input terminals	Measurement accuracy *1 ±(% of reading + digits) 23°C±5°C		
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 5 kHz
mA-COM	1000 µA	1199.9	100 nA	103 Ω	0.45+4	0.25+4	0.11+4
	10 mA	11.999	1 µA	10.5 Ω	0.45+4	0.25+4	0.11+4
	100 mA	119.99	10 µA	1.5 Ω	0.45+4	0.25+4	0.11+4
	1000 mA	1199.9	100 µA	0.5 Ω	0.45+4	0.25+4	0.11+4
	3 A	3.199	1 mA	0.5 Ω	0.45+1.2	0.25+1.2	0.15+1.2

*1: AC filter: SLOW, Integration time: 10 PLC, Display digit: 4 1/2, Sinusoidal input current

Additional error depending on the crest factor (For a non-sinusoidal input current)

1 - 2	0.05% of range
2 - 3	0.15% of range
3 - 5	0.4% of range

Additional error depending on the AC filter Integration time: 10 PLC

Frequency bandwidth		Additional error (% of reading)				Response time *2
		20 Hz - 45 Hz	45 Hz - 100Hz	100 Hz - 1 kHz	More than 1 kHz	
SLOW	20 Hz - 300kHz	0	0	0	0	3 s
MED	20 Hz - 300kHz	0.7	0.23	0.04	0	500 ms
FAST	300 Hz - 300kHz	-	-	0.1	0	100 ms

*2 Time until the measurement value reaches within 0.1% of the final value in the same range

Additional error depending on the integration time AC filter: SLOW

Integration time		Additional error(±digit)	
10 µs ≤ IT < 200 µs	-	In a 3 1/2-digit display	2
200 µs ≤ IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	4
2 ms < IT < 1 PLC	-	In a 5 1/2-digit display	20
1 PLC ≤ IT < 10 PLC	-	In a 6 1/2-digit display	120
10 PLC < IT ≤ 10 s	-	In a 6 1/2-digit display	2

Maximum allowable applied current

Between mA and COM terminals

3A (DC or ACrms)

Input protection By using the 3.15 A/250 V fast blow fuse which is compliant with IEC60127-2 sheet1
The fuse can be changed on the rear panel.

- **7451A**

Measurement method True RMS measurement, RMS display

Input range 5% or more of full scale

Crest factor 3:1 at a full scale (Except for the 3 A range)

Temperature coefficient (1/10 of measurement accuracy that includes the additional error)/°C in each range and frequency range

ACI

Input terminal	Range	Maximum display	Resolution	Resistance between input terminals	Measurement accuracy *1 ±(% of reading + digits) 23°C±5°C		
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 5 kHz
mA-COM	3000 µA	3199.99	10 nA	10.5 Ω	0.45+200	0.25+200	0.2+200
	30 mA	31.9999	100 nA	10.5 Ω	0.45+200	0.25+200	0.2+200
	300 mA	319.999	1 µA	1.5 Ω	0.45+200	0.25+200	0.2+200
	3000 mA	3199.99	10 µA	0.5 Ω	0.45+200	0.25+200	0.2+200

*1: AC filter: SLOW, Integration time: 10 PLC, Display digit: 5 1/2, Sinusoidal input current

ACI(AC+DC)

Input terminal	Range	Maximum display	Resolution	Resistance between input terminals	Measurement accuracy *1 ±(% of reading + digits) 23°C±5°C		
					20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 5 kHz
mA-COM	3000 µA	3199.9	100 nA	10.5 Ω	0.45+20	0.25+20	0.2+20
	30 mA	31.999	1 µA	10.5 Ω	0.45+20	0.25+20	0.2+20
	300 mA	319.99	10 µA	1.5 Ω	0.45+20	0.25+20	0.2+20
	3000 mA	3199.9	100 µA	0.5 Ω	0.45+20	0.25+20	0.2+20

*1: AC filter: SLOW, Integration time: 10 PLC, Display digit: 4 1/2, Sinusoidal input current

Additional error depending on the crest factor (For a non-sinusoidal input current)

Range	Crest factor ±(% of reading + % of range)	
	1 to 2	2 to 3
3000 µA	0+0.05	0+0.15
30 mA	0.01+0.05	3+0.15
300 mA	0+0.05	0.3+0.15
3000 mA	0+0.05	0.03+0.15

Additional error depending on the AC filter Integration time: 10 PLC

Frequency bandwidth		Additional error (% of reading)				Response time *1
		20 Hz - 45 Hz	45 Hz - 100Hz	100 Hz - 1 kHz	More than 1 kHz	
SLOW	20 Hz - 5 kHz	0	0	0	0	3 s
MED	20 Hz - 5 kHz	0.7	0.23	0.04	0	500 ms
FAST	300 Hz - 5 kHz	-	-	0.1	0	100 ms

*1 Time until the measurement value reaches within 0.1% of the final value in the same range

Additional error depending on the integration time AC filter: SLOW

Integration time		Additional error(±digit)	
100 µs ≤ IT < 200 µs	-	In a 3 1/2-digit display	2
200 µs ≤ IT ≤ 2 ms	Includes FAST1 and FAST2.	In a 4 1/2-digit display	10
2.01 ms < IT < 9.99 ms	-	In a 5 1/2-digit display	20
10 ms < IT < 1 PLC	-	In a 5 1/2-digit display	20
1 PLC ≤ IT < 10 PLC	-	In a 5 1/2-digit display	20
10 PLC < IT ≤ 10 s	-	In a 5 1/2-digit display	2

Maximum allowable applied current

Between mA and COM terminals
3A (DC or ACrms)

Input protection By using the 3.15 A/250 V fast blow fuse which is compliant with IEC60127-2 sheet1
The fuse can be changed on the rear panel.

7.1.6 Frequency Measurement (FREQ)

- **7451A, 7461A, 7461P**

Measurement method Reciprocal

Measurement frequency range	Measurement accuracy (Sinusoidal wave)
1 Hz - 10 Hz	0.05% of reading
10 Hz - 300 kHz	0.02% of reading

Input signal voltage range 100 mVrms to 700 Vrms (However, the input signal is restricted to the maximum allowable applied voltage.)

Gate time

Sampling rate	Gate time	Measurement frequency range	Maximum measurement period	Maximum display	
				7461A/ 61P	7451A
SLOW	1000 ms	1 Hz to 300 kHz	2.2 s	9999999	999999
MED	100 ms	10 Hz to 300 kHz	220 ms	999999	99999
FAST	10 ms	100 Hz to 300 kHz	22 ms	99999	9999

Maximum allowable applied voltage

Between V and COM terminals	700 Vrms, 1000 Vpeak, 2.2×10^7 V•Hz
Between COM terminal and chassis	500 V

7.1.7 Diode Measurement

- **7461A/61P**

Range	Maximum display	Resolution	Measurement current
1000 mV	1199.999	1 μV	1 mA

The specifications other than the above are the same as those of 1000 Ω range in 2WΩ measurement.
However, the offset error, which is calculated by the following formula; (the measurement cable resistance + internal circuit resistance error of 0.2 Ω) × 1 mA, is added to the measurement accuracy in 2WΩ measurement.

- **7451A**

Range	Maximum display	Resolution	Measurement current
3000 mV	3199.99	10 μV	1 mA

The specifications other than the above are the same as those of 3000 Ω range in 2WΩ measurement.
However, the offset error, which is calculated by the following formula; (the measurement cable resistance + internal circuit resistance error of 0.2 Ω) × 1 mA, is added to the measurement accuracy in 2WΩ measurement.

7.1.8 Continuity Measurement

- **7461A/61P**

Range	Maximum display	Resolution	Measurement current
1000 Ω	1199.999	1 mΩ	1 mA

Specifications not described above are the same as those in the 1000 Ω range in the 2WΩ measurement.
Continuity judgment setting range: 1 Ω to 1000 Ω

- **7451A**

Range	Maximum display	Resolution	Measurement current
3000 Ω	3199.99	10 mΩ	1 mA

Specifications not described above are the same as those in the 3000 Ω range in the 2WΩ measurement.
Continuity judgment setting range: 1 Ω to 1000 Ω

7.1.9 Temperature Measurement (TEMP-2W, TEMP-3W, TEMP-4W)

7461P

Platinum resistance bulb	Pt100 (compliant with JIS C1604-1997) or JPt100 (compliant with JIS C1604-1981)
Measurement range	PT100 -200 °C to +850 °C
	JPt100 -200 °C to +649 °C
Linearization	Digital calculation
Wire connection	2W, 3W or 4W
Allowable wire resistance	10 Ω or lower (except 2W)
Measurement unit	°C, °F or K

Resolution	Measurement current	Measurement accuracy ± (% of reading + °C)*1		Temperature coefficient ± (ppm of reading + °C)/°C *1
		90 days, 23±5 °C	1 year, 23±5 °C	
0.001 °C	1 mA	0.006 + 0.06	0.008 + 0.06	6 + 0.003

*1 4W, RATE: SLOW2, not including the measurement probe accuracy

7.1.10 Measurement Time and Display Digits

Sampling rate	Measurement speed (In Auto-range OFF)				Integration time	Auto-zero
	DCV Bch-DCV DCI 2WΩ LP-2WΩ	4WΩ LP-4WΩ	ACV ACI Continuity Diode	ACV(AC+DC) ACI(AC+DC)		
FAST1	1000 times/s *1	110 times/s	400 times/s	4 times/s	0.02 PLC	OFF
FAST2	300 times/s *1	80 times/s	250 times/s	4 times/s	0.1 PLC	OFF
MED	20 times/s	20 times/s	20 times/s	3 times/s	1 PLC	ON
SLOW1	4.5 times/s	4.5 times/s	4.5 times/s	2 times/s	5 PLC	ON
SLOW2	2 times/s	2 times/s	2 times/s	1 time/s	10 PLC	ON

*1 Under the following conditions
Trigger source: BUS, EXT, MAN
Sampling count: 2 or more
Complete signal output mode: SINGLE
Display: OFF, Calculation: OFF, Measurement data memory: ON

Maximum measurement speed and its conditions

	7461A/61P	7451A
Measurement speed	20,000 times/s	5,000 times/s
Target function	DCV, DCV-Bch, DCI, 2WΩ, LP-2WΩ	
Integration time	10 μs	100 μs
Sampling interval	50 μs	200 μs
Other	*1	

*1 Under the following conditions
Auto-range: OFF
Trigger source: BUS, EXT, MAN
Sampling count: 2 to 16,000
Complete signal output mode: SINGLE
Display: OFF
Measurement data memory: ON
7451A/61A: Up to 10,000 data
7461P: Up to 16,000 data

7.1.10 Measurement Time and Display Digits

Integration Time (IT) and Sampling Interval (SI)

7461A/61P		7451A	
Integration time (IT)		Integration time (IT)	
Setting range	Resolution	Setting range	Resolution
10 µs to 199 µs	1 µs	100 µs to 199 µs	1 µs
200 µs to 9.99 ms	10 µs	200 µs to 9.99 ms	10 µs
10 ms to 5999.9 ms	100 µs	10 ms to 5999.9 ms	100 µs
6 s to 10 s	200 µs	6 s to 10 s	200 µs
0.02 PLC to 100 PLC	0.01 PLC	0.02 PLC to 100 PLC	0.01 PLC
Sampling interval (SI)		Sampling interval (SI)	
Setting range		Setting range	
50 µs to 3600 s		200 µs to 3600 s	

Integration Time and Display Digits

Function / Integration time (IT)	DCV/DCI DCV-Bch 2W/4WΩ LP-2W/4WΩ ACV/ACI Diode and Continuity		ACV(AC+DC) ACI(AC+DC)	
	7461A/61P	7451A	7461A/61P	7451A
10 µs ≤ (IT) ≤ 200 µs	3 1/2-digit display	3 1/2-digit display	3 1/2-digit display	
200 µs < (IT) ≤ 2 ms	4 1/2-digit display	4 1/2-digit display	3 1/2-digit display	
2 ms < (IT) ≤ 15 ms	5 1/2-digit display	5 1/2-digit display	4 1/2-digit display	
15 ms < (IT) ≤ 10 s	6 1/2-digit display	5 1/2-digit display	4 1/2-digit display	

7.1.11 Calculation Functions

- NULL calculation
  Display value (NULL) = Measurement value - NULL constant
- Smoothing calculation
  Display value (SM) = Moving average over a specified number of measurements
- Comparator calculation
  Display (HIGH) ← HIGH setting value < Measurement value
  Display (LOW) ← Measurement value < LOW setting value
  Display (GO) ← LOW setting value ≤ Measurement value ≤ HIGH setting value
- Scaling calculation
  Display (SCL) = (Measurement value - B)/A × C
  A, B, and C are constants. (Setting values)
- MAX and MIN calculation
  Display value (MAX) = Maximum measurement value after the calculation starts
  Display value (MIN) = Minimum measurement value after the calculation starts
  Display value (AVE) = Arithmetic mean after the calculation starts (Remote output only)
- dB and dBm calculation
  dB display = 20 log (Measurement value/D)
  dBm display value = 10 log ((Measurement value)^2/D)/10^{-3}
  D is a constant. (Set by the user)
- Statistical calculation

Number of samples	Display value (SAMPLE) = Number of measurement values in the specified range of the measurement memory
Maximum value	Display value (MAX) = Maximum measurement value in the specified range of the measurement memory
Minimum value	Display value (MIN) = Minimum measurement value in the specified range of the measurement memory
Average value	Display value (AVE) = Average value in the specified range of the measurement memory
Standard deviation	Display value (SIGMA) = Standard deviation in the specified range of the measurement memory
Dispersion	Display value (P-P) = ((Maximum measurement value) - (Minimum measurement value)) in the specified range of the measurement memory

7.2 Interface Specifications

Item	Specifications
Remote control	
Remote command	Compliance with SCPI and the command format for the ADC CORPORATION's R6552 and R64 series
Interface	Either GPIB or USB can be selected.
USB	
Standard	Compliance with USB2.0 Full Speed
Connector	Type B
GPIB	
Standard	Compliance with IEEE488.2
Connector	24-pin Anphenol
Interface functions	SH1, AH1, T5, L4, SR1, RL1, PP0, DC1, DT1, C0, E2
Output format	ASCII
Addressing	31 kinds of talker/listener addresses can be specified from the front panel.
Comparator output	
Output signal	TTL Output: PASS/FAIL
Comparator output	TTL output: PASS/FAIL, Relay output: PASS/Hi/Lo (PASS/FAIL output can be set individually)
Digital output	TTL output: 2 outputs, Relay contact: 3 contacts
Connector	D-Sub9 pin
a.Optical semiconductor relay contact	
Allowable contact voltage (for break)	DC30 V
Allowable contact current	DC120 mA
Withstand voltage between contact and GND	30 V
Contact operating time	Approximately 1ms or less
b. TTL Output	
Output level	TTL Selecting the positive or negative logic
Maximum allowable applied voltage	12 Vpeak
External trigger signal	
Connector	BNC
Signal level	TTL Detecting the falling edge
Pulse width	1 μs or more
Complete signal output	
Connector	BNC
Signal level	TTL Negative pulse
Sink Current	20 mA or below
Pulse width	Approx. 5 μs or 100 μs can be selected.

7.3 General Specifications

<table>
<tr><th>Item</th><th>Specifications</th></tr>
<tr><td>Operation environment</td><td>Ambient temperature: 0°C to +50°C
Relative humidity: 85%RH or less no condensation
75% or less in resistance measurement of larger than 10 MΩ and low power resistance measurement of larger than 1 MΩ</td></tr>
<tr><td>Storage environment</td><td>Ambient temperature: -25°C to +70°C
Relative humidity: 85%RH or less no condensation</td></tr>
<tr><td>Warm-up time</td><td>60 minutes or more</td></tr>
<tr><td>Display</td><td>7-digit fluorescent display A digit is displayed by seven segments.</td></tr>
<tr><td>Range switching</td><td>Automatic and manual</td></tr>
<tr><td>Input method</td><td>Floating</td></tr>
<tr><td>Measurement method</td><td>Integration</td></tr>
<tr><td>Overload display</td><td>OL</td></tr>
<tr><td>Input switching</td><td>Front and rear (Can be switched manually and remotely)
Front: VΩ, HI(4WΩ/Bch), LO (4WΩ/Bch), mA, and COM terminals
Rear: VΩ, HI(4WΩ/Bch), LO (4WΩ/Bch), and COM terminals</td></tr>
<tr><td>Memory</td><td>Data memory: Up to 10,000 data (7451A/61A)
Up to 20,000 data (7461P)
Condition setting memory: 4 (USER0 to USER3)</td></tr>
<tr><td>Trigger function</td><td>External trigger: External trigger signal
Panel key
BUS (GPIB, USB)
Internal trigger: Level trigger
Delta trigger</td></tr>
<tr><td>Power supply</td><td>AC power supply: 100 V/120 V/220 V/240 V
(Can be switched by user)
<table><tr><td>Option number</td><td>Standard</td><td>OPT.32</td><td>OPT.42</td><td>OPT.44</td></tr><tr><td>Power supply voltage</td><td>100V</td><td>120V</td><td>220V</td><td>240V</td></tr></table>
Specify the option when ordering.
Use a power cable and fuse that are compliant with the safety standard when changing the power supply voltage.</td></tr>
<tr><td>Power supply frequency</td><td>50 Hz/60 Hz</td></tr>
<tr><td>Power consumption</td><td>20 VA or below</td></tr>
<tr><td>Dimensions</td><td>Approx. 212 (width) × 88 (height) × 340 (depth) mm</td></tr>
<tr><td>Mass</td><td>3.4 kg or less</td></tr>
<tr><td>Safety</td><td>IEC61010-1
EN61010-1
Measurement Category II</td></tr>
<tr><td>EMC</td><td>EN61326 class B</td></tr>
</table>

7.4 Options

Table 7-1 Options

Option	Description
OPT32	Changing the power supply voltage: For 120 V
OPT42	Changing the power supply voltage: For 220 V
OPT44	Changing the power supply voltage: For 240 V
OPT64	Option for ATE (7461A only)

8. MAINTENANCE

This chapter describes the following information which relates to the maintenance of this instrument.

8.1 Replacing Fuses

8.2 Cleaning

8.3 Calibration

8.4 Replacing Parts with Limited Life

8.5 Product Disposal and Recycle

8.6 Storage

8.7 Transportation

8.8 Notes Regarding Repair, Replacement, and Periodic Calibration

8.9 System Recovery Procedure

8.10 Self-Test

8.11 Error Message List

8.1 Replacing Fuses

This instrument includes a power fuse and protection fuse. Replace them according to the following procedure.

CAUTION:

1. ***To prevent fire and electrical shock, replace with a fuse listed in Table 8-1. Never use an incompatible fuse and short-circuit the fuse holder.***
2. ***Whether a fuse has blown cannot be checked by observation. Measure the resistance of the fuse and judge whether the fuse has blown. (If the resistance is 15 Ω or less, the fuse has not blown.)***
3. ***Do not cut any of the internal ground wiring in this instrument or disconnect this instrument from ground. This instrument may become unsafe to operate.***

Table 8-1 Fuse Rating and Compliant Standard

Power supply voltage	Manufacturer	Manufacturer's product code	Compliant standard	Rated current/ voltage	Application
100 V/120 V	Littelfuse, Inc.	0218.250XP	IEC60127-2 sheet3	0.25 A/250 V	Power fuse
220 V/240 V	Littelfuse, Inc.	0218.160XP	IEC60127-2 sheet3	0.16 A/250 V	Power fuse
-	Littelfuse, Inc.	02163.15XP	IEC60127-2 sheet1	3.15 A/250 V	Current terminal protection

8.1.1 Replacing a Power Fuse

The power fuse is placed in the AC power supply connector on the rear panel.

For more information on how to check and replace the power fuse, refer to Section 3.4.2, “Changing the Power Supply Voltage.”

8.1.2 Replacing a Protection Fuse

To prevent fire and burns when an overcurrent is supplied in the current input terminal, a fast-blow fuse is included on the rear panel of this instrument.

1. Rotate the fuse cap in the counter-clockwise direction by using a flathead screwdriver and remove the fuse.
2. Replace with a fuse, which is compliant with the standard and listed in Table 3-1, and secure the fuse cap.

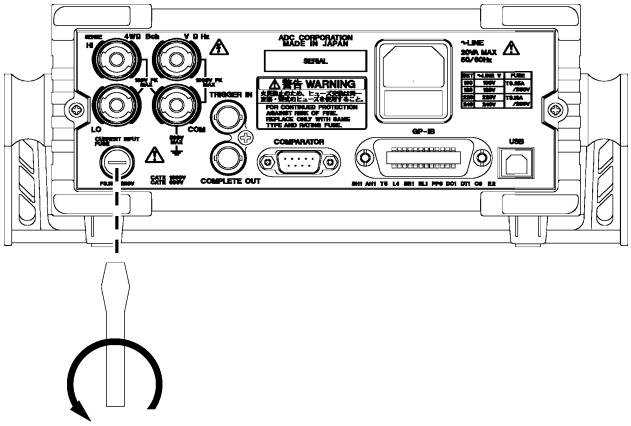


Figure 8-1 Input Terminal Protection Fuse

8.2 Cleaning

This section describes the cleaning procedure of this instrument and warning messages.

WARNING: ***To prevent accidents caused by electric shocks, remove all cables from this instrument. Never attempt to remove the cover to clean the inside of this instrument.***

8.2.1 Cleaning the Outside

Use a soft or damp cloth to clean this instrument.

NOTE:

1. ***Do not allow water inside this instrument or lint on this instrument.***
2. ***Do not use an organic solvent such as benzene or acetone for cleaning because it degrades the plastic.***

8.2.2 Cleaning Others

Take appropriate precautions to protect this instrument from dust.

WARNING: ***Periodically remove any dust which builds up on the AC power connector and plug. The tracking phenomenon, which can be caused by wet dust, may cause a fire.***

8.3 Calibration

To maintain the measurement accuracy of this instrument, perform calibration at least once a year; warranty period.

8.3.1 Preparing for Calibration

1. Power supply
   Use an AC power supply in a voltage range listed in Table 3-4 and frequency of 50 Hz or 60 Hz.
2. Environment
   Perform calibration in following conditions.
   Temperature: +23 ±1°C
   Humidity: 70% or less
   An area free from dust, vibration, wind, and noise
3. Warm-up time
   Perform calibration after allowing a warm up time of 60 minutes or more.
   Sufficient warm-up time is also required for the instrument used for calibration.

8.3.2 Calibration Standards

Calibration standard	Operation range		Accuracy
	7461A/61P	7451A	
DC voltage	100 mV - 1000 V	300 mV - 1000 V	±7 ppm
AC voltage	1 mVrms - 700 Vrms 1 kHz sinusoidal wave	3 mVrms - 700 Vrms 1 kHz sinusoidal wave	±150 ppm
DC current	1 mA - 1 A	3 mA - 300 mA	±30 ppm
	1 A - 3 A	300 mA - 3 A	±100 ppm
AC current	10 μArms - 100 mArms 1 kHz sinusoidal wave	30 μArms - 30 mArms 1 kHz sinusoidal wave	±200 ppm
	1 Arms - 3 Arms 1 kHz sinusoidal wave	300 mArms - 3 Arms 1 kHz sinusoidal wave	±600 ppm
Resistance	10 Ω - 1 MΩ	30 Ω - 3 MΩ	±9 ppm
	10 MΩ	30 MΩ	±30 ppm
	100 MΩ	-	±100 ppm
	-	300 MΩ	±1000 ppm

8.3.3 Calibration Point

1. Perform calibration for each range of each measurement function. For more information on relationships between functions and ranges, refer to Table 5-2.
2. First, calibrate the zero point in each range calibration.
3. In the DC voltage measurement (includes DCV-Bch) and DC current measurement, calibrate the zero point and full-scale point.
4. In the resistance measurement, calibrate the zero point and full-scale point for 4WΩ measurement and calibrate only zero point for 2WΩ measurement.
5. In the low-power resistance measurement, calibrate the zero point and full-scale point for 4WΩ measurement and calibrate only zero point for 2WΩ measurement.
6. In the AC voltage measurement and AC current measurement, calibrate the zero, 1/100 full-scale, 1/10 full-scale, and full-scale points.
7. In the ACV(AC+DC) and ACI(AC+DC) measurements, no calibration is required because the instrument is calibrated when the calibration is performed in the DC voltage and AC voltage measurements.
8. In the continuity, diode and temperature measurement, no calibration is required because the instrument is calibrated when the calibration is performed in the DC voltage and AC voltage measurements.
9. In the frequency measurement, no calibration is required.

Table 8-2 Calibration Point (7461A/61P) (1 of 2)

Measurement function	Range	Zero	Calibration point1	Calibration point2	Calibration point3
DCV	100 mV	Short	+100 mV	-	-
	1000 mV	Short	+1000 mV	-	-
	10 V	Short	+10 V	-	-
	100 V	Short	+100 V	-	-
	1000 V	Short	+1000 V	-	-
ACV ACV(AC+DC)	100 mV	Short	1 mV, 1 kHz	10 mV, 1 kHz	100 mV, 1 kHz
	1000 mV	Short	10 mV, 1 kHz	100 mV, 1 kHz	1000 mV, 1 kHz
	10 V	Short	100 mV, 1 kHz	1000 mV, 1 kHz	10 V, 1 kHz
	100 V	Short	1000 mV, 1 kHz	10 V, 1 kHz	100 V, 1 kHz
	700 V	Short	7 V, 1 kHz	70 V, 1 kHz	700 V, 1 kHz
DCI	1000 μA	Open	+1000 μA	-	-
	10 mA	Open	+10 mA	-	-
	100 mA	Open	+100 mA	-	-
	1000 mA	Open	+1000 mA	-	-
	3 A	Open	+3 A	-	-

Table 8-2 Calibration Point (7461A/61P) (2 of 2)

Measurement function	Range	Zero	Calibration point1	Calibration point2	Calibration point3
ACI ACI(AC+DC)	1000 µA	Open	10 µA, 1 kHz	100 µA, 1 kHz	1000 µA, 1 kHz
	10 mA	Open	100 µA, 1 kHz	1000 µA, 1 kHz	10 mA, 1 kHz
	100 mA	Open	1000 µA, 1 kHz	10 mA, 1 kHz	100 mA, 1 kHz
	1000 mA	Open	10 mA, 1 kHz	100 mA, 1 kHz	1000 mA, 1 kHz
	3 A	Open	30 mA, 1 kHz	300 mA, 1 kHz	3 A, 1 kHz
LP-4WΩ LP-2WΩ (zero)	100 Ω	Short	100 Ω	-	-
	1000 Ω	Short	1000 Ω	-	-
	10 kΩ	Short	10 kΩ	-	-
	100 kΩ	Short	100 kΩ	-	-
	1000 kΩ	Short	1000 kΩ	-	-
	10 MΩ	Short	10 MΩ	-	-
4WΩ 2WΩ (zero)	100 Ω	Short	100 Ω	-	-
	1000 Ω	Short	1000 Ω	-	-
	10 kΩ	Short	10 kΩ	-	-
	100 kΩ	Short	100 kΩ	-	-
	1000 kΩ	Short	1000 kΩ	-	-
	10 MΩ	Short	10 MΩ	-	-
	100 MΩ	Short	100 MΩ	-	-

Table 8-3 Calibration Point (7451A) (1 of 2)

Measurement function	Range	Zero	Calibration point1	Calibration point2	Calibration point3
DCV	300 mV	Short	+300 mV	-	-
	3000 mV	Short	+3000 mV	-	-
	30 V	Short	+30 V	-	-
	300 V	Short	+300 V	-	-
	1000 V	Short	+1000 V	-	-
ACV ACV(AC+DC)	300 mV	Short	3 mV, 1 kHz	30 mV, 1 kHz	300 mV, 1 kHz
	3000 mV	Short	30 mV, 1 kHz	300 mV, 1 kHz	3000 mV, 1 kHz
	30 V	Short	300 mV, 1 kHz	3000 mV, 1 kHz	30 V, 1 kHz
	300 V	Short	3000 mV, 1 kHz	30 V, 1 kHz	300 V, 1 kHz
	700 V	Short	7 V, 1 kHz	70 V, 1 kHz	700 V, 1 kHz

Table 8-3 Calibration Point (7451A) (2 of 2)

Measurement function	Range	Zero	Calibration point1	Calibration point2	Calibration point3
DCI	3000 μA	Open	+3000 μA	-	-
	30 mA	Open	+30 mA	-	-
	300 mA	Open	+300 mA	-	-
	3000 mA	Open	+3000 mA	-	-
ACI ACI(AC+DC)	3000 μA	Open	30 μA, 1 kHz	300 μA, 1 kHz	3000 μA, 1 kHz
	30 mA	Open	300 μA, 1 kHz	3000 μA, 1 kHz	30 mA, 1 kHz
	300 mA	Open	3000 μA, 1 kHz	30 mA, 1 kHz	300 mA, 1 kHz
	3000 mA	Open	30 mA, 1 kHz	300 mA, 1 kHz	3000 mA, 1 kHz
LP-4WΩ LP-2WΩ (zero)	300 Ω	Short	300 Ω	-	-
	3000 Ω	Short	3000 Ω	-	-
	30 kΩ	Short	30 kΩ	-	-
	300 kΩ	Short	300 kΩ	-	-
	3000 kΩ	Short	3000 kΩ	-	-
	30 MΩ	Short	30 MΩ	-	-
4WΩ 2WΩ (zero)	30 Ω	Short	30 Ω	-	-
	300 Ω	Short	300 Ω	-	-
	3000 Ω	Short	3000 Ω	-	-
	30 kΩ	Short	30 kΩ	-	-
	300 kΩ	Short	300 kΩ	-	-
	3000 kΩ	Short	3000 kΩ	-	-
	30 MΩ	Short	30 MΩ	-	-
	300 MΩ	Short	300 MΩ	-	-

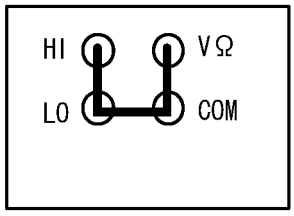
8.3.4 Calibration Procedure

Perform calibration in following procedure.

1. Set the calibration mode.
2. Short-circuit all input terminals on the rear panel.
3. Select the input terminal on the front panel.
4. Short-circuit the VΩ, COM, HI, and LO terminals and then open the mA terminal.
5. Calibrate the zero point in a function and range to be calibrated.
6. Connect the calibration standard to the input terminal on the front panel.
7. Calibrate the full-scale point in a function and range to be calibrated.
8. Calibrate the zero point for the input terminal on the rear panel. (The input terminal on the front panel is still selected.)
9. Cancel the calibration mode.

Setting the calibration mode

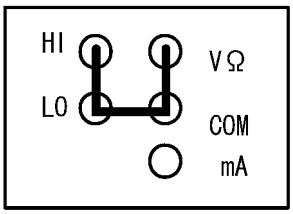
1. Select ON from 'CAL' in **9 SYS** in **MENU**.
2. Press **ENTER** to determine the setting.
   The CAL indicator is turned on and indicates that the calibration mode is set.
3. Press **EXIT** to exit from MENU.
4. Short-circuit between all input terminals on the rear panel.



5. Select FRONT from 'TERM' in **9 SYS** in **MENU**.
6. Set a function to be calibrated.

Calibrating the zero point

7. Short-circuit between the VΩ, COM, HI, and LO terminals and open the mA terminal.



8. Set a range to be calibrated.
9. Pressing the **HOLD** key displays the measurement value as a setting value.
10. Adjust the setting value to the calibration value by using the △, ▽, ◁, ▷ keys.

NOTE: ***Press the* EXIT *key if calibration must be cancelled due to a mistake such as entering an incorrect number.***

---

11. Press the **ENTER** key to execute the calibration.
12. Repeat steps 8 to 11 when calibrating other ranges.

Calibrating the full-scale point

13. Connect the calibration standard to the input terminal on the front panel.
14. Set a range to be calibrated.
15. Input a voltage equivalent to the calibration point.
16. Pressing the **HOLD** key displays the measurement value as a setting value.
17. Adjust the setting value to the calibration value by using the △, ▽, ◁, ▷ keys.

NOTE: ***Press the* EXIT *key if the calibration must be cancelled due to a mistake such as entering an incorrect number.***

---

18. Press the **ENTER** key to execute the calibration.
19. Repeat steps 14 to 18 when calibrating other ranges.
20. Repeat steps 6 to 19 when calibrating other functions.

Calibrating the zero point for the input terminal on the rear panel

21. Select 'R.CAL' from **9 SYS** in **MENU**.

22. Press the **ENTER** key to start the calibration.
23. Return to step 21 after displaying “PASS” or “FAIL”

    If “PASS” is displayed, press the **EXIT** key to exit.

    If “FAIL” is displayed, restart the calibration after checking that all input terminals on the rear panel are short-circuited.

    If FAIL occurs, the error is registered in the record and the ERR indicator lights.

Canceling the calibration mode

24. Select OFF from ‘CAL’ in **9 SYS** in **MENU**.
25. Press **ENTER** to determine the setting.

    The CAL indicator is turned off and indicates that the calibration mode is canceled.
26. Press **EXIT** to exit from MENU.

NOTE:

1. ***The* AUTO, RATE, TRIG, NULL, *and* SM *keys are disabled in the calibration mode.***
   ***Only* 8 I/F *and* 9 SYS *can be selected in* MENU.**
2. ***Before turning off the power, cancel the calibration mode.***
   Calibration data is stored in internal nonvolatile memory and written in it at once when the calibration mode is canceled.

8.4 Replacing Parts with Limited Life

Parts with limited life used in this instrument are listed in the following table.

Contact an ADC CORPORATION sales representative to replace any parts, referring to the limited lives listed in the following table.

Note that the life span may be shorter than that described depending on the operating environment, frequency of use, and storage environment.

MEMO: ***The table shows the life spans of parts and the number of times parts can be used before requiring replacement, but they are not guaranteed.***

Table 8-4 Parts with Limited Life

Name	Expected lifespan
Relays for switching functions and ranges	1,000,000 operations
Relays for switching input terminals between front and rear panels	1,000,000 operations
Panel keys	500,000 operations
Fluorescent display tube	20,000 hours

8.5 Product Disposal and Recycle

This product should be disposed of according to the regulations and laws that are established by your country and municipality.

When treating this product, separately collect components according to this section to prevent the spread of substances, which may be harmful to humans, and to protect the global environment.

Components, which must be separately collected, are shown in the following table.

The treatment of this product should comply with the relevant laws of your country and waste-disposal regulations of your company.

Table 8-5 Disposal (1 of 2)

Substance/Component	Used/ Not used	Location	Unit	Component	Quantity in maximum configuration
Polychlorinated biphenyls (PCB) containing capacitors	Not used	-	-	-	-
Mercury containing components	Not used	-	-	-	-
Batteries	Not used	-	-	-	-
Printed circuit boards	Used	Main frame	MAIN	Printed circuit board	1
			PANEL	Printed circuit board	1
			TERMINAL	Printed circuit board	1
			TERMINAL	Printed circuit board	1
Toner cartridges	Not used	-	-	-	-
Plastic containing brominated flame retardants	Used	Main frame	BLL-031484X05/04/03 BLC-031485	Connector, zener diode, photo coupler, FET, comparator, logic IC, FLASH, and transistor	15
Asbestos waste and components which contain asbestos	Not used	-	-	-	-
Cathode ray tubes	Not used	-	-	-	-
Chlorofluorocarbons (CFC), Hydrochlorofluorocarbons (HCFC), Hydrofluorocarbons (HFC) or Hydrocarbons (HC)	Not used	-	-	-	-
Gas discharge lamps	Not used	-	-	-	-
Liquid crystal displays of a surface greater than 100 square centimeters	Not used	-	-	-	-

Table 8-5 Disposal (2 of 2)

Substance/Component	Used/ Not used	Location	Unit	Component	Quantity in maximum configuration
External electric cables	Used	Main frame	Accessories	Power cable Input cable	1 2
Components containing refractory ceramic fibers	Not used	-	-	-	-
Components containing radioactive substances	Not used	-	-	-	-
Electrolyte capacitors containing substances of concern (height > 25 mm, diameter > 25 mm or proportionately similar volume)	Not used	-	-	-	-
Arsenic and Arsenic compounds	Used	Main frame	BLL-031484X05/04/03	Photo coupler and logic IC	2
Nickel and Nickel compounds	Used	Main frame		Electronic components and mechanical parts	Nickel-plated parts
Lead and Lead compounds	Used	Main frame	BLL-031484X05/04/03 BLC-031485	Lead solder used for assembling electronic components on printed circuit boards	41
Polyvinyl chloride (PVC)	Used	Main frame	Main frame electronics BLL-031484X05/04/03	PVC resin components	2
Antimony and Antimony compounds	Used	Main frame	BLL-031484X05/04/03 BLC-031485	Electronic components	26

8.6 Storage

If not using this instrument for a long time, put it in a carton case and store it in an area away from dust and direct sunlight.

Storage temperature: -25°C to +70°C

8.7 Transportation

When transporting this instrument, use the packing materials used for the shipping of this instrument or equivalent materials (thickness of carton case is 5 mm or more).

Packing procedure

1. Place the cushioning materials inside the carton case and cover all sides of this instrument with cushioning materials.
2. Place accessories in the carton case and then place the cushioning materials.
3. Close the carton case and bind it with packing strings.

8.8 Notes Regarding Repair, Replacement, and Periodic Calibration

8.8.1 Contacting ADC CORPORATION for Repair or Calibration

When sending this instrument to ADC CORPORATION sales representative, attach a tag that indicates the following information.

- Your company name and address
- Name of the person in charge
- Serial number (on the rear panel)
- What work to request (Repair or periodic calibration)

8.8.2 Address and Phone Number

Contact an ADC CORPORATION sales representative.

8.9 System Recovery Procedure

Initialization

The following shows how to initialize the setting parameters.

1. Select **6 INIT** from the **MENU** mode and press ▽ to enter the selection layer.
2. The 'Param' parameter is displayed. Press ▽ to enter the entry layer.
3. Press **ENTER** while 'Dflt' is displayed.

The parameters can be initialized by selecting and executing 'DFLT' when loading the parameter setting.

For more information on how to initialize the parameters, refer to Section 5.10, "Saving and Loading the Measurement Condition Settings."

Initial value when shipping

The following shows how to set all parameters except for the calibration value and calibration memo to the values when shipping.

1. Turn on the power while pressing the **NULL** and **UP** keys.

Quitting MENU

Press **EXIT** to quit the operation in MENU.

MENU is closed even if any layer in MENU is operated.

NOTE: ***Changing parameters, which are not determined by pressing*** **ENTER** ***, is canceled.***

Initial value

The list of conditions, under which the setting parameters are initialized, and initial values are shown in Table 8-6.

Table 8-6 Initial Values of Setting Parameters (1 of 3)

Parameter	When shipping	Selecting DFLT from 6INIT in MENU	Selecting P.OFF from 'PON.LD'	'LOAD' in 5PARAM		*RST	*CLS	DCL
				DFLT	USER setting			
Function	DCV	DCV	Saved value	DCV	Selecting USER0-3	DCV		
Selecting Bch	HI-COM	HI-COM	Saved value	HI-COM	Selecting USER0-3	HI-COM		
Range	Highest	Highest	Saved value	Highest	Selecting USER0-3	Highest		
Auto-range	ON	ON	Saved value	ON	Selecting USER0-3	ON		
Number of display digits.	Maximum	Maximum	Saved value	Maximum	Selecting USER0-3	Maximum		
Auto-zero	ON	ON	Saved value	ON	Selecting USER0-3	ON		
AC filter	MED	MED	Saved value	MED	Selecting USER0-3	MED		
Integration time	10 PLC	10 PLC	Saved value	10 PLC	Selecting USER0-3			
NULL calculation	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
NULL constant	0	0	Saved value	0	Selecting USER0-3	0		
COMP calculation	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
HIGH value	0	0	Saved value	0	Selecting USER0-3	0		
LOW value	0	0	Saved value	0	Selecting USER0-3	0		
MAX/MIN calculation	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Selecting the MAXIMUM and MINIMUM display	NORMAL	NORMAL	Saved value	NORMAL	Selecting USER0-3	NORMAL		
Smoothing calculation	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Smoothing count	10	10	Saved value	10	Selecting USER0-3	10		
Scaling calculation	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Constant A	1	1	Saved value	1	Selecting USER0-3	1		
Constant B	0	0	Saved value	0	Selecting USER0-3	0		
Constant C	1	1	Saved value	1	Selecting USER0-3	1		
dB calculation	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Constant D	1	1	Saved value	1	Selecting USER0-3	1		
Sampling HOLD	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Sampling rate	SLOW2	SLOW2	Saved value	SLOW2	Selecting USER0-3	SLOW2		
Sampling interval	500 ms	500 ms	Saved value	500ms	Selecting USER0-3	500 ms		
Trigger delay	0	0	Saved value	0	Selecting USER0-3	0		
Sampling count	1	1	Saved value	1	Selecting USER0-3	1		
Level trigger LEVEL	0	0	Saved value	0	Selecting USER0-3	0		

Blank: No Action

Saved value: Last setting value when the power is turned off.

Table 8-6 Initial Values of Setting Parameters (2 of 3)

Parameter	When shipping	Selecting DFLT from 6INIT in MENU	Selecting P.OFF from 'PON.LD'	'LOAD' in 5PARAM		*RST	*CLS	DCL
				DFLT	USER setting			
Level trigger SLOPE	POSITIVE	POSITIVE	Saved value	POSITIVE	Selecting USER0-3	POSITIVE		
Delta trigger DELTA	1	1	Saved value	1	Selecting USER0-3	1		
Delta trigger SLOPE	POSITIVE	POSITIVE	Saved value	POSITIVE	Selecting USER0-3	POSITIVE		
Delta trigger PATRN	EMISSION	EMISSION	Saved value	EMISSION	Selecting USER0-3	EMISSION		
Memory store	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Continuity threshold constant	10 Ω	10 Ω	Saved value	10 Ω	Selecting USER0-3	10 Ω		
DCV input resistance	HiZ	HiZ	Saved value	HiZ	Selecting USER0-3	HiZ		
Trigger count	1	1	Saved value	1	Selecting USER0-3	1		
Trigger source	IMMediate	IMMediate	Saved value	IMMediate	Selecting USER0-3	IMMediate		
Complete signal pulse width	100 µs	100 µs	Saved value	100 µs	Selecting USER0-3	100 µs		
Complete signal output mode	SINGLE	SINGLE	Saved value	SINGLE	Selecting USER0-3	SINGLE		
Input terminal	Front	Front	Saved value	Front	Selecting USER0-3	Front		
PASS/FAIL buzzer	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Buzzer	OFF	OFF	Saved value	OFF	Selecting USER0-3	OFF		
Interface	GPIB	GPIB	Saved value					
GPIB address	1	1	Saved value					
USB ID	1	1	Saved value					
Talk only	OFF	OFF	Saved value					
Header output	ON	ON	Saved value					
Panel lock	OFF		Saved value					
Disabling function	ENA	ENA	Saved value					
PSC flag	1	1	Saved value					
Programming language	ADC	ADC	Saved value					
Continue setting	OFF	OFF	Saved value			*1		
Password	0000		Saved value					
Measurement memory:	Empty	Empty	Empty	Empty	Empty	Empty		Empty
GPIB input output buffer	Empty	Empty	Empty	Empty	Empty			Empty

Blank: No Action

Saved value: Last setting value when the power is turned off.

*1: By selecting the initial value for continue setting
(The initial value selection of the Continuous setting is applied to software revision B00 and later.)

Table 8-6 Initial Values of Setting Parameters (3 of 3)

Parameter	When shipping	Selecting DFLT from 6INIT in MENU	Selecting P.OFF from 'PON.LD'	'LOAD' in 5PARAM		*RST	*CLS	DCL
				DFLT	USER setting			
Block delimiter	CR/LF+EOI	CR/LF+EOI	CR/LF+EOI	CR/LF+EOI	CR/LF+EOI	CR/LF+EOI		
Memory readout range	0,0	0,0	0,0	OFF	0,0			0,0
SRQ	ON	ON	ON	ON	ON			
Display ON/OFF	ON	ON	ON	ON	ON	ON		
Calibration mode	OFF	OFF	OFF			OFF		
Status register QEER	0	0	0					
Status register OEER	0	0	0					
Status register SESER	0	0	0					
Status register MEER	0	0	0					
Status register SRER	0	0	0					
Status register QER	0	0	0				0	
Status register SESR	0	0	0				0	
Status register STB	0	0	0				0	
Status register OER	0	0	0				0	
Status register MER	0	0	0				0	
Error queue	Empty	Empty	Empty				Empty	
Calibration memo	""							

Blank: No Action

Saved value: Last setting value when the power is turned off.

8.10 Self-Test

8.10.1 Self-Test

The self-test can be performed from the panel operations or by executing the remote command.

The self-test is performed by selecting 'TEST' from **9 SYS** in **MENU** or executing the *TST? remote command.

The test items are shown in Table 8-7.

Table 8-7 Self-Test Items

Item	Can be performed from the Panel or by Remote	Can be performed when the power is turned on
ROM check sum	-	○
RAM read/write check	-	○
Panel display check	-	○
Alarm status check	-	○
Calibration data check	○	○
Parameter check	○	○
Analog to logic communication check	○	○
AD operation check in analog	○	○
Range operation check in analog	○	○
Reference voltage check in analog	○	○
Reference current check in analog	○ (except 7451A)	○ (except 7451A)

8.10.2 Display and Key Tests

Turn on all segments of the display and check whether they light.

Pressing panel keys display their names.

Test according to the following procedure.

	Operations	Display
1.	Press **MENU** and select **9 SYS** by using ◁ and ▷.	MENU SYS
2.	Press ▽.	SYS BEEP

	Operations	Display
3.	Select ‘P/KEY’ by using ▷.	SYS P/KEY
4.	Press ▽.	P/KEY ENTER
5.	Press **ENTER**.	All segments of the display are turned on.
	The display changes after a few seconds.	KEY Push
6.	Press a key to be checked.	KEY Key name to be pressed
7.	To quit the key test, press **SHIFT**.	KEY SHIFT
	The display changes from “SHIFT” to “Done”.	KEY DONE
8.	Press **EXIT** to exit from MENU.	Measurement state

8.11 Error Message List

This section describes the error number and how to deal with error displayed on this instrument.

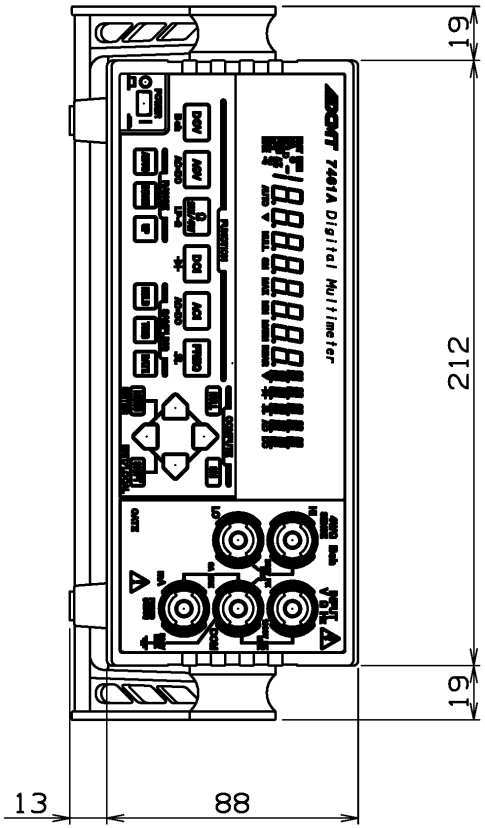
No.	Error display	Error description [ ]: Response to the ERR? command	Solution
1	Err1	RAM read/write error [RAM read/write failed]	No solution
2	Err2	Display RAM test error or Communication error [Front panel does not respond]	No solution
3	Err3	Calibration data check SUM error [Calibration memory lost]	No solution
4	Err4	ROM check SUM error [ROM Chk SUM]	No solution
5	Err5	Alarm check error when the power is turned on [PON Alarm Check]	No solution
6	Err6	Parameter check SUM error [Save/recall memory lost]	No solution
7	Err16	FLASH memory writing error [FLASH memory error]	No solution
8	Err101	The function cannot be set. [Inhibited function]	Solution1
9	Err150	USB communication error [Illegal packet received]	Solution2
10	Err161	HI ALARM occurs. [+Overload ALARM for OHM circuit]	Solution3
11	Err162	LO ALARM occurs. [-Overload ALARM for OHM circuit]	Solution3
12	Err200	Ratio test error of IR1 to IR2 in AD [AD Ratio 1-2]	No solution
13	Err201	Ratio test error of IR2 to IR3 in AD [AD Ratio 2-3]	No solution
14	Err202	Ratio test error of IR3 to IR4 in AD [AD Ratio 3-4]	No solution
15	Err203	Ratio test error of IR4 to IR5 in AD [AD Ratio 4-5]	No solution
16	Err205	HV×1 zero measurement test error in analog operation [HV ×1 Zero Check]	No solution
17	Err206	DC filter zero measurement test error in analog operation [LV ×1 Zero Check]	No solution
18	Err207	LV×10 zero measurement test error in analog operation [LV ×10 Zero Check]	No solution

No.	Error display	Error description [ ]: Response to the ERR? command	Solution
19	Err208	LV×100 zero measurement test error in analog operation [LV ×100 Zero Check]	No solution
20	Err210	Communication error in Analog [Communication error (Analog)]	No solution
21	Err215	Ref5 V measurement test error in analog operation [Ref 5V Check]	No solution
22	Err216	HV×1 ATT measurement test error in analog operation [Ref 5V(ATT) HV×1 Check]	No solution
23	Err217	LV×10 ATT measurement test error in analog operation [Ref 5V(ATT) LV×10 Check]	No solution
24	Err218	LV×100 ATT measurement test error in analog operation [Ref 5V(ATT) LV×100 Check]	No solution
25	Err220	Is10 mA measurement test error in analog operation [OhmIs 10mA Check]	No solution
26	Err221	Is5 mA measurement test error in analog operation [OhmIs 5mA Check]	No solution
27	Err222	Is1 mA(1) measurement test error in analog operation [OhmIs 1mA_1 Check]	No solution
28	Err223	Is1 mA(2) measurement test error in analog operation [OhmIs 1mA_2 Check]	No solution
29	Err224	Is100 μA measurement test error in analog operation [OhmIs 100μA Check]	No solution
30	Err225	Is10 μA measurement test error in analog operation [OhmIs 10μA Check]	No solution
31	Err229	Leak measurement test error in analog operation [Leak Check]	No solution
32	Err500	Calibration data error [CAL data]	Solution4
33	Err501	ZERO calibration is not performed. [before ZERO]	Solution5
34	Err-102	Command syntax error [Cmd Syntax]	Solution6
35	Err-113	The command cannot be used. [Undefined header]	Solution6
36	Err-121	Unrecognized character as the numerical value [Invalid character in number]	Solution6
37	Err-141	Invalid character string parameter is entered. [Invalid character data]	Solution6
38	Err-151	Error as the character string data (Quotation mark error) [Invalid string data]	Solution6

No.	Error display	Error description [ ]: Response to the ERR? command	Solution
39	Err-200	Execution error (Command which cannot be executed now) [Execution error]	Solution7
40	Err-213	The INITiate command is ignored because the measurement is performing. [Init ignored]	Solution7
41	Err-214	The READ? or MEAS? command is received when any trigger source except for IMM is set. [Trigger deadlock]	Solution7
42	Err-221	The command cannot be executed in the current setting condition. [Settings conflict]	Solution7
43	Err-222	Entered value is outside the setting range or lacking for necessary parameters [Data out of range]	Solution7
44	Err-313	Calibration data is lost. [Calibration memory lost]	Solution8
45	Err-314	Parameters saved by “*SAV” command are lost. [Save/recall memory lost]	Solution9
46	Err-315	Saved parameters are lost. [Configuration memory lost]	Solution7
47	Err-330	Self-test error [Self-test failed]	No solution
48	Err-350	Error queue overflows. [Queue overflow]	Solution10

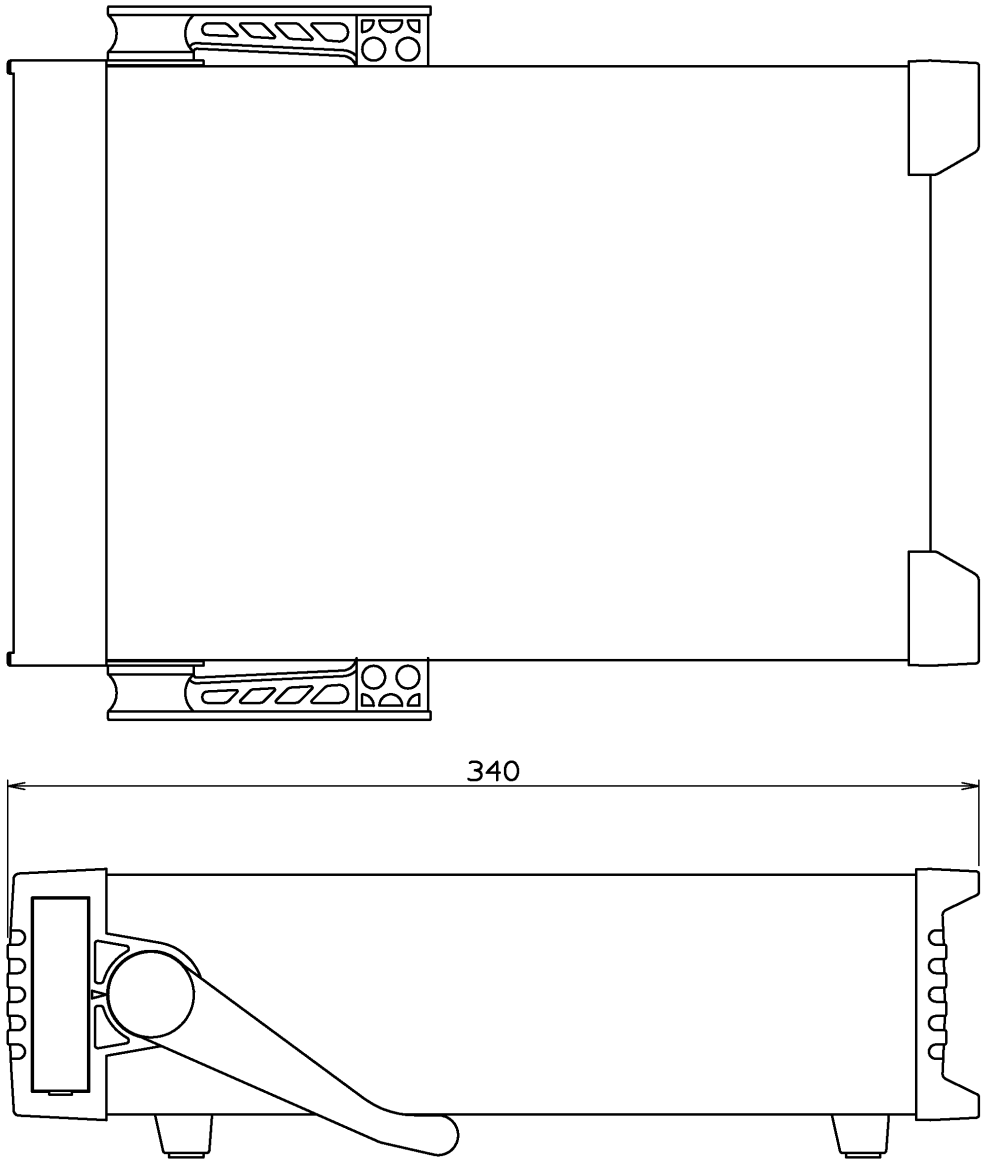
Solution1	Set the correct function.
Solution2	Check the USB cable connection
Solution3	Check the entered value.
Solution4	Calibrate according to the correct procedure.
Solution5	Perform the ZERO cancellation.
Solution6	Send the correct command
Solution7	Check the settings.
Solution8	Re-calibrate.
Solution9	Re-set.
Solution10	Read out the error queue.

If the error cannot be fixed by the described solution, contact an ADC CORPORATION sales representative this instrument.



Unit : mm

NOTE
This drawing shows external dimensions of this instrument.
The difference in products and options used can cause a change in the appearance of the instrument.



DIMENSIONAL OUTLINE DRAWING

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